

From Experimental Modeling of Shotcrete to Numerical Simulations of Tunneling

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Abstract

Shotcrete shells constitute an essential part of modern tunnel construction according to the New Austrian Tunneling Method (NATM): Immediately after excavation of a tunnel segment, layers of shotcrete are applied, forming a supporting shell which is loaded already at a very early age of the material. In the past decades, huge effort was spent on the improvement of wet-mix shotcrete compositions, which nowadays are characterized by a fast setting and hydration process, ductile behavior in compression, and good adhesion on rock surfaces even for thick layers applied overhead – making shotcrete an ideal securing measure in tunneling. Shotcrete exhibits a highly nonlinear and time-dependent material behavior, in particular due to the hydration-dependent evolution of stiffness and strength. In the present review, an advanced constitutive model for shotcrete representing the time-dependent and nonlinear material behavior is discussed. It accounts for the time-dependent evolution of material properties due to hydration of shotcrete, hardening and softening material behavior, the latter resulting from cracking of shotcrete, based on the combination of the theories of plasticity and continuum damage mechanics, as well as nonlinear creep and shrinkage. Since softening material behavior results in pathological mesh dependence in finite element simulations, a sophisticated extension of the material model for obtaining mesh-objective results is presented, based on the implicit gradient enhanced continuum. For calibrating the material model, a comprehensive experimental program was conducted, for which specimens were sampled directly on site during the construction of the Brenner Base Tunnel. It focused on the evolution of stiffness and strength, the creep and shrinkage behavior, as well as the specific mode I fracture energy. Finally, the calibrated shotcrete model is applied in a finite element study of the construction of an emergency stretch of the Brenner Base Tunnel. For this specific stretch, a single, permanent single shotcrete lining is used throughout the complete structural life time, which motivates for a detailed analysis of the load bearing capacity and the assessment of the degree of utilization.

Contents

1	Introduction	5
1.1	Time-dependent Nonlinear Material Behavior of Shotcrete	7
1.2	State of the Art – Constitutive Models for Shotcrete	8
1.2.1	Simplified Early Approaches	8
1.3	Advanced Approaches	9
1.4	Aim and Focus	10
2	The Shotcrete Damage Plasticity (SCDP) Model	12
2.1	Evolution of Instantaneous Elastic and Creep Strain	13
2.2	Evolution of the Shrinkage Strain	15
2.3	Evolution of the Plastic Strain	15
2.4	Evolution of Material Strength	17
2.5	Softening Material Behavior	17
2.6	Mesh Objective Representation of Softening Material Behavior in the SCDP Model .	18
2.7	Over-Nonlocal Gradient-Enhanced Formulation of the SCDP Model	22
3	Numerical Algorithm of the SCDP Model at Finite Element Level	24
3.1	Weak Form of the Governing Equations	24
3.2	Spatial Discretization	24
3.3	Discretization in Time	25
4	Stress Update Algorithm of the SCDP model at Integration Point Level	27
4.1	Shrinkage Strain Increment	28
4.2	Viscoelastic Strain Increment	28
4.3	Flow Strain Increment	31
4.4	Plastic Strain Increment	31
4.5	Damage Increment	35
4.6	Determination of the Parameters for the Dirichlet Series	35
4.7	Algorithmic Tangents	37
4.8	Summary of the Stress Update Algorithm for the Gradient-Enhanced SCDP Model .	38
5	Experimental Program on Young Shotcrete and Calibration of the SCDP model	40
5.1	Evolution of the Young's Modulus and the Uniaxial Compressive Strength	42
5.2	Shrinkage and Creep Behavior of Young Shotcrete	42
5.3	Specific Mode I Fracture Energy of Shotcrete	44
5.4	Summary of the Material Parameters of the SCDP model	47
6	Application of the SCDP Model in a Numerical Study	48
6.1	Finite Element Simulation of Tunnel Advance	49
6.2	Time-dependent construction process	51
6.3	Linear-Elastic Perfectly-Plastic Hoek-Brown Model for the Rock Mass	51
6.4	Calibration of the Finite Element Model	53

6.5	Case study I: Predicted Time-dependent Displacements and Utilization of the Shotcrete Shell	54
6.6	Case study II: Influence of the Construction Procedure of the Shotcrete Shell on the Predicted Time-dependent Structural Response	57
6.7	Case study III: Influence of Nonlinear Creep Behavior of Shotcrete	59
6.8	Case study IV: Prediction of Potential Structural Collapse	61
7	Summary, Conclusions and Outlook	64

1 Introduction

The *New Austrian Tunneling Method* (NATM) is a modern method for designing and constructing tunnels, taking advantage of the inherent geological strength of the surrounding rock mass or soil for stabilizing the tunnel structure. Its name goes back to Rabcewicz (1964), one of the pioneers who developed the method during the 1950s. A clear definition is constituted in (ÖIAV, 1980) in terms of the 22 elementary laws originally formulated by L. Müller and Fecker (1978).

The basic principle of the NATM can be described as follows: During excavation, the strength of the surrounding soil or rock mass is sustained to the greatest possible extent, and the excavation surface is supported by a layer of shotcrete and optional reinforcements like rock anchors and steel arches. The process is accompanied by continuous exploration and monitoring in order to control the stabilization of the structure, and for optimizing the construction process.

The use of shotcrete is essential to the NATM, as one aspect of tunnel construction is the transmission of stresses from the surrounding rock mass to the tunnel support structure. In this context, shotcrete is considered as the *ideal* securing material (B. Maidl, Heimbecher, Schmid, Jodl, and Petri, 1997), since it allows to create a continuous bond between the excavation surface and the tunnel support structure. Shotcrete is conveyed under pressure through a pneumatic hose or pipe, and sprayed into place at high velocity with simultaneous compaction due to impact, forming a supporting shell (DIN 18851, 2005). Due to an added chemical accelerator, fast setting of the young material is achieved, resulting in good adhesion on excavation surfaces, and making it possible to apply thick layers of shotcrete even at overhead areas.

Shotcrete can be applied immediately after an excavation step, requires no form work, allows to form complex geometries like those present at junctions, and the application in partial areas is possible (B. Maidl, 1992; B. Maidl, Thewes, and U. Maidl, 2013). Due to the spraying process, the thickness of the shotcrete shell can be adjusted to meet specific local requirements, and shotcrete can be combined with additional securing measures like rock anchors or steel arches, as shown in Figure 1. In contrast to normal concrete structures, which are loaded commonly at matured material stages, shotcrete shells are loaded already at a very early age of the material, i.e., usually a couple of hours after application due to the continuing tunnel advance. One important property of shotcrete is the ability to sustain large deformations, imposed by movements of surrounding rock mass due to stress rearrangement especially at early ages of the material (B. Maidl, Heimbecher, Schmid, Jodl, and Petri, 1997). In this context, its very ductile behavior is commonly denoted as the *good-natured* behavior of shotcrete (Pöttler, 1990a).

Basically, two different techniques are employed for creating shotcrete compositions (Thomas, 2008): *Dry-mix* shotcrete and *wet-mix* shotcrete. For the former technique, a dry composition, consisting of sand, cement and aggregate is conveyed pneumatically to the spraying nozzle, and water and a chemical accelerator are added directly at the nozzle. For the latter technique, a wet ready-mixed composition is conveyed to the nozzle, where solely the accelerator is added. Both techniques are widely used, but in general wet-mix shotcrete became more popular in the past 20 years for large scale applications like tunnel construction, since the wet-mix process allows for a better quality control, produces less rebound, i.e., blown back non-adhering shotcrete, and the spraying process is associated with a lower dust pollution. Compared to normal concrete, modern wet-mix shotcrete compositions are characterized by a higher water to cement ratio, a



Figure 1: Application of shotcrete as securing measure during tunnel advance at the Brenner Base Tunnel, in combination with rock bolts and steel arches.

smaller maximum aggregate size, the chemical accelerator, which is added for an increased hydration speed, and additional admixtures for improving properties like adhesion or workability of shotcrete (Thomas, 2008). Optional reinforcements consisting of steel meshes or short steel or synthetic fibres are used to increase the tensile strength of shotcrete.

For non-mechanized tunneling, the conventional construction methods of tunnel linings are distinguishing between *primary and secondary linings* (Thomas, 2008): The shotcrete shell applied as a securing measure during the excavation process is employed exclusively as a temporary securing measure, denoted as the *primary* shell, whereas a later on installed second shell, the inner *secondary* shell, made either from cast in place concrete or shotcrete, is used as part of the permanent support structure. Usually, the inner shell and the outer shell are separated by waterproofing measures like sheet membranes or sprayed-on membranes. Following this design approach, for dimensioning of the inner shell it is assumed that the outer shotcrete shell deteriorates completely with ongoing time and accordingly, the complete load is carried exclusively by the inner shell. However, the assumption of a complete deterioration of the inner shell is in contrast to results of long-term observations (Galler and Lorenz, 2017).

Alternatively, the *single shell lining* approach aims at permanent usage of the primary shotcrete shell as part of the tunnel support structure (B. Maidl, 1992; Golser and Kienberger, 1997; Thomas, 2008). This permanent shotcrete shell can consist of several layers of shotcrete, which can be applied at different times during tunnel construction. Compared to primary and secondary linings, considerable cost reductions can be achieved by this approach. A comprehensive summary of tunnels with single shell linings can be found in (Franzen, Garshol, and Tomisawa, 2001).

For numerical analyses of tunnel construction and tunnel advance, constitutive models for representing the behavior of the involved materials are necessary. Compared to normal concrete, material modeling of shotcrete has received considerably less attention in the literature. While this is clearly related to the narrow field of application, to a certain extent this might be attributed also to the complex nature of numerical simulations of tunnel advance: Although often simplified by means of two dimensional models, tunnel advance is a three dimensional mechanical problem, which is time-dependent due to the sequential construction process, and due to the time-dependent behavior

of the involved materials, in particular that of shotcrete. Since material properties of shotcrete are continuously developing due to the hydration process, constitutive modeling of shotcrete can be performed only in a realistic manner by considering its material properties as time-dependent. In addition to the complex construction process and the nonlinear, time-dependent material behavior, the actual loads imposed on the tunnel support structure can be estimated only roughly. This is due to unknown in-situ stresses prevailing in the surrounding rock mass, due to the complex and often anisotropic material behavior of rock, due to possible joints and fault zones within the surrounding rock mass, and due to limited exploration techniques for estimating the geometric and mechanical properties of those joints and fault zones.

1.1 Time-dependent Nonlinear Material Behavior of Shotcrete

Similar to concrete, the description of the material behavior of shotcrete is a complex issue since the material exhibits a number of nonlinear and time-dependent phenomena. These phenomena are summarized according to Meschke (1996) and Thomas (2008) as:

- The material properties are dependent on the degree of hydration, and thus dependent on the material age. For instance, stiffness and strength are increasing during hydration, while ductility is decreasing. Compared to normal concrete, due to the added chemical accelerator material properties are evolving considerably faster at early material ages, i.e., in the first hours after application of the shotcrete shell.
- The theory of linear elasticity for describing the stress-strain relation for an instant of time is admissible only for comparatively low stress levels: In uniaxial compression, approximately linear-elastic material behavior is observed for ratios of the applied stress to the uniaxial compressive strength of about up to 30 %. For higher load levels, inelastic, i.e., plastic, strains are observed after unloading. Furthermore, shotcrete exhibits hardening material behavior in compression, and behaves increasingly more ductile with increasing confining pressure (Sezaki, Kibe, Ichikiwa, and Kawamoto, 1989). After attaining the material strength, shotcrete exhibits softening material behavior, accompanied by stiffness degradation. Material failure in compression is observed in form of cracking or crushing, dependent on the level of confinement.
- By contrast, in uniaxial tension the material behavior of shotcrete is approximately linear-elastic nearly up to peak stress, followed by quasi-brittle failure due to cracking. Compared to its uniaxial compressive strength, shotcrete exhibits a considerably lower uniaxial tensile strength. Experimental tests indicate a ratio of the uniaxial tensile strength to the uniaxial compressive strength of 10 % to 20 %. As demonstrated by means of experimental tests reported by Hammer (2018), this ratio is also dependent on the degree of hydration, with higher ratios observed at younger material ages.
- Shotcrete exhibits shrinkage, which denotes the phenomenon of a time-dependent decrease in volume, independent of the applied load. According to the ACI Committee 209 (1992), shrinkage can be subdivided into autogenous shrinkage, which is the part of shrinkage which is exclusively related to hydration in absence of moisture exchange, and drying shrinkage, which is related to moisture exchange.

- Under sustained loading, shotcrete tends to creep, which denotes the phenomenon of a load-dependent, time-dependent increase of deformations. Creep is subdivided into basic creep, which is the part of creep independent of any moisture exchange with the ambient environment, and drying creep, which is the additional part of creep related to simultaneous drying. Similar to the evolution of stiffness and strength, the creep behavior of shotcrete is dependent on the degree of hydration, and thus dependent on the material age.
- Compared to normal concrete, nonlinear creep is an important issue in the construction of shotcrete shells: Linear creep is characterized by the linear relation between the creep strain accumulated in a certain time period and an applied constant stress. Experimental results on concrete, e.g. by Mazzotti and Savoia (2002), indicate that for uniaxial compression the assumption of linear creep is valid up to ratios of the applied stress to the compressive strength of approximately 30 % to 40 %, whereas for higher load levels a disproportionally higher increase of the creep strain rate is observed. However, whereas the assumption of linear creep is usually admissible for normal concrete structures, shotcrete shells may undergo considerably higher degrees of loading, resulting in nonlinear creep.
- The coefficient of thermal expansion is dependent on the degree of hydration, and thus dependent on the age of the material.

Advanced constitutive models aim at the realistic representation of the described nonlinear, time-dependent material phenomena. For calibrating such advanced material models, experimental results on the material behavior are necessary. However, conducting experimental tests on shotcrete is considerably more complex compared to normal concrete: While common, casted concrete specimens can be sampled usually under controlled laboratory conditions, shotcrete specimens must be sampled employing the same spraying process as used at the construction site since the spraying process has a substantial influence on the resulting properties of shotcrete. An attempt to simulate the spraying process under controlled laboratory conditions is reported in (Thewes and Vollmann, 2007; Nakashima, Hammer, Thewes, Elshafie, and Soga, 2015), employing a test rig equipped with a robot arm from the automotive industry for the application of shotcrete.

1.2 State of the Art – Constitutive Models for Shotcrete

Unlike to normal concrete, only few constitutive models explicitly developed for shotcrete have been proposed in the literature so far. Due to the considerable utilization of shotcrete shells at young material ages in tunnel construction, advanced material models for shotcrete need to represent simultaneously the time-dependent material behavior of shotcrete, i.e., hydration dependent evolution of stiffness, strength and ductility, creep and shrinkage, and the complex hardening and softening material behavior in compression and tension. This is in contrast to most constitutive models for normal concrete, which usually focus only on one or a few of those aspects.

1.2.1 Simplified Early Approaches

Simplified early approaches focused only on a rough approximation of the material behavior of shotcrete. The most simple approach for representing the material behavior of shotcrete in numerical

simulations is the assumption of linear-elastic material behavior (Thomas, 2008). Commonly, a time-dependent Young's modulus is assumed to account for the evolution of material stiffness. Linear-elastic models are still widely used in engineering practice for describing the material behavior of shotcrete in tunneling simulations, and reports of application can be found in, e.g., (Schuller and Schweiger, 2002; Ng, Lee, and Tang, 2004; Yeo et al., 2009; Kivi, Sadaghiani, and Ahmadi, 2012).

An early attempt to account for the time-dependent material behaviour of shotcrete in an approximate manner was proposed by Pöttler (1990b). In this relatively simple approach, a reduced Young's modulus for shotcrete is used for considering effects like time-dependent stiffness, creep, shrinkage and even the time-dependent loading during the excavation process. By combining these effects in a hypothetical modulus of elasticity (HME), approximate solutions for problems with axisymmetric boundary conditions can be obtained. Problematic is the lack of calibration parameters for the model in the literature, and thus, the HME is based more or less only on experience.

A model for describing the evolution of stiffness and creep of shotcrete was published as the rate of flow method. The model was originally proposed by Schubert (1988) for shotcrete, and extended by Aldrian (1991) to account for nonlinear creep. In (M. Müller, 2001), the calibration of the material model, based on a comprehensive experimental program on a dry-mix shotcrete composition, was presented. A brief summary of the model and a critical discussion are presented in (Thomas, 2008).

1.3 Advanced Approaches

Compared to the early, simple approaches, advanced models for shotcrete take into account simultaneously the time-dependent and the nonlinear hardening and softening material behavior, i.e., plastic or damaging material behavior due to loading.

A viscoplastic constitutive model for shotcrete was presented by Meschke (1996). The model represents time-dependent stiffness and strength of shotcrete, creep and shrinkage as well as plastic material behavior in compression and tension. To this end, it is based on a hardening Drucker-Prager yield criterion for describing the material behaviour for compressive and mixed stress states, and a softening Rankine criterion is used for describing the material behaviour in tension. Associated flow rules are employed for describing the evolution of viscoplastic strains. The model takes into account time-dependent stiffness due to hydration based on hyperelastic relations, and the evolution of material strength and ductility is considered by means of empirical time functions. An application of the model to 3D simulations of shallow tunnel advance is reported in (Meschke, Kropik, and Mang, 1996).

Shotcrete models based on the theory of thermo-chemo-plasticity consider a dependency of material properties on the degree of hydration. In (Ulm and Coussy, 1995), a general strategy to develop such constitutive models is documented, based on the theory of porous media proposed by Coussy (1995). A constitutive model based on thermo-chemo-plasticity theory was proposed by Hellmich, Ulm, and Mang (1999a) and was further developed by Sercombe, Hellmich, Ulm, and Mang (2001) and Lackner, Hellmich, and Mang (2002). In (Hellmich, Ulm, and Mang, 1999b), a report of the application of the thermo-chemo-plastic model can be found. For considering early age cracking of shotcrete, the model was further extended by Lackner and Mang (2003, 2004). The final model represents hydration-dependent evolution of stiffness and strength, creep and shrinkage as well as plastic material behavior in compression and tension.

A constitutive model based on the theories of viscoelasticity and plasticity, representing time-

dependent stiffness and strength, nonlinear strain hardening and softening, creep and shrinkage, was presented by Schütz, Potts, and Zdravkovic (2011). Partially based on this model, an extended approach was presented by Schädlich and Schweiger (2014). The model by Schädlich and Schweiger includes strain hardening and strain softening material behavior, time-dependent evolution of stiffness and strength, and nonlinear creep within the framework of viscoelasticity. To delimit the elastic domain, the Mohr-Coulomb and the Rankine yield criteria are employed, and plastic flow is modeled by means of non-associated flow rules. The model was validated and applied in (Schädlich, Schweiger, Marcher, and Saurer, 2014a) and (Schädlich, Schweiger, Marcher, and Saurer, 2014b).

A novel damage plasticity model for shotcrete, the SCDP model, was proposed in (Neuner, Gamnitzer, and Hofstetter, 2017) and extended in (Neuner, Schreter, and Hofstetter, 2018). It is based on the damage plasticity model for normal concrete by Grassl and Jirásek (2006a) for representing hardening and softening material behavior, supplemented by a modified version of the solidification theory (Bažant and Prasannan, 1989a) for representing the evolution of stiffness and nonlinear creep. In addition, empirical time functions are used for describing the evolution of material strength, ductility as well as shrinkage strains. A comparison of the SCDP model with the shotcrete models by Schädlich and Schweiger (2014) and Meschke (1996) based on experimental results from the literature is presented in (Neuner, Gamnitzer, and Hofstetter, 2017), and based on a comparison at structural level by means of a finite element study in (Neuner, Schreter, Unteregger, and Hofstetter, 2017). The model was applied successfully in several 2D and 3D numerical studies on stretches of the Brenner Base Tunnel in (Schreter, Neuner, Unteregger, and Hofstetter, 2018; Neuner, Schreter, Gamnitzer, and Hofstetter, 2020; Neuner, Dummer, Cordes, Schreter, and Hofstetter, 2020).

The so far mentioned constitutive models for shotcrete represent the time-dependent, nonlinear material behavior of shotcrete in a phenomenological manner. However, the complex material behavior of shotcrete is a consequence of the strongly heterogeneous micro structure of the material, consisting of different constituents, e.g., water, unbound cement, hardened cement paste, aggregates, and the chemical reaction between water, cement and potential chemical admixtures, i.e, the hydration process. Taking into account this micro mechanical structure and the chemical mechanisms at micro level by means of multi-scale approaches (Horstemeyer, 2009; Budarapu, Zhuang, Rabczuk, and S. P. Bordas, 2019) allows to estimate material properties depending on the shotcrete composition. For instance, approaches for predicting the evolution of stiffness and strength of shotcrete depending on the employed shotcrete composition by means of multi-scale models have been presented by Hellmich and Mang (2005) and Pichler, Scheiner, and Hellmich (2008).

1.4 Aim and Focus

In the present chapter, a concise summary of the recent work by the authors focusing on numerical and experimental modeling of shotcrete is presented. This work emerged mainly in the context of the Brenner Base Tunnel project, currently constructed between Austria and Italy (Bergmeister, 2011). The paper is structured as follows: First, the SCDP model for shotcrete is described, and an extension of the model to the implicit gradient-enhanced framework for a mesh objective representation of material failure in numerical simulations is presented. Furthermore, details on the implementation of the SCDP model into a finite element framework are explained. Next, a recently performed experimental program (Neuner, Cordes, Drexel, and Hofstetter, 2017) is presented, which was used for calibrating the SCDP model, and the application of the calibrated SCDP model to a

numerical study of a stretch of the Brenner Base Tunnel is demonstrated. The review concludes with a discussion on the numerical results. Finally, an outlook over currently ongoing research efforts on numerical and experimental modeling of shotcrete is presented.

2 The Shotcrete Damage Plasticity (SCDP) Model

The SCDP model (Neuner, Gammitzer, and Hofstetter, 2017; Neuner, Schreter, and Hofstetter, 2018) is derived from several material models for normal concrete, including the concrete damage plasticity (CDP) model by Grassl and Jirásek (2006a) for representing hardening and softening material behavior, the solidification theory (Bažant and Prasannan, 1989a) for representing time-dependent stiffness and nonlinear creep, and empirical relations for representing shrinkage (ACI Committee 209, 1992; Bažant and Panula, 1978). The stress-strain relation of the SCDP model is expressed in total form as

$$\begin{aligned}\boldsymbol{\sigma} &= (1 - \omega) \bar{\boldsymbol{\sigma}} \\ &= (1 - \omega) \mathbb{C} : \boldsymbol{\varepsilon}^{\text{el}} \\ &= (1 - \omega) \mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\text{p}} - \boldsymbol{\varepsilon}^{\text{ve}} - \boldsymbol{\varepsilon}^{\text{f}} - \boldsymbol{\varepsilon}^{\text{shr}}),\end{aligned}\tag{1}$$

in which $\boldsymbol{\sigma}$ denotes the nominal stress, i.e., force per total area, $\bar{\boldsymbol{\sigma}}$ the effective stress, i.e., force per undamaged area, ω is the isotropic scalar damage parameter and $\mathbb{C}$ is the fourth order stiffness tensor. The total strain $\boldsymbol{\varepsilon}$ is decomposed into the instantaneous elastic strain $\boldsymbol{\varepsilon}^{\text{el}} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^{\text{p}} - \boldsymbol{\varepsilon}^{\text{ve}} - \boldsymbol{\varepsilon}^{\text{f}} - \boldsymbol{\varepsilon}^{\text{shr}}$, the plastic strain $\boldsymbol{\varepsilon}^{\text{p}}$, the viscoelastic strain $\boldsymbol{\varepsilon}^{\text{ve}}$, the flow (viscous) strain $\boldsymbol{\varepsilon}^{\text{f}}$ and the shrinkage strain $\boldsymbol{\varepsilon}^{\text{shr}}$.

The effective stress-strain relation according to the SCDP model can be illustrated by means of an equivalent uniaxial rheological model, as depicted in Figure 2. While this uniaxial rheological model is a simplification of the actual structure of the three-dimensional SCDP model, it presents the basic mechanisms of the model in an intuitive manner. Therein,

- the instantaneous elastic strain is represented by a spring with time-independent stiffness,
- the plastic strain by a hardening Saint-Venant unit,
- the viscoelastic strain by an aging Kelvin chain,
- the flow strain by a dashpot with time-dependent viscosity and
- the shrinkage strain by a stress-independent, time-dependent portion of the total strain.

The representation of the viscoelastic strain in terms of an aging Kelvin chain corresponds to the numerical approximation of the viscoelastic formulation, as explained in the subsequent sections. The evolution of the individual strain components is described in the following.

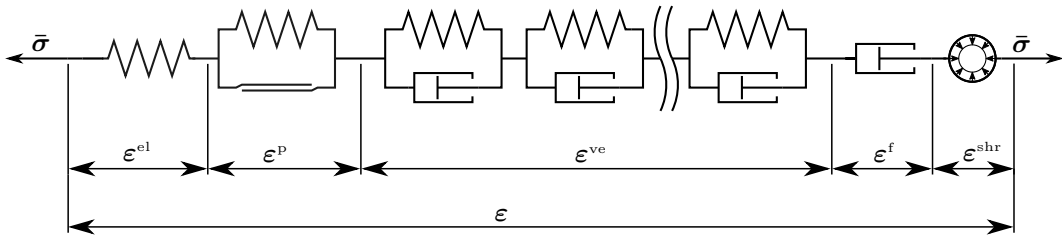


Figure 2: Uniaxial rheological model representing the relation between the effective stress $\bar{\sigma}$ and the total strain ε according to the SCDP model, based on the decomposition of ε into the elastic strain ε^{el} , the plastic strain ε^{p} , the viscoelastic strain ε^{ve} , the flow strain ε^{f} and the shrinkage strain ε^{shr} .

2.1 Evolution of Instantaneous Elastic and Creep Strain

The rates of the instantaneous elastic strain $\dot{\boldsymbol{\varepsilon}}^{\text{el}}$, the viscoelastic strain $\dot{\boldsymbol{\varepsilon}}^{\text{ve}}$ and the flow strain $\dot{\boldsymbol{\varepsilon}}^{\text{f}}$ are computed according to a modified version of the solidification theory (Bažant and Prasannan, 1989a; Neuner, Gamnitzer, and Hofstetter, 2017; Neuner, Schreter, and Hofstetter, 2018) as

$$\dot{\boldsymbol{\varepsilon}}^{\text{el}}(t) = \mathbb{C}^{-1} : \dot{\boldsymbol{\sigma}}(t) = q_1 \mathbb{C}_\nu^{-1} : \dot{\boldsymbol{\sigma}}(t), \quad (2)$$

$$\dot{\boldsymbol{\varepsilon}}^{\text{ve}}(t) = \frac{F(\bar{\boldsymbol{\sigma}}(t), t)}{v_{\text{sc}}(t)} \int_0^t \frac{d}{dt} \Phi(t-t') \mathbb{C}_\nu^{-1} : d\bar{\boldsymbol{\sigma}}(t'), \quad (3)$$

$$\dot{\boldsymbol{\varepsilon}}^{\text{f}}(t) = \frac{q_4 F(\bar{\boldsymbol{\sigma}}(t), t)}{(t)^{e_{\text{f}}}} \mathbb{C}_\nu^{-1} : \bar{\boldsymbol{\sigma}}(t). \quad (4)$$

Therein q_1 is the instantaneous compliance parameter, $\mathbb{C}_\nu$ is the dimensionless fourth order unit stiffness tensor (Jirásek and Bažant, 2002), $v_{\text{sc}}(t)$ denotes the solidified volume at time t , $\frac{d}{dt} \Phi(t-t')$ is the viscoelastic compliance rate with its antiderivative defined as

$$\Phi(t-t') = q_2 \ln(1 + (t-t')^{0.1}), \quad (5)$$

by means of the viscoelastic compliance parameter q_2 , $F(\bar{\boldsymbol{\sigma}}(t), t)$ is an amplifying function dependent on the effective stress to account for nonlinear creep behavior, and parameters q_4 and e_{f} are material parameters which govern the evolution of the long-term creep strain, i.e., the flow strain.

Function $v_{\text{sc}}(t)$ describes the portion of solidified volume at time t ranging from 0 to 1. It was introduced by Neuner, Schreter, and Hofstetter (2018) as

$$v_{\text{sc}}(t) = \begin{cases} v_{\text{sc}}^{\text{I}}(t) = r_v + a_v t + b_v t^2 & \text{if } t < t_{\text{T}}, \\ v_{\text{sc}}^{\text{II}}(t) = \frac{c_v}{(t-t_{\text{D}})^{e_v} + d_v} & \text{otherwise.} \end{cases} \quad (6)$$

Therein, r_v is the initial value of $v_{\text{sc}}(t)$ at $t = 0$, t_{T} is the time of transition between the two phases v_{sc}^{I} and $v_{\text{sc}}^{\text{II}}$, $t - t_{\text{D}}$ is a time period to account for a delayed start of hydration, and a_v , b_v , c_v , d_v , and e_v are model parameters.

In contrast to the original solidification theory (Bažant and Prasannan, 1989a), for the SCDP model the solidified volume function $v_{\text{sc}}(t)$ is related to $E^{(1)}$ and $E^{(28)}$, i.e., the experimentally determined values of the Young's moduli at the age of 1 day and 28 days, as follows: According to the solidification theory, the compliance function for a constant unit stress $\bar{\boldsymbol{\sigma}}$ applied at time t' , neglecting any plastic strain and shrinkage, is defined as

$$J(t, t') = q_1 + F(\bar{\boldsymbol{\sigma}}) \int_{t'}^t \frac{d}{d\tau} \Phi(\tau - t') \frac{1}{v_{\text{sc}}(\tau)} d\tau + F(\bar{\boldsymbol{\sigma}}) q_4 \ln \left(\frac{t}{t'} \right). \quad (7)$$

Departing from $J(t, t')$, the experimentally determined Young's modulus can be approximated by computing the effective stiffness for a short time period Δt as $E(t) = J(t + \Delta t, t)^{-1}$ (Bažant and Baweja, 1995). Approximating the integral in (7) for a short Δt as

$$\int_t^{t+\Delta t} \frac{d}{d\tau} \Phi(\tau - t) \frac{1}{v_{\text{sc}}(\tau)} d\tau \approx \frac{\Phi(\Delta t)}{v_{\text{sc}}(t)}, \quad (8)$$

assuming linear viscoelastic behavior, i.e., $F(\boldsymbol{\sigma}(t)) = 1$, and neglecting the effect of flow creep for

short Δt , $E(t)$ can be approximated as

$$E(t) = (q_1 + q_2 v_{\text{sc}}(t)^{-1} \ln(1 + \Delta t^{0.1}))^{-1}. \quad (9)$$

From this approximation together with $v_{\text{sc}}(28 \text{ d}) = 1$, $E^{(1)}$ and $E^{(28)}$ follow as

$$\begin{aligned} E^{(1)} &= (q_1 + q_2 v_{\text{sc}}(1 \text{ d})^{-1} \chi)^{-1}, \\ E^{(28)} &= (q_1 + q_2 \chi)^{-1}, \end{aligned} \quad (10)$$

with $\chi = \ln(1 + \Delta t^{0.1})$.

For experimentally determined values of $E^{(1)}$ and $E^{(28)}$, q_2 and $v_{\text{sc}}(1 \text{ d})$ follow accordingly as

$$\begin{aligned} q_2 &= \frac{1/E^{(28)} - q_1}{\chi}, \\ v_{\text{sc}}(1 \text{ d}) &= \frac{q_2 \chi}{1/E^{(1)} - q_1}. \end{aligned} \quad (11)$$

Following (Bažant and Baweja, 1995), the remaining unknown compliance parameter q_1 is estimated as $q_1 = 0.6/E^{(28)}$ and Δt is chosen as 10^{-3} d . For ensuring a smooth transition at $t = t_{\text{T}}$, parameters a_{v} and b_{v} in Eq. (6) are computed as

$$a_{\text{v}} = \left. \frac{dv_{\text{sc}}^{\text{II}}}{dt} \right|_{t=t_{\text{T}}} - 2 b_{\text{v}} t_{\text{T}} \quad (12)$$

and

$$b_{\text{v}} = \left(r_{\text{v}} + t_{\text{T}} \left. \frac{dv_{\text{sc}}^{\text{II}}}{dt} \right|_{t=t_{\text{T}}} - v_{\text{sc}}^{\text{II}} \Big|_{t=t_{\text{T}}} \right) / t_{\text{T}}^2, \quad (13)$$

and for satisfying $v_{\text{sc}}(28 \text{ d}) = 1$ in combination with the prescribed value of $v_{\text{sc}}(1 \text{ d})$, parameters c_{v} and d_{v} follow as

$$c_{\text{v}} = (28 \text{ d} - t_{\text{D}})^{e_{\text{v}}} + d_{\text{v}} \quad (14)$$

and

$$d_{\text{v}} = \frac{v_{\text{sc}}(1 \text{ d}) (1 \text{ d} - t_{\text{D}})^{e_{\text{v}}} - (28 \text{ d} - t_{\text{D}})^{e_{\text{v}}}}{1 - v_{\text{sc}}(1 \text{ d})}. \quad (15)$$

The remaining model parameters t_{D} , t_{T} , r_{v} and e_{v} are proposed as $t_{\text{D}} = 0.3 \text{ d}$, $t_{\text{T}} = 0.5 \text{ d}$, $r_{\text{v}} = 0.01$, and $e_{\text{v}} = -0.65$, calibrated based on the experimentally observed evolution of the Young's modulus of young shotcrete.

Nonlinear creep behavior under high compressive stress levels is represented by means of amplification function $F(\bar{\boldsymbol{\sigma}}(t), t)$ in Eq. (3) and Eq. (4), as proposed by Bažant and Prasannan (1989a) for uniaxial stress states. To determine the degree of utilization for multiaxial stress states within $F(\bar{\boldsymbol{\sigma}}(t), t)$, the invariant $\bar{J}_2$ of the effective stress tensor $\bar{\boldsymbol{\sigma}}$ and the current uniaxial compressive strength $f_{\text{cu}}(t)$ are used:

$$F(\bar{\boldsymbol{\sigma}}(t), t) = 1 + s(\bar{\boldsymbol{\sigma}}(t), t)^2, \quad \text{with} \quad s(\bar{\boldsymbol{\sigma}}(t), t) = \min \left(\frac{\sqrt{3 \bar{J}_2}}{f_{\text{cu}}(t)}, 1 \right). \quad (16)$$

2.2 Evolution of the Shrinkage Strain

The shrinkage strain is computed by means of the empirical law for concrete, proposed by Bažant and Panula (Bažant and Panula, 1978), as

$$\boldsymbol{\varepsilon}^{\text{shr}}(t) = \mathbf{I} \varepsilon_{\infty}^{\text{shr}} k_h S(t - t_0, \tau_{\text{shr}}), \quad (17)$$

in which $\varepsilon_{\infty}^{\text{shr}}$ denotes the ultimate shrinkage strain, k_h a humidity dependent scaling factor, $S(t - t_0, \tau_{\text{shr}})$ the square-root hyperbolic law, dependent on the current time t , age of the material at the beginning of drying, t_0 , and shrinkage half time τ_{shr} , and $\mathbf{I}$, the second order unit tensor. An alternative estimate of the shrinkage strain may be obtained by the empirical law according to the ACI Committee 209 (1992) :

$$\boldsymbol{\varepsilon}^{\text{shr}}(t) = \mathbf{I} \varepsilon_{\infty}^{\text{shr}} \frac{t}{t + t_{50}^{\text{shr}}}, \quad (18)$$

with t_{50}^{shr} denoting the shrinkage half time according to the ACI code.

2.3 Evolution of the Plastic Strain

For delimiting the elastic domain and for describing the evolution of the plastic strain, the yield function f_p and the plastic potential function g_p are adopted from the CDP model by Grassl and Jirásek (2006a). In contrast to the CDP model, the uniaxial compressive strength $f_{\text{cu}}(t)$, the uniaxial yield stress $f_{\text{cy}}(t)$, the biaxial compressive strength $f_{\text{cb}}(t)$, the uniaxial tensile strength $f_{\text{tu}}(t)$ are assumed to be dependent on the shotcrete age t :

$$\begin{aligned} f_p(\bar{\sigma}_m, \bar{\rho}, \theta, q_h(\alpha_p), t) = & \left((1 - q_h(\alpha_p)) \left(\frac{\bar{\rho}}{\sqrt{6} f_{\text{cu}}(t)} + \frac{\bar{\sigma}_m}{f_{\text{cu}}(t)} \right)^2 + \sqrt{\frac{3}{2}} \frac{\bar{\rho}}{f_{\text{cu}}(t)} \right)^2 \\ & + m_0 q_h^2(\alpha_p) \left(\frac{\bar{\rho}}{\sqrt{6} f_{\text{cu}}(t)} r(\theta) + \frac{\bar{\sigma}_m}{f_{\text{cu}}(t)} \right) - q_h^2(\alpha_p), \end{aligned} \quad (19)$$

$$\begin{aligned} g_p(\bar{\sigma}_m, \bar{\rho}, q_h(\alpha_p), t) = & \left((1 - q_h(\alpha_p)) \left(\frac{\bar{\rho}}{\sqrt{6} f_{\text{cu}}(t)} + \frac{\bar{\sigma}_m}{f_{\text{cu}}(t)} \right)^2 + \sqrt{\frac{3}{2}} \frac{\bar{\rho}}{f_{\text{cu}}(t)} \right)^2 \\ & + q_h^2(\alpha_p) \left(\frac{m_0 \bar{\rho}}{\sqrt{6} f_{\text{cu}}(t)} + \frac{m_g(\bar{\sigma}_m)}{f_{\text{cu}}(t)} \right). \end{aligned} \quad (20)$$

Both functions are formulated in terms of three invariants of the effective stress tensor, i.e., the effective mean stress $\bar{\sigma}_m$, the effective deviatoric radius $\bar{\rho}$ and the Lode angle θ , and the stress-like internal variable q_h , which is a function of the strain-like internal variable α_p . Function $r(\theta)$ is the Willam-Warnke function (Willam and Warnke, 1975), which describes the shape of the yield surface in deviatoric planes, and m_0 is a friction-like parameter defined as

$$m_0 = 3 \frac{f_{\text{cu}}(t)^2 - f_{\text{tu}}(t)^2}{f_{\text{cu}}(t) f_{\text{tu}}(t)} \frac{e}{e + 1}, \quad e = \frac{1 + \epsilon}{2 - \epsilon}, \quad \epsilon = \frac{f_{\text{tu}}(t) (f_{\text{cb}}(t)^2 - f_{\text{cu}}(t)^2)}{f_{\text{cb}}(t) (f_{\text{cu}}(t)^2 - f_{\text{tu}}(t)^2)}, \quad (21)$$

and the dilation function

$$m_g(\bar{\sigma}_m) = A_g B_g f_{\text{cu}}(t) \exp \left(\frac{\bar{\sigma}_m - f_{\text{tu}}(t)/3}{B_g f_{\text{cu}}(t)} \right) \quad (22)$$

with the model parameters A_g and B_g , which controls the evolution of the volumetric plastic strain.

The stress-like internal hardening variable q_h is defined as

$$q_h(\alpha_p) = \begin{cases} f_{cy}(t)/f_{cu}(t) + (1 - f_{cy}(t)/f_{cu}(t)) \alpha_p (\alpha_p^2 - 3\alpha_p + 3) & \text{if } \alpha_p < 1, \\ 1 & \text{otherwise.} \end{cases} \quad (23)$$

From the plastic potential, the rate of the plastic strain $\dot{\epsilon}^p$ is computed as

$$\dot{\epsilon}^p = \dot{\lambda} \frac{\partial g_p(\bar{\sigma}_m, \bar{\rho}, q_h(\alpha_p), t)}{\partial \bar{\sigma}} \quad (24)$$

and for describing the evolution of the strain-like internal hardening variable α_p , the hardening law is adopted from (Unteregger, 2015):

$$\dot{\alpha}_p = \|\dot{\epsilon}^p(t)\| \frac{1}{x_h(\bar{\sigma}_m, t)} \left(1 + 3 \frac{\bar{\rho}^2}{\bar{\rho}^2 + \vartheta_h} \cos^2(1.5\theta) \right). \quad (25)$$

Therein, ϑ_h is a disturbance parameter to avoid division by zero in case of a purely hydrostatic stress, and $x_h(\bar{\sigma}_m, t)$ is a ductility function for describing the magnitude of $\dot{\alpha}_p$ dependent on the acting hydrostatic stress:

$$x_h(\bar{\sigma}_m, t) = \begin{cases} A_h - (A_h - B_h) \exp\left(\frac{-R_h(\bar{\sigma}_m, t)}{C_h}\right) & \text{if } R_h(\bar{\sigma}_m, t) \geq 0, \\ (B_h - D_h) \exp\left(\frac{R_h(\bar{\sigma}_m, t)(A_h - B_h)}{(B_h - D_h) C_h}\right) + D_h & \text{otherwise,} \end{cases} \quad (26)$$

$$R_h(\bar{\sigma}_m, t) = -\frac{\bar{\sigma}_m}{f_{cu}(t)} - \frac{1}{3}.$$

Model parameters A_h , B_h , C_h and D_h are calibrated from uniaxial tensile and uniaxial and triaxial compression tests. Assuming fixed values of A_h , C_h and D_h , the material ductility in the pre-peak regime of the stress strain relation is controlled by model parameter B_h . Although the actual relation between ϵ_{cpu}^p , the plastic strain at peak stress in uniaxial compression, and B_h is nonlinear, an approximative linear relation has been proposed by Grassl and Jirásek (2006a). However, this linear relation is valid only for a limited range of ϵ_{cpu}^p from 0 to approximately -0.002 . Hence, to extend the valid range of the approximative law a bilinear extension was proposed by Neuner, Gamnitzer, and Hofstetter (2017) as

$$B_h(\epsilon_{cpu}^p) = \begin{cases} -2.29 \epsilon_{cpu}^p + 0.00046 & \text{if } \epsilon_{cpu}^p > -0.002, \\ -1.55 (\epsilon_{cpu}^p + 0.002) + 0.00504 & \text{if } \epsilon_{cpu}^p \leq -0.002. \end{cases} \quad (27)$$

The evolution law for $\epsilon_{cpu}^p(t)$ is adopted from the model by Schädlich and Schweiger (2014): It is based on a linear interpolation of ϵ_{cpu}^p between the values at the age of 1 hour, 8 hours and 24 hours, denoted as $\epsilon_{cpu}^{p(1)}$, $\epsilon_{cpu}^{p(8)}$ and $\epsilon_{cpu}^{p(24)}$, respectively. For $t < 1$ h and $t > 24$ h, ϵ_{cpu}^p is assumed constant.

2.4 Evolution of Material Strength

The evolution of the uniaxial compressive strength $f_{\text{cu}}(t)$ is described by a modified law originally proposed by Meschke, Kropik, and Mang (1996) for the evolution of the Young's modulus:

$$f_{\text{cu}}(t) = f_{\text{cu}}^{(28)} \beta_{\text{f}}(t), \quad \beta_{\text{f}}(t) = \begin{cases} \beta_{\text{f}}^{\text{I}} = r_{\text{f}} + c_{\text{f}}t + d_{\text{f}}t^2 & \text{if } t \leq t_{\text{f}}, \\ \beta_{\text{f}}^{\text{II}} = \left(a_{\text{f}} + \frac{b_{\text{f}}}{t - \Delta t_{\text{f}}}\right)^{-0.5} & \text{if } t_{\text{f}} < t \leq 28 \text{ d}, \\ \beta_{\text{f}}^{\text{III}} = 1 & \text{otherwise.} \end{cases} \quad (28)$$

Therein, the residual parameter r_{f} ensures a non-zero compressive strength already at *zero* age. A default value is proposed as $r_{\text{f}} = 10^{-2}$, resulting in a uniaxial compressive strength of 1 % of $f_{\text{cu}}^{(28)}$ at $t = 0$. The remaining model parameters are defined as

$$\begin{aligned} a_{\text{f}} &= \frac{1 - \frac{28 - \Delta t_{\text{f}}}{1 - \Delta t_{\text{f}}} (f_{\text{cu}}^{(1)} / f_{\text{cu}}^{(28)})^2}{(1 - \frac{28 - \Delta t_{\text{f}}}{1 - \Delta t_{\text{f}}}) (f_{\text{cu}}^{(1)} / f_{\text{cu}}^{(28)})^2}, & b_{\text{f}} &= (28 - \Delta t_{\text{f}})(1 - a_{\text{f}}), \\ c_{\text{f}} &= \left. \frac{d\beta_{\text{f}}^{\text{II}}}{dt} \right|_{t=t_{\text{f}}} - 2d_{\text{f}}t_{\text{f}}, & d_{\text{f}} &= \left(\left. \frac{d\beta_{\text{f}}^{\text{II}}}{dt} \right|_{t=t_{\text{f}}} t_{\text{f}} - (\beta_{\text{f}}^{\text{II}} - r_{\text{f}}) \right) / t_{\text{f}}^2. \end{aligned} \quad (29)$$

$f_{\text{cu}}^{(1)}$ and $f_{\text{cu}}^{(28)}$ denote the experimentally determined values of $f_{\text{cu}}(t)$ at the age of 1 day and 28 days, respectively. Parameters t_{f} and Δt_{f} account for the delayed start of the evolution of compressive strength. Default values are proposed as $t_{\text{f}} = 0.25 \text{ d}$ and $\Delta t_{\text{f}} = 0.18 \text{ d}$. For these values, a monotonic growth of $f_{\text{cu}}(t)$ is ensured only for ratios $f_{\text{cu}}^{(1)} / f_{\text{cu}}^{(28)} \geq 0.16$, which are usually met in practical applications.

Regarding $f_{\text{cy}}(t)$, $f_{\text{cb}}(t)$ and $f_{\text{tu}}(t)$, it is assumed that the ratios $f_{\text{cy}}(t)/f_{\text{cu}}(t)$, $f_{\text{cb}}(t)/f_{\text{cu}}(t)$ and $f_{\text{tu}}(t)/f_{\text{cu}}(t)$ remain constant throughout the hydration process. Hence, $f_{\text{cy}}(t)$, $f_{\text{cb}}(t)$ and $f_{\text{tu}}(t)$ are computed directly from $f_{\text{cu}}(t)$ and the respective ratios.

2.5 Softening Material Behavior

According to Eq. (1), softening material behavior of the SCDP model is described by means of the scalar isotropic damage parameter ω :

$$\boldsymbol{\sigma} = (1 - \omega) \bar{\boldsymbol{\sigma}}. \quad (30)$$

The evolution of ω is initiated once a fully hardened material state is attained, i.e., $\alpha_{\text{p}} \geq 1.0$ in Eq. (23). For the SCDP model introduced in (Neuner, Gamnitzer, and Hofstetter, 2017), a relation between ω and a strain-like internal variable α_{d} was introduced, adopted from (Grassl and Jirásek, 2006a) as

$$\omega = 1 - \exp\left(-\frac{\alpha_{\text{d}}}{\epsilon_{\text{f}}}\right), \quad (31)$$

in which ϵ_{f} is the softening modulus, controlling the slope of the softening curve. The evolution of the strain-like internal variable α_{d} is related to the rate of the volumetric plastic strain $\dot{\epsilon}_{\text{V}}^{\text{p}}$ by

$$\dot{\alpha}_{\text{d}} = \begin{cases} 0 & \text{if } \alpha_{\text{p}} < 1, \\ \dot{\epsilon}_{\text{V}}^{\text{p}} / x_{\text{s}}(\dot{\epsilon}^{\text{p}}) & \text{otherwise,} \end{cases} \quad (32)$$

with $x_s(\dot{\epsilon}^p)$ denoting a ductility measure, defined as

$$x_s(\dot{\epsilon}^p) = \begin{cases} 1 + A_s R_s^2(\dot{\epsilon}^p) & \text{if } R_s(\dot{\epsilon}^p) < 1, \\ 1 - 3 A_s + 4 A_s \sqrt{R_s(\dot{\epsilon}^p)} & \text{otherwise.} \end{cases} \quad (33)$$

Therein, A_s is a ductility material parameter, which is calibrated by means of uniaxial compression tests, and $R_s(\dot{\epsilon}^p)$ denotes the ratio of the *negative* volumetric plastic strain rate to the total volumetric plastic strain rate:

$$R_s(\dot{\epsilon}^p) = \frac{\dot{\epsilon}_V^{p\ominus}}{\dot{\epsilon}_V^p} \quad \text{with} \quad \dot{\epsilon}_V^{p\ominus} = \sum_{i=1}^{\text{III}} \left\langle -\dot{\epsilon}_{(i)}^p \right\rangle, \quad (34)$$

the latter computed from the principal values of the plastic strain rate tensor, with $\langle \bullet \rangle$ denoting the Macaulay brackets.

It is well known that standard continuum models describing softening material behavior, e.g., models based on the theory of plasticity or the theory of continuum damage mechanics, exhibit several theoretical deficiencies, as reported and summarized by Bažant (1976): (i) The softening process zone is infinitesimally small, i.e., it has the size of a material point, (ii) at structural level, snapback behavior due to the infinitesimally small softening zone is observed, (iii) and the amount of dissipated energy during the failure process is zero due to the infinitesimal zone in which energy is dissipated.

In a mathematical sense, those issues are related to the loss of ellipticity of the initial boundary value problem (IBVP) in static and quasi-static analyses. In finite element analyses, those deficiencies lead to mesh-dependent results, commonly denoted as *pathological* mesh sensitivity. Failure patterns like cracks tend to localize into the smallest possible bandwidth, i.e., usually a single layer of finite elements, and upon mesh refinement the localization zone is decreasing into an arbitrarily small domain. Accordingly, the obtained results are not objective with respect to the finite element mesh, i.e., the results are sensitive to the numerical discretization scheme.

2.6 Mesh Objective Representation of Softening Material Behavior in the SCDP Model

The mesh objective representation of softening material behavior is a vast research topic, and in the past decades different approaches for regularizing softening material behavior in finite element simulations have been proposed so far.

A popular and widely used method for continuum models is the crack band approach (Pietruszczak and Mróz, 1981; Bažant and Oh, 1983), based on a mesh adjusted softening modulus, which was also employed for the SCDP model presented by Neuner, Gamnitzer, and Hofstetter (2017) and Neuner, Schreter, and Hofstetter (2018). Following the crack band approach, the softening modulus is adjusted by means of a characteristic size of a finite element and the specific mode I fracture energy, which is considered as a material parameter. However, as pointed out by Jirásek and Bauer (2012), the crack band approach involves several issues: (i) In finite element simulations employing continuum models based on the crack band approach, crack paths tend to be sensitive to the orientation of the finite element mesh, i.e., crack paths tend to be aligned with the grid lines of the finite element mesh, (ii) for higher order finite elements, localization into subdomains

may occur, and hence, adjusting the softening modulus based on the size of the complete element yields inaccurate results, and (iii) determining the characteristic length of an element is complicated, especially for distorted or elongated elements, as the proper characteristic length depends on the orientation of the crack. Issues (i) and (ii) are related to the fact that deformations in consequence of material failure localize usually into the smallest possible domain if no proper countermeasures are employed. Concerning issue (iii), different methods of different complexity for estimating the characteristic length of a finite element were proposed, for instance by Oliver (1989). Furthermore, due to the assumption of a softening process zone attributed to the size of a finite element, the crack band approach is not suitable for diffuse material failure patterns, i.e., failure patterns which do not localize into a single layer of finite elements. A discussion on that issue was presented by Grassl and Jirásek (2006b), based on an example of compressive failure of an eccentrically loaded concrete column. For these reasons, an improved regularization technique for the SCDP model is presented in the following.

For preventing mesh dependent localization into arbitrarily small domains, several advanced techniques were proposed in the literature (Jirásek and Bažant, 2002), commonly denoted as localization limiters. Examples are nonlocal models of the integral type (Eringen, 1966; Pijaudier-Cabot and Bažant, 1987), gradient-enhanced models (Peerlings, Geers, de Borst, and Brekelmans, 2001), models based on Cosserat continua (de Borst, 1991), or viscoplastic models (Needleman, 1988). Nonlocal models of the integral type and gradient-enhanced models enrich the constitutive relations for an infinitesimal material point by nonlocal terms, i.e., quantities which are influenced by a finite or even infinite neighborhood of the considered material point. Both approaches are widely used and are therefore presented briefly in the subsequent subsections. Models based on the Cosserat continuum (E. Cosserat and F. Cosserat, 1909) incorporate an additional field of micropolar rotations, and were applied successfully to mode II failure (de Borst, 1991). However, these models are not suitable for pure mode I failure. Viscoplastic models enrich the constitutive relations by a rate-dependent viscous term which has a regularizing effect, but which also introduces a dependency on the rate of loading. All those approaches have in common that a length scale is incorporated into the constitutive relations. This holds also for the viscoplastic approach, as explained by Needleman (1988).

According to nonlocal models of the integral type, the stress-strain relation for a material point is governed not exclusively by its local state, but also by its neighborhood (Jirásek and Bažant, 2002). The nonlocal character is introduced in the constitutive relations by replacing one or more quantities by their nonlocal counterparts. The nonlocal counterpart of a quantity is defined as the spatial average of the respective local quantity, which is computed in terms of an integral expression employing a nonlocal weight function. For instance, assuming a local quantity $f(\mathbf{x})$ for each material point $\mathbf{x}$ of a domain Ω , its nonlocal counterpart $\bar{f}(\mathbf{x})$ is computed as

$$\bar{f}(\mathbf{x}) = \int_{\Omega} w(\mathbf{x}, \boldsymbol{\xi}) f(\boldsymbol{\xi}) \mathrm{d}\boldsymbol{\xi}. \quad (35)$$

Therein, the nonlocal weight function $w(\mathbf{x}, \boldsymbol{\xi})$ defines the interaction between two material points $\mathbf{x}$ and $\boldsymbol{\xi}$, dependent on their spatial distance. Although the zone of interaction can embrace the complete domain, for numerical implementations usually only a finite neighborhood is considered in the nonlocal weight function. Commonly employed weight functions are Gaussian bell functions or parabolic functions. The integral nonlocal formulation is basically applicable to any type of

constitutive model, including models based on the theory of plasticity or the theory of continuum damage mechanics. A drawback of integral nonlocal models is that in finite element simulations due to the increased interaction radius of each degree of freedom the global finite element stiffness matrix has an increased bandwidth, and is nonsymmetric in general.

Gradient-enhanced approaches are considered as the differential complement to nonlocal models of the integral type: The nonlocal character is introduced in the constitutive relations by incorporating higher order gradients of one or more quantities. Basically, two families exist within this class of models: *Explicit* and *implicit* gradient-enhanced models. Explicit gradient-enhanced models incorporate higher order gradients directly into the constitutive relations, whereas for implicit gradient-enhanced models the constitutive relations are enriched by nonlocal quantities which are defined implicitly as the solution of a higher order partial differential equation, with the local quantity acting as the driving force. For instance, by analogy to Eq. (35), the nonlocal counterpart $\bar{f}(\mathbf{x})$ of a local quantity $f(\mathbf{x})$ may be described according to the explicit approach in terms of a second order expression as

$$\bar{f}(\mathbf{x}) = f(\mathbf{x}) + l^2 \nabla^2 f(\mathbf{x}), \quad (36)$$

and according to the implicit approach employing a second order partial differential equation as

$$\bar{f}(\mathbf{x}) - l^2 \nabla^2 \bar{f}(\mathbf{x}) = f(\mathbf{x}). \quad (37)$$

In both expressions, l denotes the interaction radius, which is considered as an internal length scale material parameter.

Despite their superficial similarities, the regularization and localization properties of both approaches are substantially different. A comparison of explicit and implicit gradient-enhanced models based on continuum damage mechanics was presented by Peerlings, Geers, de Borst, and Brekelmans (2001). It is demonstrated that despite their theoretical simplicity, the numerical implementation of explicit gradient-enhanced models is complicated, and in general, higher order finite elements are required. For the explicit approach, boundary conditions must be formulated explicitly in terms of higher order derivatives of the displacement field. In contrast, implicit gradient-enhanced models are amenable to an easier and robust implementation, at the price of a fully coupled partial differential equation system. Furthermore, Peerlings, Geers, de Borst, and Brekelmans (2001) demonstrated that the implicit approach is equivalent to the nonlocal approach of the integral type if the Green's function of the higher order partial differential equation is employed as the nonlocal weight function of the integral approach.

In this context, phase-field models are mentioned, which have been developed extensively in the past years Miehe, Welschinger, and Hofacker (2010), Miehe, Schänzel, and Ulmer (2015), Wu (2017), Mandal, Nguyen, and Heidarpour (2019), and Wu et al. (2019): Phase-field models regularize sharp crack discontinuities by means of pure continuum formulations. Although phase-field models originate from a completely different motivation, they might be considered as localization limiters in the aforementioned sense. In fact, phase-field models are closely related to implicit gradient-enhanced damage models in a mathematical sense, as demonstrated by means of a comparison of both approaches by de Borst and C. Verhoosel (2016).

For particular continuum models representing softening material behavior, it is reported that a mere nonlocal formulation, employing either the integral approach or the gradient-enhanced formu-

lation, fails to achieve a proper regularization (Geers, de Borst, Brekelmans, and Peerlings, 1998; Simone, Askes, and Sluys, 2004). Di Luzio and Bažant (2005) investigated the localization properties of different nonlocal plasticity and damage models by means of spectral analyses of wave propagation in a body exhibiting softening material behavior. They demonstrated a necessary requirement for a proper regularization of softening material behavior. It is given by the fact that the propagation speed of stress acceleration waves must not become imaginary. In case of plasticity models formulated in terms of strain-like internal damage driving variables, this requirement is fulfilled only if the so called *over-nonlocal* formulation is employed. The over-nonlocal approach was proposed by Vermeer and Brinkgreve (1994), introducing a weighted nonlocal quantity $\hat{f}(\mathbf{x})$ for formulating the constitutive relations. The weighted variable $\hat{f}(\mathbf{x})$ is computed by means of a scalar weighting parameter m from both the local field $f(\mathbf{x})$ and the nonlocal field $\bar{f}(\mathbf{x})$:

$$\hat{f}(\mathbf{x}) = m \bar{f}(\mathbf{x}) + (1 - m) f(\mathbf{x}). \quad (38)$$

Choosing $m = 0$ leads to a pure local formulation, i.e., $\hat{f}(\mathbf{x}) = f(\mathbf{x})$, and choosing $m = 1$ leads to a pure nonlocal formulation, i.e., $\hat{f}(\mathbf{x}) = \bar{f}(\mathbf{x})$. To obtain the over-nonlocal formulation, weighting parameter m is chosen larger than unity, i.e., $m > 1$. Examples of application of the over-nonlocal formulation to damage plasticity models were presented by Poh and Swaddiwudhipong (2009a,b) and Hosseini, Horák, Zysset, and Jirásek (2015).

Grassl and Jirásek (2006b) presented an application of the nonlocal formulation of the integral type to their damage plasticity model for concrete (Grassl and Jirásek, 2006a). It is shown that the integral nonlocal approach yields mesh-insensitive results for both cracking in tension and diffuse failure in compression. Poh and Swaddiwudhipong (2009b) proposed an implicit gradient-enhanced version of the respective concrete model, using the over-nonlocal weighting for computing the damage driving variable. For simplicity and for a clear distinction, the nonlocal approach of the integral type is henceforth simply denoted as the *integral* approach, and the implicit gradient-enhanced formulation is denoted simply as the *gradient* approach.

Comparing both approaches, the following differences are observed:

- According to the gradient approach, the nonlocal field is computed implicitly by solving a fully coupled partial differential equation system, whereas for the integral approach the computation of the spatially averaged quantity in terms of the spatial distance between individual material points requires a specific algorithmic treatment.
- In finite element simulations, no additional degrees of freedom are introduced for models based on the integral approach. From a computational viewpoint, this is advantageous for constitutive models with higher order tensorial nonlocal quantities and for models based on multiple nonlocal quantities. For instance, in (Xenos and Grassl, 2016) an extension of the concrete model by Grassl and Jirásek to cyclic material behavior adopting the integral approach is proposed, based on altogether six scalar nonlocal variables.
- Models based on the integral approach require a special treatment of the nonlocal weight function near boundaries, i.e., if the support of the nonlocal weight function is located outside the domain. In contrast, boundary conditions are treated in a natural and mathematical rigorous sense within the gradient approach (Peerlings, Geers, de Borst, and Brekelmans, 2001).

- Models based on the integral approach predict spurious interactions between material points near non-convex boundaries, for instance across notches. These spurious interactions are not predicted by models based on the gradient approach.
- For models based on the gradient approach, the computation of the algorithmic tangent stiffness by means of consistent linearization of the constitutive relations is straightforward, and the global system matrix in finite element simulations preserves its sparse structure.

2.7 Over-Nonlocal Gradient-Enhanced Formulation of the SCDP Model

Because of the mentioned advantages of the gradient approach over the integral approach, and considering that the SCDP model is formulated in terms of a single scalar damage driving internal variable, an extension of the SCDP model adopting the over-nonlocal implicit gradient-enhanced formulation proposed by Poh and Swaddiwudhipong (2009b) and investigated in (Schreter, Neuner, and Hofstetter, 2018) is presented. Following this approach, the weighted *over-nonlocal* damage driving variable $\hat{\alpha}_d$ is introduced, which is computed from the local internal strain-like variable α_d (32) and its nonlocal counterpart $\bar{\alpha}_d$:

$$\hat{\alpha}_d = m \bar{\alpha}_d + (1 - m) \alpha_d, \quad (39)$$

with weighting parameter m usually chosen slightly larger than 1. Accordingly, relation (31) is replaced by

$$\omega = 1 - \exp\left(-\frac{\hat{\alpha}_d}{\epsilon_f}\right). \quad (40)$$

As explained by Grassl and Jirásek (2006b), for over-nonlocal formulations negative values of ω can arise due to the negative weighting of α_d in (39), which is thermodynamically inadmissible. Hence, for the SCDP model an additional constraint ensuring a strictly non-negative damage parameter is employed. Following the implicit approach (37), the field of the nonlocal variable $\bar{\alpha}_d$ is defined by the second order partial differential equation of the Helmholtz type

$$\bar{\alpha}_d - l^2 \nabla^2 \bar{\alpha}_d = \alpha_d, \quad (41)$$

with α_d denoting the local strain-like internal variable acting as the driving force. Together with the equilibrium equation

$$\nabla \boldsymbol{\sigma} + \mathbf{f} = \mathbf{0}, \quad (42)$$

with $\mathbf{f}$ denoting the body forces, (41) forms a fully coupled system of partial differential equations.

The standard boundary conditions for the equilibrium equation are given as

$$\begin{aligned} \mathbf{u} &= \bar{\mathbf{u}} & \text{on } \Gamma_u \\ \mathbf{n} \cdot \boldsymbol{\sigma} &= \bar{\mathbf{t}} & \text{on } \Gamma_t, \end{aligned} \quad (43)$$

in which $\mathbf{n}$ denotes the unit vector perpendicular on the boundary $\Gamma = \Gamma_u \cup \Gamma_t$ of the domain of interest, $\mathbf{u}$ the displacement vector, $\bar{\mathbf{u}}$ the prescribed displacement vector on Γ_u and $\bar{\mathbf{t}}$ the prescribed traction vector on Γ_t .

For the Helmholtz type equation (41), homogeneous Neumann boundary conditions are assumed

for the complete domain:

$$\boldsymbol{n} \cdot \nabla \bar{\alpha}_d = 0 \quad \text{on } \Gamma. \quad (44)$$

An interpretation of this boundary condition in the context of phase-field models was presented by de Borst and C. Verhoosel (2016) in terms of the condition that cracks are oriented perpendicular to the surface of a body.

3 Numerical Algorithm of the SCDP Model at Finite Element Level

In the following, the implementation of the SCDP model, embedded in the implicit gradient-enhanced framework, into a finite element code is explained. To this end, the governing equations of the time-dependent, quasi-static problem are converted to the weak form, discretized in space employing the finite element method and subsequently discretized in time. The standard incremental-iterative procedure based on the Newton-Raphson scheme is employed for solving the IBVP. The here presented procedure follows the methods proposed by de Borst, Crisfield, Remmers, and C. Verhoosel (2012) and Belytschko, Liu, Moran, and Elkhodary (2013). At integration point level of a finite element, the stress is computed by means of a stress update algorithm, which is described in Section 4. For implementing the model, the C++ programming language was used, following an object-oriented design approach. Since C++ provides no linear algebra functionality like vector-matrix operations, the *Eigen template library* (Guennebaud, Jacob, et al., 2010) for linear algebra is used.

3.1 Weak Form of the Governing Equations

The governing partial differential equations in the strong form (42) and (41) are converted to the weak form by taking the integral over the complete domain Ω and applying test functions $\delta \mathbf{u}$ and $\delta \bar{\alpha}_d$ for the displacement field and for the field of the nonlocal damage driving internal strain-like variable, respectively. The latter is henceforth simply denoted as the *nonlocal* field. The framework is formulated within the regime of small strains and displacements, with the small strain tensor $\boldsymbol{\varepsilon}$ given as the symmetric part of the displacement gradient:

$$\boldsymbol{\varepsilon} = \nabla_{\text{sym}} \mathbf{u}. \quad (45)$$

For the equilibrium equation (42), the weak form is given as

$$\int_{\Omega} \delta \boldsymbol{\varepsilon} : \boldsymbol{\sigma} \, dV = \int_{\Gamma_t} \delta \mathbf{u} \cdot \bar{\mathbf{t}} \, dS + \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{f} \, dV, \quad (46)$$

in which use of the Gauss theorem and the boundary conditions (43) was made. In a similar manner, departing from (41), conversion to the weak form and application of the Gauss theorem yields

$$\int_{\Omega} \delta \bar{\alpha}_d \bar{\alpha}_d \, dV + \int_{\Omega} l^2 \nabla \delta \bar{\alpha}_d \cdot \nabla \bar{\alpha}_d \, dV - \int_{\Omega} \delta \bar{\alpha}_d \alpha_d \, dV = 0, \quad (47)$$

in which use of the homogeneous Neumann boundary conditions (44) was made.

3.2 Spatial Discretization

The weak form (46) and (47) is spatially discretized by the finite element method following the Bubnov-Galerkin approach, i.e., using the same discretization scheme for trial and test functions. The displacement field and the nonlocal field are interpolated in terms of global interpolation operators $\mathbf{N}_u$ and $\mathbf{N}_{\bar{\alpha}}$ for the displacement field and for the nonlocal field, respectively:

$$\mathbf{u} = \mathbf{N}_u \mathbf{q}_u, \quad (48)$$

$$\bar{\alpha}_d = \mathbf{N}_{\bar{\alpha}} \mathbf{q}_{\bar{\alpha}}, \quad (49)$$

in which $\mathbf{q}_u$ and $\mathbf{q}_{\bar{\alpha}}$ denote the vectors of the nodal displacements and the nodal nonlocal values, respectively. Furthermore, the global strain-displacement operator $\mathbf{B}_u$ for the displacement field and the global gradient operator $\mathbf{B}_{\bar{\alpha}}$ for the nonlocal field are introduced as:

$$\boldsymbol{\varepsilon} = \mathbf{B}_u \mathbf{q}_u, \quad (50)$$

$$\nabla \bar{\alpha}_d = \mathbf{B}_{\bar{\alpha}} \mathbf{q}_{\bar{\alpha}}. \quad (51)$$

Departing from (46) and (47), the discretized weak form expressions are obtained as

$$\begin{aligned} \int_{\Omega} \delta \mathbf{q}_u^T \mathbf{B}_u^T \boldsymbol{\sigma} dV &= \int_{\Gamma_t} \delta \mathbf{q}_u^T \mathbf{N}_u^T \bar{\mathbf{t}} dS + \int_{\Omega} \delta \mathbf{q}_u^T \mathbf{N}_u^T \mathbf{f} dV \\ \delta \mathbf{q}_u^T \mathbf{f}_u^{\text{int}} &= \delta \mathbf{q}_u^T \mathbf{f}_u^{\text{ext}} \end{aligned} \quad (52)$$

and

$$\begin{aligned} \int_{\Omega} \delta \mathbf{q}_{\bar{\alpha}}^T \mathbf{N}_{\bar{\alpha}}^T \mathbf{N}_{\bar{\alpha}} \mathbf{q}_{\bar{\alpha}} dV + \int_{\Omega} l^2 \delta \mathbf{q}_{\bar{\alpha}}^T \mathbf{B}_{\bar{\alpha}}^T \mathbf{B}_{\bar{\alpha}} \mathbf{q}_{\bar{\alpha}} dV - \int_{\Omega} \delta \mathbf{q}_{\bar{\alpha}}^T \mathbf{N}_{\bar{\alpha}}^T \alpha_d dV &= 0 \\ \delta \mathbf{q}_{\bar{\alpha}}^T \mathbf{f}_{\bar{\alpha}}^{\text{int}} &= 0, \end{aligned} \quad (53)$$

with $\mathbf{f}_u^{\text{int}}$ and $\mathbf{f}_u^{\text{ext}}$ denoting the standard internal and external nodal force vectors associated with the displacement field, and $\mathbf{f}_{\bar{\alpha}}^{\text{int}}$ the non-standard internal force vector associated with the nonlocal field:

$$\mathbf{f}_u^{\text{int}} = \int_{\Omega} \mathbf{B}_u^T \boldsymbol{\sigma} dV, \quad (54)$$

$$\mathbf{f}_u^{\text{ext}} = \int_{\Gamma_t} \mathbf{N}_u^T \bar{\mathbf{t}} dS + \int_{\Omega} \mathbf{N}_u^T \mathbf{f} dV, \quad (55)$$

$$\mathbf{f}_{\bar{\alpha}}^{\text{int}} = \int_{\Omega} \mathbf{N}_{\bar{\alpha}}^T \mathbf{N}_{\bar{\alpha}} \mathbf{q}_{\bar{\alpha}} dV + \int_{\Omega} l^2 \mathbf{B}_{\bar{\alpha}}^T \mathbf{B}_{\bar{\alpha}} \mathbf{q}_{\bar{\alpha}} dV - \int_{\Omega} \mathbf{N}_{\bar{\alpha}}^T \alpha_d dV. \quad (56)$$

The virtual work expressions (52) and (53) hold for arbitrary, kinematically admissible values of $\delta \mathbf{q}_u$ and $\delta \mathbf{q}_{\bar{\alpha}}$, which leads to the system of nonlinear algebraic equations

$$\begin{Bmatrix} \mathbf{f}_u^{\text{int}} - \mathbf{f}_u^{\text{ext}} \\ \mathbf{f}_{\bar{\alpha}}^{\text{int}} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{0} \end{Bmatrix}. \quad (57)$$

Following the standard finite element procedure, the internal and external force vectors are obtained by applying the standard assembly procedure on the respective element load vectors.

3.3 Discretization in Time

The system of nonlinear algebraic equations (57) holds for any time t , and for any prescribed external load $\mathbf{f}_u^{\text{ext}}$. Due to the time- and path-dependent character of the considered IBVP, the solution is computed employing an incremental-iterative procedure. Following the standard incremental-iterative procedure for time-dependent quasi-static problems, a discretization in time is introduced, and the respective solution for a particular time increment is computed in an iterative manner, starting in each increment from the computed solution of the preceding increment. For a particular increment, the system of nonlinear equations is formulated in terms of the unknown nodal increments

$\Delta \mathbf{q}_u$ and $\Delta \mathbf{q}_{\bar{\alpha}}$, and solved employing the Newton-Raphson scheme. In particular, considering the increment defined by the interval $[t_n, t_{n+1}]$ with $t_{n+1} = t_n + \Delta t$ and departing from the known state at t_n , the internal force vectors at the end of the interval expressed in terms of the increments $\Delta \mathbf{q}_u$ and $\Delta \mathbf{q}_{\bar{\alpha}}$ are given as

$$\mathbf{f}_u^{\text{int}}(t_{n+1}, \mathbf{q}_{u_n}, \Delta \mathbf{q}_u, \mathbf{q}_{\bar{\alpha}_n}, \Delta \mathbf{q}_{\bar{\alpha}}) = \int_{\Omega} \mathbf{B}_u^T \boldsymbol{\sigma}_{n+1} dV, \quad (58)$$

$$\mathbf{f}_{\bar{\alpha}}^{\text{int}}(t_{n+1}, \mathbf{q}_{u_n}, \Delta \mathbf{q}_u, \mathbf{q}_{\bar{\alpha}_n}, \Delta \mathbf{q}_{\bar{\alpha}}) = \int_{\Omega} \mathbf{N}_{\bar{\alpha}}^T \mathbf{N}_{\bar{\alpha}} (\mathbf{q}_{\bar{\alpha}_n} + \Delta \mathbf{q}_{\bar{\alpha}}) + l^2 \mathbf{B}_{\bar{\alpha}}^T \mathbf{B}_{\bar{\alpha}} (\mathbf{q}_{\bar{\alpha}_n} + \Delta \mathbf{q}_{\bar{\alpha}}) - \mathbf{N}_{\bar{\alpha}}^T \alpha_{d_{n+1}} dV. \quad (59)$$

Therein, $\mathbf{q}_{\bar{\alpha}_n}$ denotes the known nodal values of the nonlocal field at the beginning of the interval, and $\boldsymbol{\sigma}_{n+1}$ and $\alpha_{d_{n+1}}$ denote the values of the stress and of the local damage driving variable the end of the interval $[t_n, t_{n+1}]$. They are computed by means of the stress update algorithm presented in Section 4 based on the constitutive relations, the prescribed strain increment $\Delta \boldsymbol{\varepsilon} = \mathbf{B}_u \Delta \mathbf{q}_u$ and the value of the nonlocal damage driving variable $\bar{\alpha}_{d_{n+1}}$, departing from the known stress $\boldsymbol{\sigma}_n$, and the known set of internal variables $\boldsymbol{\alpha}_n$ at the beginning of the increment. Accordingly, $\boldsymbol{\sigma}_{n+1}$ is expressed as a function of those quantities:

$$\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}(\Delta \boldsymbol{\varepsilon}, \bar{\alpha}_{d_{n+1}}, t_n, t_{n+1}, \boldsymbol{\sigma}_n, \boldsymbol{\alpha}_n), \quad (60)$$

and in a similar manner $\alpha_{d_{n+1}}$ is expressed as

$$\alpha_{d_{n+1}} = \alpha_d(\Delta \boldsymbol{\varepsilon}, t_n, t_{n+1}, \boldsymbol{\sigma}_n, \boldsymbol{\alpha}_n), \quad (61)$$

which is however, independent of $\bar{\alpha}_{d_{n+1}}$. As a consequence of the employed numerical integration schemes and the time- and path-dependent character of the considered problem, the accuracy of the solution crucially depends on the size of the increments, i.e., on the resolution of the time-discretization.

For solving iteratively the system of nonlinear equations for the unknown nodal increments $\Delta \mathbf{q}_u$ and $\Delta \mathbf{q}_{\bar{\alpha}}$ by means of the Newton-Raphson method, the Jacobian of the nonlinear system is required. It is obtained by consistent linearization of (57) with respect to the nodal increments $\Delta \mathbf{q}_u$ and $\Delta \mathbf{q}_{\bar{\alpha}}$, considering that the external forces are independent of the nodal values:

$$\begin{bmatrix} \frac{\partial \mathbf{f}_u^{\text{int}}}{\partial \Delta \mathbf{q}_u} & \frac{\partial \mathbf{f}_u^{\text{int}}}{\partial \Delta \mathbf{q}_{\bar{\alpha}}} \\ \frac{\partial \mathbf{f}_{\bar{\alpha}}^{\text{int}}}{\partial \Delta \mathbf{q}_u} & \frac{\partial \mathbf{f}_{\bar{\alpha}}^{\text{int}}}{\partial \Delta \mathbf{q}_{\bar{\alpha}}} \end{bmatrix} = \begin{bmatrix} \int_{\Omega} \mathbf{B}_u^T \frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \Delta \boldsymbol{\varepsilon}} \mathbf{B}_u dV & \int_{\Omega} \mathbf{B}_u^T \frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \bar{\alpha}_{d_{n+1}}} \mathbf{N}_{\bar{\alpha}} dV \\ - \int_{\Omega} \mathbf{N}_{\bar{\alpha}}^T \frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \boldsymbol{\varepsilon}} \mathbf{B}_u dV & \int_{\Omega} \mathbf{N}_{\bar{\alpha}}^T \mathbf{N}_{\bar{\alpha}} + l^2 \mathbf{B}_{\bar{\alpha}}^T \mathbf{B}_{\bar{\alpha}} dV \end{bmatrix}. \quad (62)$$

Therein, $\frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \Delta \boldsymbol{\varepsilon}}$ is the algorithmic tangent stiffness, and $\frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \bar{\alpha}_{d_{n+1}}}$ and $\frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \boldsymbol{\varepsilon}}$ are tangents describing the coupling between the displacement field and the nonlocal field, which are computed simultaneously to $\boldsymbol{\sigma}_{n+1}$, $\alpha_{d_{n+1}}$ and $\frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \Delta \boldsymbol{\varepsilon}}$ within the stress update algorithm. It is apparent that the Jacobian is nonsymmetric. Similar to the internal and external force vectors, the Jacobian is obtained by assembling the respective element stiffness matrices.

4 Stress Update Algorithm of the SCDP model at Integration Point Level

At integration point level of a finite element, the SCDP model is implemented in terms of a strain-driven stress update algorithm, computing for a given time interval $[t_n, t_{n+1} = t_n + \Delta t]$, a given strain increment $\Delta \boldsymbol{\varepsilon}$, a given value of the nonlocal damage driving variable $\hat{\alpha}_{d_{n+1}}$ at the end of the increment, and based on the known state of the material at time t_n the updated stress $\boldsymbol{\sigma}(t_{n+1})$ and the updated set of internal variables $\boldsymbol{\alpha}(t_{n+1})$. The set of internal variables $\boldsymbol{\alpha}$ of the SCDP model comprises the hardening variable α_p , the damage driving variable α_d , the damage parameter ω , and the internal variables $\gamma_1, \dots, \gamma_M$ of the viscoelastic part of the model:

$$\boldsymbol{\alpha} = \{\alpha_p, \alpha_d, \omega, \gamma_1, \dots, \gamma_M\}. \quad (63)$$

Variables $\gamma_1, \dots, \gamma_M$ are related to the numerical approximation scheme for the viscoelastic formulation of the SCDP model, and their purpose is explained in the subsequent sections.

In addition, the algorithmic tangent stiffness matrices $\frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \Delta \boldsymbol{\varepsilon}}$, $\frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \alpha_{d_{n+1}}}$ and $\frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \boldsymbol{\varepsilon}}$ are computed within the stress update algorithm. In the following, subscript $(\bullet)_{n+1}$ denotes the value of a variable at the end of the interval, whereas subscript $(\bullet)_n$ denotes the respective value at the beginning of the interval.

Summarizing, the input parameters of the stress update algorithm comprise

- the interval $[t_n, t_{n+1}]$ with time increment $\Delta t = t_{n+1} - t_n$,
- the total strain increment $\Delta \boldsymbol{\varepsilon}$,
- the value of the nonlocal damage driving scalar variable $\bar{\alpha}_{d_{n+1}}$,
- the nominal stress $\boldsymbol{\sigma}_n$ at the beginning of the interval,
- the set of internal state variables $\boldsymbol{\alpha}_n = \{\alpha_{p_n}, \alpha_{d_n}, \omega_n, \gamma_{1_n}, \dots, \gamma_{M_n}\}$ at the beginning of the interval,

whereas the output parameters of the algorithm comprise

- the updated nominal stress $\boldsymbol{\sigma}_{n+1}$,
- the updated set of internal state variables $\boldsymbol{\alpha}_{n+1} = \{\alpha_{p_{n+1}}, \alpha_{d_{n+1}}, \omega_{n+1}, \gamma_{1_{n+1}}, \dots, \gamma_{M_{n+1}}\}$,
- the algorithmic tangent matrices: $\frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \Delta \boldsymbol{\varepsilon}}$, $\frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \bar{\alpha}_{d_{n+1}}}$ and $\frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \boldsymbol{\varepsilon}}$.

Departing from (1), the nominal stress $\boldsymbol{\sigma}_n$ at the beginning of the interval is given as

$$\boldsymbol{\sigma}_n = (1 - \omega_n) \bar{\boldsymbol{\sigma}}_n \quad (64)$$

and the nominal stress $\boldsymbol{\sigma}_{n+1}$ at the end of the interval is computed as

$$\begin{aligned} \boldsymbol{\sigma}_{n+1} &= (1 - \omega_{n+1}) \bar{\boldsymbol{\sigma}}_{n+1} \\ &= (1 - \omega_{n+1}) (\bar{\boldsymbol{\sigma}}_n + \Delta \bar{\boldsymbol{\sigma}}). \end{aligned} \quad (65)$$

For computing the effective stress increment $\Delta\bar{\sigma}$ in (65) the relation between the effective stress rate $\dot{\bar{\sigma}}$ and the strain rate $\dot{\epsilon}$ is expressed as

$$\dot{\bar{\sigma}} = \mathbb{C} : (\dot{\epsilon} - \dot{\epsilon}^p - \dot{\epsilon}^{ve} - \dot{\epsilon}^f - \dot{\epsilon}^{shr}), \quad (66)$$

which is converted to the incremental form by integrating over the interval $[t_n, t_{n+1}]$:

$$\Delta\bar{\sigma} = \mathbb{C} : (\Delta\epsilon - \Delta\epsilon^p - \Delta\epsilon^{ve} - \Delta\epsilon^f - \Delta\epsilon^{shr}). \quad (67)$$

Hence, for computing the nominal stress σ_{n+1} within the stress update algorithm for a given interval $[t_n, t_{n+1}]$ and a given strain increment $\Delta\epsilon$, the increment of the effective stress $\bar{\sigma}$ is computed based on the partial strain increments $\Delta\epsilon^p, \Delta\epsilon^{ve}, \Delta\epsilon^f$ and $\Delta\epsilon^{shr}$, and the updated damage parameter ω_{n+1} is calculated. The involved individual steps are presented in the following.

4.1 Shrinkage Strain Increment

The increment of the shrinkage strain is independent of the stress, and it is directly computed according to the time function of the Bažant-Panula model (17) as

$$\Delta\epsilon^{shr} = \epsilon^{shr}(t_{n+1}) - \epsilon^{shr}(t_n) = \mathbf{I} \epsilon_{\infty}^{shr} k_h (S(t + \Delta t, \tau_{shr}) - S(t, \tau_{shr})). \quad (68)$$

Alternatively, the shrinkage strain increment can be evaluated in a similar manner for the shrinkage law according to (ACI Committee 209, 1992) in (18).

4.2 Viscoelastic Strain Increment

For computing the viscoelastic strain increment, the evolution equation according to the modified solidification theory (3), is integrated over $[t_n, t_{n+1}]$. However, since the integral expression in (3) is formulated in terms of the complete stress history, its direct numerical implementation for computing $\dot{\epsilon}^{ve}(t)$ is not reasonable, as the number of arithmetic operations and the required memory rapidly increase with increasing number of increments. Instead, for an efficient numerical implementation a differential expression is preferable, for which the complete history can be represented by a finite number of internal variables. To this end, the approach proposed by Bažant and Prasannan (1989b) and reformulated by Jirásek and Bažant (2002) is adopted, which is based on two steps, (i) and (ii):

(i) The compliance function $\Phi(t - t')$ in (5) is approximated by means of a Dirichlet series, consisting of a finite number of terms:

$$\Phi(t - t') \approx \sum_{i=1}^M \frac{1}{D_i} \left(1 - \exp\left(\frac{-(t - t')}{\tau_i}\right) \right). \quad (69)$$

Therein, M is the number of Dirichlet terms, D_i are the moduli, and τ_i are the retardation times of the Dirichlet series. The determination of D_i and τ_i is presented in Section 4.6.

Accordingly, the viscoelastic strain rate (3) is rewritten as

$$\dot{\epsilon}^{ve}(t) = \frac{F(\bar{\sigma}(t))}{v_{sc}(t)} \sum_{i=1}^M \gamma_i(t) \quad (70)$$

with

$$\gamma_i(t) = \int_{\bar{\sigma}(0)}^{\bar{\sigma}(t)} \frac{1}{\tau_i D_i} \exp\left(\frac{-(t-t')}{\tau_i}\right) \mathbb{C}_\nu^{-1} : d\bar{\sigma}(t'), \quad (71)$$

in which $\gamma_1(t), \dots, \gamma_M(t)$ are the internal history variables of the Dirichlet series.

(ii) The aforementioned preferable differential expression for describing the evolution of the internal variables is obtained in terms of the solution of a first order differential equation. For constituting the first order differential equation, the first derivative of $\gamma_i(t)$ is computed by application of the Leibniz rule as

$$\dot{\gamma}_i(t) = -\frac{1}{\tau_i} \int_{\bar{\sigma}(0)}^{\bar{\sigma}(t)} \frac{1}{\tau_i D_i} \exp\left(\frac{-(t-t')}{\tau_i}\right) \mathbb{C}_\nu^{-1} : d\bar{\sigma}(t') + \frac{1}{\tau_i D_i} \mathbb{C}_\nu^{-1} : \dot{\bar{\sigma}}(t). \quad (72)$$

Combining (71) and (72) yields the first order differential equation

$$\frac{1}{\tau_i} \gamma_i(t) + \dot{\gamma}_i(t) = \frac{1}{\tau_i D_i} \mathbb{C}_\nu^{-1} : \dot{\bar{\sigma}}. \quad (73)$$

If the right hand side forms a time-independent expression, an analytical solution for (73) can be computed by application of the exponential algorithm (Zienkiewicz, Watson, and King, 1968; Bažant, 1971). This can be exploited by assuming a constant stress rate within the interval $[t_n, t_{n+1}]$ given as $\dot{\bar{\sigma}}(t) = \Delta\bar{\sigma}/\Delta t$, and consequently, the analytical solution for the interval $[t_n, t_{n+1}]$ is obtained as

$$\gamma_i(t) = \mathbb{C}_\nu^{-1} : \frac{\Delta\bar{\sigma}}{\Delta t D_i} + \mathbf{K}_i \exp\left(\frac{-(t-t_n)}{\tau_i}\right). \quad (74)$$

Constant $\mathbf{K}_i$ is determined from the initial condition $\gamma_i(t_n) = \gamma_{i_n}$ within in the interval $[t_n, t_{n+1}]$:

$$\mathbf{K}_i = \left(\gamma_{i_n} - \mathbb{C}_\nu^{-1} : \frac{\Delta\bar{\sigma}}{\Delta t D_i} \right) \exp\left(\frac{t_n}{\tau_i}\right). \quad (75)$$

Substitution into (74) finally yields

$$\gamma_i(t) = \gamma_{i_n} \exp\left(\frac{-(t-t_n)}{\tau_i}\right) + \mathbb{C}_\nu^{-1} : \frac{\Delta\bar{\sigma}}{\Delta t D_i} \left(1 - \exp\left(\frac{-(t-t_n)}{\tau_i}\right)\right), \quad (76)$$

which, in contrast to (71), is not formulated in terms of the complete stress history, and allows to describe the evolution of γ_i for each interval $[t_n, t_{n+1}]$ starting from the known state γ_{i_n} .

The viscoelastic strain increment is computed by integrating (70) over the interval $[t_n, t_{n+1}]$:

$$\Delta\epsilon^{\text{ve}} = \int_{t_n}^{t_{n+1}} \frac{F(\bar{\sigma}(t))}{v_{\text{sc}}(t)} \sum_{i=1}^M \gamma_i(t) dt. \quad (77)$$

For evaluating the integral, the solidified volume is approximated by its value at midstep of the interval as $v_{\text{sc}}(t_n + \Delta t/2)$, and the nonlinear creep amplification function $F(\bar{\sigma}(t))$ is approximated in terms of the effective stress at the beginning of the increment as $F(\bar{\sigma}_n)$:

$$\Delta\epsilon^{\text{ve}} \approx \frac{F(\bar{\sigma}_n)}{v_{\text{sc}}(t_n + \Delta t/2)} \sum_{i=1}^M \int_{t_n}^{t_{n+1}} \gamma_i(t) dt. \quad (78)$$

Substitution of (76) into (78) and evaluation of the integral yields

$$\Delta \epsilon^{\text{ve}} = \frac{F(\bar{\sigma}_n)}{v_{\text{sc}}(t_n + \Delta t/2)} \sum_{i=1}^M \left(\mathbb{C}_\nu^{-1} : \frac{\Delta \bar{\sigma}}{D_i} (1 - \lambda_i) + \gamma_{i_n} (1 - \beta_i) \right), \quad (79)$$

with

$$\beta_i = \exp \left(\frac{-\Delta t}{\tau_i} \right), \quad (80)$$

$$\lambda_i = \frac{\tau_i}{\Delta t} (1 - \beta_i). \quad (81)$$

As apparent from (79), the increment of the viscoelastic strain $\Delta \epsilon^{\text{ve}}$ is an inhomogeneous linear function of the effective stress increment $\Delta \bar{\sigma}$. If a different rule for approximating the amplification function $F(\bar{\sigma}(t))$ is employed, e.g., the midpoint rule as $F(\bar{\sigma}(t_n + \Delta t/2))$, (79) becomes nonlinear in the stress increment, and hence, an iterative solution procedure for computing the viscoelastic strain increment must be employed.

It can be shown that the approximation in terms of a Dirichlet series is equivalent to the representation of the viscoelastic part of the model by means of an aging Kelvin-chain, i.e., a serial coupling of a finite number of parallel spring-dashpot units with time-dependent stiffness and viscosity, as shown in Figure 2. To demonstrate equivalence, the stress-strain relation for a single Kelvin unit in one dimension with time-dependent properties according to Bažant (1979)

$$E(t) \dot{\epsilon}(t) + \eta(t) \dot{\epsilon}(t) + \eta(t) \ddot{\epsilon}(t) = \dot{\sigma}(t) \quad (82)$$

is considered, with $E(t)$ denoting the stiffness of the spring and $\eta(t)$ denoting the dashpot viscosity. For the viscoelastic part of the SCDP model, the contribution of a single term of the Dirichlet series (70) to the viscoelastic strain rate is given as

$$\dot{\epsilon}_i^{\text{ve}}(t) = \frac{F(\bar{\sigma}(t))}{v_{\text{sc}}(t)} \gamma_i(t), \quad (83)$$

from which $\gamma_i(t)$ and $\dot{\gamma}_i(t)$ follow as

$$\gamma_i(t) = \frac{v_{\text{sc}}(t)}{F(\bar{\sigma}(t))} \dot{\epsilon}_i^{\text{ve}}(t), \quad (84)$$

$$\dot{\gamma}_i(t) = \frac{d}{dt} \left(\frac{v_{\text{sc}}(t)}{F(\bar{\sigma}(t))} \right) \dot{\epsilon}_i^{\text{ve}}(t) + \frac{v_{\text{sc}}(t)}{F(\bar{\sigma}(t))} \ddot{\epsilon}_i^{\text{ve}}(t). \quad (85)$$

Subsequent substitution into (73) yields

$$\frac{1}{\tau_i} \frac{v_{\text{sc}}(t)}{F(\bar{\sigma}(t))} \dot{\epsilon}_i^{\text{ve}}(t) + \frac{d}{dt} \left(\frac{v_{\text{sc}}(t)}{F(\bar{\sigma}(t))} \right) \dot{\epsilon}_i^{\text{ve}}(t) + \frac{v_{\text{sc}}(t)}{F(\bar{\sigma}(t))} \ddot{\epsilon}_i^{\text{ve}}(t) = \frac{1}{\tau_i D_i} \mathbb{C}_\nu^{-1} : \dot{\bar{\sigma}}(t), \quad (86)$$

which has a structure similar to (82).

4.3 Flow Strain Increment

For integrating the flow strain rate of the SCDP model (4) over the interval $[t_n, t_{n+1}]$, the numerical approximation

$$\Delta \epsilon^f = \int_{t_n}^{t_{n+1}} q_4 \frac{F(\bar{\sigma}(t))}{t^{e_{\text{flow}}}} \mathbb{C}_\nu^{-1} : \bar{\sigma}(t) dt \approx q_4 \frac{F(\bar{\sigma}_n) \Delta t}{(t_n + \Delta t/2)^{e_{\text{flow}}}} \mathbb{C}_\nu^{-1} : (\bar{\sigma}_n + \Delta \bar{\sigma}/2) \quad (87)$$

is used. Similar as for the viscoelastic strain increment, the amplification function $F(\bar{\sigma}(t))$ is evaluated at t_n as $F(\bar{\sigma}_n)$, in order to preserve the inhomogeneous linear relationship between the flow strain increment $\Delta \epsilon^f$ and the effective stress increment $\Delta \bar{\sigma}$.

4.4 Plastic Strain Increment

For computing the increment of the plastic strain, the fully implicit return mapping algorithm is employed, as presented by Simo and T. J. R. Hughes (1998) and in particular by Valentini (2011) for the damage plasticity model for concrete. For the sake of brevity, the following description is limited to the application of the return mapping algorithm for the stress update of the SCDP model and the thereby resulting nonlinear algebraic equation system, while reference is made to (Valentini, 2011) for a detailed description of the computational treatment of the return mapping algorithm. In addition, issues related to the yield function and the plastic potential of the SCDP model, as well as a substepping scheme are discussed briefly in the following.

Following the return mapping algorithm, a trial effective stress $\bar{\sigma}_{n+1}^{\text{tr}}$ assuming $\Delta \epsilon^p = \mathbf{0}$, denoted as the elastic predictor, is computed to determine if the yield condition is violated, i.e., $f_p(\bar{\sigma}_{n+1}^{\text{tr}}, \alpha_{p_n}, t_{n+1}) > 0$.

If $f_p(\bar{\sigma}_{n+1}^{\text{tr}}, \alpha_{p_n}, t_{n+1}) \leq 0$, no plastic flow occurs, i.e., $\Delta \epsilon^p = \mathbf{0}$, and accordingly the strain-like internal variables α_p and α_d do not change within the current interval $[t_n, t_{n+1}]$, i.e., $\alpha_{p_{n+1}} = \alpha_{p_n}$, $\alpha_{d_{n+1}} = \alpha_{d_n}$ and consequently, $\omega_{n+1} = \omega_n$. Otherwise, i.e., if $f_p(\bar{\sigma}_{n+1}^{\text{tr}}, \alpha_{p_n}, t_{n+1}) > 0$, the return mapping algorithm is performed.

For computing the trial effective stress, departing from (67) the relation between the effective stress $\bar{\sigma}_{n+1}$ and the total strain increment $\Delta \epsilon$ is given as

$$\bar{\sigma}_{n+1} = \bar{\sigma}_n + \mathbb{C} : (\Delta \epsilon - \Delta \epsilon^{\text{shr}} - \Delta \epsilon^{\text{ve}}(\Delta \bar{\sigma}) - \Delta \epsilon^f(\Delta \bar{\sigma}) - \Delta \epsilon^p(\Delta \bar{\sigma})), \quad (88)$$

and accordingly, the effective trial stress is computed as

$$\bar{\sigma}_{n+1}^{\text{tr}} = \bar{\sigma}_n + \mathbb{C} : (\Delta \epsilon - \Delta \epsilon^{\text{shr}} - \Delta \epsilon^{\text{ve}}(\Delta \bar{\sigma}) - \Delta \epsilon^f(\Delta \bar{\sigma})). \quad (89)$$

As indicated in (88) and (89), both the viscoelastic strain and the flow strain increment are dependent on the effective stress increment $\Delta \bar{\sigma}$, which is not known a priori in case of plastic flow, and consequently, the structure of (88) differs from the standard problem of small strain plasticity.

However, the inhomogeneous linear relation between the increments of the viscoelastic strain and the flow strain on the one hand, and the increment of the effective stress on the other hand can be exploited by means of an effective stiffness tensor, leading to a computationally efficient algorithm. This is accomplished in two steps: (i) The effective stiffness tensor is derived, and (ii) (88) and (89) are reformulated in order to obtain the standard problem of small strain plasticity, expressed in

terms of the effective stiffness tensor.

(i) For deducing the effective stiffness tensor, the effective stiffness $\tilde{E}$ for the interval $[t_n, t_{n+1}]$ is computed according to the approach presented by Jirásek and Bažant (2002): As apparent from (79) and (87), both the viscoelastic strain increment and the flow strain increment are composed of terms which are independent of the stress increment, and terms which are linear in the effective stress increment. Departing from this observation, a decomposition of the total strain increment is introduced as

$$\Delta \epsilon = \Delta \epsilon'' + \Delta \hat{\epsilon} + \Delta \epsilon^p, \quad (90)$$

with

$$\Delta \epsilon'' = \Delta \epsilon^{\text{shr}} + \frac{F(\bar{\sigma}_n)}{v_{\text{sc}}(t_n + \Delta t/2)} \sum_{i=1}^M \gamma_{i_n} (1 - \beta_i) + q_4 \frac{F(\bar{\sigma}_n) \Delta t}{(t_n + \Delta t/2)^{e_{\text{flow}}}} \mathbb{C}_\nu^{-1} : \bar{\sigma}_n \quad (91)$$

and

$$\Delta \hat{\epsilon} = \left(q_1 + \frac{F(\bar{\sigma}_n)}{v_{\text{sc}}(t_n + \Delta t/2)} \sum_{i=1}^M \frac{1}{D_i} (1 - \lambda_i) + \frac{1}{2} q_4 \frac{F(\bar{\sigma}_n) \Delta t}{(t_n + \Delta t/2)^{e_{\text{flow}}}} \right) \mathbb{C}_\nu^{-1} : \Delta \bar{\sigma}, \quad (92)$$

based on a decomposition of the viscoelastic strain increment and the flow strain increment, and based on the definition of $\mathbb{C}$ in (2):

$$\mathbb{C}^{-1} = q_1 \mathbb{C}_\nu^{-1}. \quad (93)$$

The expression in parentheses in (92),

$$\frac{1}{\tilde{E}} = q_1 + \frac{F(\bar{\sigma}_n)}{v_{\text{sc}}(t_n + \Delta t/2)} \sum_{i=1}^M \frac{1}{D_i} (1 - \lambda_i) + \frac{1}{2} q_4 \frac{F(\bar{\sigma}_n) \Delta t}{(t_n + \Delta t/2)^{e_{\text{flow}}}}, \quad (94)$$

defines the effective stiffness $\tilde{E}$ for the interval $[t_n, t_{n+1}]$, from which the effective stiffness tensor $\tilde{\mathbb{C}}$ is computed as

$$\tilde{\mathbb{C}} = \mathbb{C}_\nu \tilde{E}. \quad (95)$$

(ii) Expressions (88) and (89) are reformulated in order to obtain a structure amenable to the standard return mapping algorithm. Departing from the stress-strain relation in (92), and substitution of the effective stiffness tensor (95) into (92) and considering the decomposition of the total strain (90) yields after rearrangement

$$\Delta \bar{\sigma} = \tilde{\mathbb{C}} : \Delta \hat{\epsilon} = \tilde{\mathbb{C}} : (\Delta \epsilon - \Delta \epsilon'' - \Delta \epsilon^p) \quad (96)$$

and finally

$$\begin{aligned} \bar{\sigma}_{n+1} &= \bar{\sigma}_n + \tilde{\mathbb{C}} : (\Delta \epsilon - \Delta \epsilon'' - \Delta \epsilon^p) \\ &= \bar{\sigma}_n + \tilde{\mathbb{C}} : (\Delta \tilde{\epsilon} - \Delta \epsilon^p) \end{aligned} \quad (97)$$

with

$$\Delta \tilde{\epsilon} = \Delta \epsilon - \Delta \epsilon'', \quad (98)$$

for which the identical structure to the standard problem of small strain plasticity is immediately apparent. Accordingly, the trial effective stress is computed as

$$\bar{\sigma}_{n+1}^{\text{tr}} = \bar{\sigma}_n + \tilde{\mathbb{C}} : \Delta \tilde{\epsilon}, \quad (99)$$

and the return mapping algorithm is performed based on the effective stiffness tensor $\tilde{\mathbb{C}}$.

Further discussion is required on the yield function and the plastic potential function: For both the yield function and the plastic potential function the partial derivatives with respect to the effective stress are not defined uniquely for stress states located on the hydrostatic axis. For the yield surface, this issue is manifested by two vertices at its intersection with the hydrostatic axis. Consequently, the standard return mapping algorithm fails for trial stress states located in those domains of the effective stress space, for which the back projected stress state is located at one of those vertices. As a remedy, the approach described in (Hofstetter and Taylor, 1991; Valentini, 2011) is followed: A second cut-off yield function is employed, defining two plane yield surfaces orthogonal to the hydrostatic axis, and cutting the main yield function slightly beneath the vertices.

The cut-off yield function is given as

$$f_{\text{cut-off}}(\bar{\sigma}_m, q_h(\alpha_p), t) = \left(\frac{1 - q_h(\alpha_p)}{f_{\text{cu}}^2(t)} \right)^2 (\bar{\sigma}_m + \text{sgn}(\bar{\sigma}_m) \sigma_m^{\text{shift}})^4 + \frac{m_0 q_h^2(\alpha_p)}{f_{\text{cu}}(t)} (\bar{\sigma}_m + \text{sgn}(\bar{\sigma}_m) \bar{\sigma}_m^{\text{shift}}) - q_h^2(\alpha_p) \quad (100)$$

with $\bar{\sigma}_m^{\text{shift}}$ denoting a numerical parameter for shifting the cut-off yield function a small amount beneath the vertices of the main yield surface, defined as $\bar{\sigma}_m^{\text{shift}} = 10^{-4}$ MPa. For the cut-off yield surface, an associated plastic potential

$$g_{\text{cut-off}}(\bar{\sigma}_m, q_h(\alpha_p), t) = f_{\text{cut-off}}(\bar{\sigma}_m, q_h(\alpha_p), t) \quad (101)$$

is employed.

Consequently, the return mapping algorithm is formulated within the framework of multi-surface plasticity, based on Koiter's generalized flow rule (Koiter, 1953, 1964). The evolution equation for the plastic strain (24) is extended to account for the cut-off yield surface as

$$\dot{\epsilon}^p = \sum_{j \in I_{\text{act}}} \dot{\lambda}_j \frac{\partial g_j(\bar{\sigma}, q_h(\alpha_p), t)}{\partial \bar{\sigma}}, \quad (102)$$

with

$$I_{\text{act}} = \{j \in \{p, \text{cut-off}\} \mid f_j(\bar{\sigma}, q_h(\alpha_p), t) = 0\} \quad (103)$$

denoting the active set of yield functions, complemented by the Kuhn-Tucker conditions

$$\dot{\lambda}_j \geq 0, \quad f_j(\bar{\sigma}, q_h(\alpha_p), t) \leq 0, \quad \dot{\lambda}_j f_j(\bar{\sigma}, q_h(\alpha_p), t) = 0 \quad (104)$$

and the consistency condition

$$\dot{\lambda}_j \dot{f}_j(\bar{\sigma}, q_h(\alpha_p), t) = 0. \quad (105)$$

Fully implicit integration of the rate equations yields the incremental form of (102) and (25) as

$$\Delta \epsilon^p(\bar{\sigma}_{n+1}, \alpha_{p_{n+1}}, \Delta \lambda_j, t_{n+1}) = \sum_{j \in I_{\text{act}}} \Delta \lambda_j \frac{\partial g_i(\bar{\sigma}_{n+1}, q_h(\alpha_{p_{n+1}}), t_{n+1})}{\partial \bar{\sigma}_{n+1}}, \quad (106)$$

$$\Delta \alpha_p(\bar{\sigma}_{n+1}, \alpha_{p_{n+1}}, \Delta \lambda_j, t_{n+1}) = \|\Delta \epsilon^p(\bar{\sigma}_{n+1}, \alpha_{p_{n+1}}, \Delta \lambda_j)\| \frac{1 + 3 \frac{\bar{\rho}_{n+1}^2}{\bar{\rho}_{n+1}^2 + \vartheta_h} \cos^2(1.5 \theta)}{x_h((\bar{\sigma}_m)_{n+1}, t_{n+1})}, \quad (107)$$

expressed as nonlinear functions of the unknowns $\{\boldsymbol{\sigma}_{n+1}, \alpha_{p_{n+1}}, \Delta\lambda_j\}$.

Finally, based on those incremental evolution equations, the system of nonlinear algebraic equations of the return mapping algorithm is formulated as:

$$\begin{Bmatrix} \bar{\boldsymbol{\sigma}}_{n+1}^{\text{tr}} \\ \alpha_{p_n} \\ \mathbf{0} \end{Bmatrix} = \begin{Bmatrix} \bar{\boldsymbol{\sigma}}_{n+1} + \tilde{\mathbb{C}} : \Delta\boldsymbol{\varepsilon}^p(\bar{\boldsymbol{\sigma}}_{n+1}, \alpha_{p_{n+1}}, \Delta\lambda_j, t_{n+1}) \\ \alpha_{p_{n+1}} - \Delta\alpha_p(\bar{\boldsymbol{\sigma}}_{n+1}, \alpha_{p_{n+1}}, \Delta\lambda_j, t_{n+1}) \\ \mathbf{f}(\bar{\boldsymbol{\sigma}}_{n+1}, q_h(\alpha_{p_{n+1}}), t_{n+1}) \end{Bmatrix}, \quad (108)$$

in which $\mathbf{f}(\bar{\boldsymbol{\sigma}}_{n+1}, q_h(\alpha_{p_{n+1}}), t_{n+1})$ is the vector of all active yield functions:

$$\mathbf{f}(\bar{\boldsymbol{\sigma}}_{n+1}, q_h(\alpha_{p_{n+1}}), t_{n+1}) = \begin{cases} \left\{ f_p(\bar{\boldsymbol{\sigma}}_{n+1}, q_h(\alpha_{p_{n+1}}), t_{n+1}) \right\} & \text{if } I_{\text{act}} = \{p\}, \\ \left\{ f_{\text{cut-off}}(\bar{\boldsymbol{\sigma}}_{n+1}, q_h(\alpha_{p_{n+1}}), t_{n+1}) \right\} & \text{if } I_{\text{act}} = \{\text{cut-off}\}, \\ \left\{ \begin{array}{l} f_p(\bar{\boldsymbol{\sigma}}_{n+1}, q_h(\alpha_{p_{n+1}}), t_{n+1}) \\ f_{\text{cut-off}}(\bar{\boldsymbol{\sigma}}_{n+1}, q_h(\alpha_{p_{n+1}}), t_{n+1}) \end{array} \right\} & \text{if } I_{\text{act}} = \{p, \text{cut-off}\}. \end{cases} \quad (109)$$

Within the stress update algorithm, the nonlinear system (108) is solved using the Newton-Raphson method.

For improving the robustness of the return mapping algorithm, the substepping scheme presented by Pérez-Foguet, Rodríguez-Ferran, and Huerta (2001) for plasticity models and extended by Valentini (2011) to the damage plasticity model for concrete is employed: If the return mapping procedure fails, i.e., if the Newton-Raphson procedure does not converge, the trial strain increment $\Delta\tilde{\boldsymbol{\varepsilon}}$ is additively split into N subincrements $\Delta\tilde{\boldsymbol{\varepsilon}}_k$, defined by subincrement ratios α_k given as

$$\Delta\tilde{\boldsymbol{\varepsilon}} = \sum_{k=1}^N \Delta\tilde{\boldsymbol{\varepsilon}}_k = \sum_{k=1}^N \alpha_k \Delta\tilde{\boldsymbol{\varepsilon}}. \quad (110)$$

Following the substepping scheme, the return mapping algorithm is applied in a sequential manner for N subincrements. The stress at the end of each subincrement is given as

$$(\bar{\boldsymbol{\sigma}}_{n+1})_k = (\bar{\boldsymbol{\sigma}}_{n+1})_{k-1} + \tilde{\mathbb{C}} : (\Delta\tilde{\boldsymbol{\varepsilon}}_k - \Delta\boldsymbol{\varepsilon}_k^p), \quad (111)$$

with $(\bar{\boldsymbol{\sigma}}_{n+1})_0 = \bar{\boldsymbol{\sigma}}_n$, and with the respective trial effective stress computed as

$$(\bar{\boldsymbol{\sigma}}_{n+1}^{\text{tr}})_k = (\bar{\boldsymbol{\sigma}}_{n+1})_{k-1} + \tilde{\mathbb{C}} : \Delta\tilde{\boldsymbol{\varepsilon}}_k. \quad (112)$$

Consequently, the total plastic strain increment $\Delta\boldsymbol{\varepsilon}^p$ is computed as the sum of the plastic subincrements as

$$\Delta\boldsymbol{\varepsilon}^p = \sum_{k=1}^N \Delta\boldsymbol{\varepsilon}_k^p. \quad (113)$$

The total number of subincrements N is not known a priori, but follows from the adaptive scheme

which becomes active if the return mapping fails. For the adaptive scheme, a cutback ratio of 10 is employed: If the return mapping algorithm fails, the current subincrement size is divided by a factor of 10, and the return mapping procedure is repeated with the reduced subincrement. As soon as a number of 5 successful subincrements are computed, the subincrement size is increased again by a factor of 1.5 in order to minimize the overall number of involved computational steps. If a specified minimum size of the subincrement is attained, i.e., if $\alpha_k \leq 1 \times 10^{-3}$, the complete stress update procedure for the present strain increment $\Delta \boldsymbol{\varepsilon}$ is abandoned, and the global displacement increment size must be decreased.

4.5 Damage Increment

With the increment of the plastic strain at hand, departing from the rate form evolution law for the damage-driving internal variable α_d (25) the increment $\Delta \alpha_d$ is computed as

$$\Delta \alpha_d = \begin{cases} 0 & \text{if } \alpha_{p_{n+1}} < 1, \\ \Delta \boldsymbol{\varepsilon}_V^p / x_s(\Delta \boldsymbol{\varepsilon}^p) & \text{otherwise,} \end{cases} \quad (114)$$

and consequently, weighted damage driving variable $\hat{\alpha}_d$ is updated

$$\hat{\alpha}_{d_{n+1}} = m \bar{\alpha}_{d_{n+1}} + (1 - m) (\alpha_{d_n} + \Delta \alpha_d) \quad (115)$$

and damage parameter ω is updated as

$$\omega_{n+1} = 1 - \exp \left(-\frac{\hat{\alpha}_{d_{n+1}}}{\epsilon_f} \right). \quad (116)$$

4.6 Determination of the Parameters for the Dirichlet Series

The approximation of $\Phi(t - t')$ in (69) by means of a Dirichlet series composed of M terms allows to describe the viscoelastic strain rate in terms of first order ordinary differential equations, corresponding to an aging Kelvin chain. To this end, the parameters of the Dirichlet series, i.e., the moduli D_i and the retardation times τ_i , are chosen in order to approximate the continuous retardation spectrum of the exact compliance function (5) by means of the discrete retardation spectrum of the Dirichlet series.

The parameters of the Dirichlet series are determined according to the approach proposed by Bažant and Xi (1995) and extended by Jirásek and Havlásek (2014). The basic idea behind that approach is that any compliance function can be computed from its continuous retardation spectrum $L(\tau)$ by means of transformation

$$\Phi(t) = \int_{\tau=0}^{\infty} L(\tau) (1 - \exp(-t/\tau)) d(\ln \tau), \quad (117)$$

and that the continuous retardation spectrum $L(\tau)$ can be approximated by exploiting the Post-Widder inversion formula for Laplace transforms (Widder, 1971), given as

$$L(\tau) \approx L_k(\tau) = \frac{(-k\tau)^k}{(k-1)!} \Phi^{(k)}(k\tau), \quad \text{for } k \in \{1, 2, \dots\}. \quad (118)$$

Therein, k denotes the order of approximation and $\Phi^{(k)}$ the k th derivate of the compliance function. The Post-Widder inversion formula has the property

$$L(\tau) = \lim_{k \rightarrow \infty} L_k(\tau), \quad (119)$$

i.e., the approximation $L_k(\tau)$ converges to the exact continuous spectrum $L(\tau)$ with increasing order k .

For the discrete spectrum of the Dirichlet series, the retardation times τ_i are chosen such that they form a geometric progression with common ratio 10, i.e., $\tau_{i+1} = 10 \tau_i$. The shortest retardation time τ_1 is chosen such that any duration of load of interest t in the numerical analyses is longer than τ_1 , i.e., $\tau_1 \ll t$, whereas τ_M is chosen such that $\tau_M \gg t$.

For a chosen set of retardation times and a chosen order of approximation of the continuous spectrum $L(t)$ of the exact compliance function $\Phi(t)$ in (5), integral (117) is approximated as

$$\Phi(t) \approx \int_{\tau=0}^{\tau_1/\sqrt{10}} L_k d(\ln(\tau)) + \ln(10) \sum_{i=1}^M L_k(\tau_i) (1 - \exp(-t/\tau_i)). \quad (120)$$

The second term on the right hand side is obtained by application of the midpoint rule with a step size of $\ln(10)$ for the interval $[\tau_1/\sqrt{10}, \tau_M]$, with each retardation time τ_i covering one order of magnitude of the time scale, defined by the interval $[\tau_i/\sqrt{10}, \tau_i \cdot \sqrt{10}]$ (Jirásek and Havlásek, 2014). This term corresponds directly to the presented approximation of the compliance function by means of a Dirichlet series, as a comparison of (120) with (69) reveals. The interval $[\tau_M \cdot \sqrt{10}, \infty]$ is neglected in the approximation. The first term on the right hand side requires further explanations: In (117), the expression $(1 - \exp(-t/\tau))$ is very close to 1 for all $\tau \ll \tau_1$ and $t \gg \tau_1$. This is exploited in (120) by means of the approximation

$$\int_{\tau=0}^{\tau_1/\sqrt{10}} L_k (1 - \exp(-t/\tau)) d(\ln(\tau)) \approx \int_{\tau=0}^{\tau_1/\sqrt{10}} L_k d(\ln(\tau)). \quad (121)$$

Accordingly, the approximation of $\Phi(t - t')$ in (69) can be rewritten as

$$\Phi(t - t') = \frac{1}{D_0} + \sum_{i=1}^M \frac{1}{D_i} \left(1 - \exp\left(\frac{-(t - t')}{\tau_i}\right) \right), \quad (122)$$

with the moduli expressed as

$$\frac{1}{D_0} = \int_{\tau=0}^{\tau_1/\sqrt{10}} L_k(\tau) d(\ln(\tau)), \quad (123)$$

$$\frac{1}{D_i} = \ln(10) L_k(\tau_i) \quad \text{for } i = 1, 2, \dots, M. \quad (124)$$

As pointed out by Jirásek and Havlásek (2014), for practical applications the second order approximation $L_2(\tau)$ of the continuous retardation spectrum of $\Phi(t)$ is sufficiently accurate, which is

computed by choosing $k = 2$ and substituting (5) into (118) as

$$\begin{aligned} L_2(\tau) &= \frac{(-2\tau)^2}{(2-1)!} \Phi^{(2)}(2\tau) \\ &= q_2 \frac{0.1(2\tau)^{0.1}(0.9 + (2\tau)^{0.1})}{(1 + (2\tau)^{0.1})^2}. \end{aligned} \quad (125)$$

Based on the second order approximation, the moduli of the Dirichlet series are computed as

$$\frac{1}{D_0} = q_2 \left(\ln(1 + \tilde{\tau}_0) - \frac{\tilde{\tau}_0}{10(1 + \tilde{\tau}_0)} \right), \quad (126)$$

$$\frac{1}{D_i} = q_2 \ln(10) \frac{\tilde{\tau}_i(0.9 + \tilde{\tau}_i)}{10(1 + \tilde{\tau}_i)^2} \quad \text{for } i = 1, 2, \dots, M, \quad (127)$$

with

$$\tilde{\tau}_0 = \left(\frac{2\tau_1}{\sqrt{10}} \right)^{0.1}, \quad (128)$$

$$\tilde{\tau}_i = (2\tau_i)^{0.1} \quad \text{for } i = 1, 2, \dots, M. \quad (129)$$

To account for the additional time-independent term $\frac{1}{D_0}$ within the stress update algorithm of the SCDP model, it is incorporated into the expression for the effective stiffness (94):

$$\frac{1}{\tilde{E}} = q_1 + \frac{F(\bar{\sigma}_n)}{v_{sc}(t_n + \Delta t/2)} \left(\frac{1}{D_0} + \sum_{i=1}^M \frac{1}{D_i} (1 - \lambda_i) \right) + \frac{1}{2} q_4 \frac{F(\bar{\sigma}_n) \Delta t}{(t_n + \Delta t/2)^{e_{flow}}}. \quad (130)$$

In addition, Jirásek and Havlásek (2014) proposed two improvements in a heuristic manner for better agreement of the second order approximation with the exact compliance function: (i) After determining the moduli D_i , all retardation times are multiplied by a factor of 1.35, and (ii) the longest retardation time τ_M is additionally multiplied a factor of 1.2. A justification for those heuristic improvements despite the present convergence properties (119) of the Post-Widder formula is given also in (Jirásek and Havlásek, 2014).

4.7 Algorithmic Tangents

For preserving the quadratic rate of convergence within the global Newton-Raphson scheme, the finite element stiffness matrices are computed by means of the algorithmic tangent stiffness matrices. They are computed by consistent linearization of the stress update algorithm, as presented in the following. The first algorithmic tangent describes the change of σ_{n+1} with respect to the strain increment $\Delta \epsilon$:

$$\frac{\partial \sigma_{n+1}}{\partial \Delta \epsilon} = \frac{\partial \Delta \sigma}{\partial \Delta \epsilon}. \quad (131)$$

Taking the stress-strain relation of the stress update algorithm (65), and deriving the latter with respect to $\Delta \epsilon$ yields

$$\frac{\partial \sigma_{n+1}}{\partial \Delta \epsilon} = -\bar{\sigma}_{n+1} \otimes \frac{\partial \omega_{n+1}}{\partial \Delta \epsilon} + (1 - \omega_{n+1}) \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon}. \quad (132)$$

Therein, the first term on the right hand side represents the change of the nominal stress with respect to the strain increment due to an increase of the damage, whereas the second term represents the change of the nominal stress due to plastic material behavior. Departing from (97), the latter is

expressed as

$$\frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon} = \frac{\partial \Delta \bar{\sigma}}{\partial \Delta \epsilon} = \tilde{\mathbb{C}} : \frac{\partial}{\partial \Delta \epsilon} (\Delta \epsilon - \Delta \epsilon'' - \Delta \epsilon^p) = \tilde{\mathbb{C}} : \left(\mathbb{I} - \frac{\partial \Delta \epsilon^p}{\partial \Delta \epsilon} \right), \quad (133)$$

with $\mathbb{I}$ denoting the fourth order unit tensor. This expression represents the standard algorithmic elasto-plastic tangent stiffness in the context of the implicit return mapping algorithm and in terms of the effective stiffness tensor $\tilde{\mathbb{C}}$. The computation of $\frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon}$ is performed directly within the return mapping algorithm. Expanding the first term on the right hand side in (132) by application of the chain rule and substitution of (133) yields

$$\begin{aligned} \bar{\sigma}_{n+1} \otimes \frac{\partial \omega_{n+1}}{\partial \Delta \epsilon} &= \bar{\sigma}_{n+1} \otimes \frac{\partial \omega_{n+1}}{\partial \hat{\alpha}_{d_{n+1}}} \frac{\partial \hat{\alpha}_{d_{n+1}}}{\partial \alpha_{d_{n+1}}} \frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \epsilon} \\ &= \bar{\sigma}_{n+1} \otimes \frac{\partial \omega_{n+1}}{\partial \hat{\alpha}_{d_{n+1}}} \frac{\partial \hat{\alpha}_{d_{n+1}}}{\partial \alpha_{d_{n+1}}} \frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \epsilon^p} : \frac{\partial \Delta \epsilon^p}{\partial \Delta \epsilon} \\ &= -\bar{\sigma}_{n+1} \otimes \frac{\partial \omega_{n+1}}{\partial \hat{\alpha}_{d_{n+1}}} \frac{\partial \hat{\alpha}_{d_{n+1}}}{\partial \alpha_{d_{n+1}}} \frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \epsilon^p} : (\tilde{\mathbb{C}}^{-1} : \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon} - \mathbb{I}). \end{aligned} \quad (134)$$

Expression $\frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \epsilon^p}$ is obtained by consistent linearization of (114), and vanishes for $\alpha_p < 1.0$. Substituting expressions (134) and (133) into (132), algorithmic tangent stiffness $\frac{\partial \sigma_{n+1}}{\partial \Delta \epsilon}$ is obtained as

$$\begin{aligned} \frac{\partial \sigma_{n+1}}{\partial \Delta \epsilon} &= -\bar{\sigma}_{n+1} \otimes \frac{\partial \omega_{n+1}}{\partial \hat{\alpha}_{d_{n+1}}} \frac{\partial \hat{\alpha}_{d_{n+1}}}{\partial \alpha_{d_{n+1}}} \frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \epsilon^p} : (\tilde{\mathbb{C}}^{-1} : \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon} - \mathbb{I}) + (1 - \omega_{n+1}) \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon} \\ &= -(1 - m) \bar{\sigma}_{n+1} \otimes \frac{\partial \omega_{n+1}}{\partial \hat{\alpha}_{d_{n+1}}} \frac{\partial \hat{\alpha}_{d_{n+1}}}{\partial \alpha_{d_{n+1}}} \frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \epsilon^p} : (\tilde{\mathbb{C}}^{-1} : \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon} - \mathbb{I}) + (1 - \omega_{n+1}) \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon}. \end{aligned} \quad (135)$$

Equation (134) already contains also the second required algorithmic tangent

$$\frac{\partial \hat{\alpha}_{d_{n+1}}}{\partial \Delta \epsilon} = -\frac{\partial \alpha_{d_{n+1}}}{\partial \Delta \epsilon^p} : (\tilde{\mathbb{C}}^{-1} : \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta \epsilon} - \mathbb{I}). \quad (136)$$

The third required algorithmic tangent $\frac{\partial \sigma_{n+1}}{\partial \bar{\alpha}_{d_{n+1}}}$ is obtained as

$$\begin{aligned} \frac{\partial \sigma_{n+1}}{\partial \bar{\alpha}_{d_{n+1}}} &= \bar{\sigma}_{n+1} \left(-\frac{\partial \omega_{n+1}}{\partial \hat{\alpha}_{d_{n+1}}} \frac{\partial \hat{\alpha}_{d_{n+1}}}{\partial \bar{\alpha}_{d_{n+1}}} \right) \\ &= -m \bar{\sigma}_{n+1} \frac{\partial \omega_{n+1}}{\partial \hat{\alpha}_{d_{n+1}}}. \end{aligned} \quad (137)$$

4.8 Summary of the Stress Update Algorithm for the Gradient-Enhanced SCDP Model

After computing the nominal stress σ_{n+1} and the algorithmic tangent stiffness matrices, the internal variables $\gamma_{1_n}, \dots, \gamma_{M_n}$ of the Dirichlet series are updated according to (76), and the updated stress σ_{n+1} , the algorithmic tangent stiffness matrices, and the updated internal variables α_{n+1} are passed back to the finite element framework. The presented stress update algorithm is summarized in Algorithm 1. For the sake of simplicity, the substepping scheme is omitted.

Algorithm 1 Stress update algorithm for the gradient-enhanced SCDP model at integration point level

```

1: function Stress Update ( $\Delta\epsilon, \bar{\alpha}_{dn+1}, t_n, t_{n+1}, \sigma_n, \alpha_{pn}, \alpha_{dn}, \omega_n, \gamma_{1n}, \dots, \gamma_{Mn}$ )
2:    $\bar{\sigma}_n \leftarrow \frac{1}{(1-\omega_n)} \sigma_n$  ▷ Compute effective stress at  $t_n$ 
3:    $\Delta\epsilon^{shr} \leftarrow \epsilon^{shr}(t_n, t_{n+1})$ 
4:    $\Delta\epsilon'' \leftarrow \Delta\epsilon''(\bar{\sigma}_n, \gamma_{1n}, \dots, \gamma_{Mn}, t_n, t_{n+1}, \Delta\epsilon^{shr})$ 
5:    $\bar{E} \leftarrow \bar{E}(\bar{\sigma}_n, \gamma_{1n}, \dots, \gamma_{Mn}, t_n, t_{n+1})$  ▷ Compute effective stiffness
6:    $\tilde{C} \leftarrow C_\nu \bar{E}$ 
7:    $\bar{\sigma}_{n+1}^{tr} \leftarrow \bar{\sigma}_n + \tilde{C} : (\Delta\epsilon - \Delta\epsilon'')$  ▷ Compute trial effective stress
8:   if  $f(\bar{\sigma}_{n+1}^{tr}, \alpha_n, t_{n+1}) < 0$  then ▷ Yield condition not violated
9:      $\bar{\sigma}_{n+1} \leftarrow \bar{\sigma}_{n+1}^{tr}$ 
10:     $\alpha_{pn+1} \leftarrow \alpha_{pn}$ 
11:     $\alpha_{dn+1} \leftarrow \alpha_{dn}$ 
12:  else ▷ Yield condition violated
13:     $\bar{\sigma}_{n+1}, \Delta\epsilon^p, \alpha_{pn+1}, \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta\epsilon} \leftarrow \text{Return Mapping}(\sigma_n, \bar{\sigma}_{n+1}^{tr}, \tilde{C}, \alpha_{pn}, t_{n+1})$ 
14:     $\alpha_{dn+1} \leftarrow \alpha_d(\alpha_{dn}, \Delta\epsilon^p)$ 
15:  end if
16:   $\hat{\alpha}_{dn+1} = m \bar{\alpha}_{dn+1} + (1 - m) \alpha_{dn+1}$  ▷ Update weighted variable
17:   $\omega_{n+1} \leftarrow \omega(\hat{\alpha}_{dn+1})$  ▷ Update damage parameter
18:   $\frac{\partial \sigma_{n+1}}{\partial \bar{\alpha}_{dn+1}} \leftarrow \frac{\partial \sigma_{n+1}}{\partial \bar{\alpha}_{dn+1}}(\bar{\sigma}_{n+1}, \hat{\alpha}_{dn+1}, \bar{\alpha}_{dn+1})$  ▷ Algorithmic tangents
19:   $\frac{\partial \alpha_{dn+1}}{\partial \Delta\epsilon} \leftarrow \frac{\partial \alpha_{dn+1}}{\partial \Delta\epsilon}(\Delta\epsilon^p, \alpha_{dn+1})$ 
20:   $\frac{\partial \sigma_{n+1}}{\partial \Delta\epsilon} \leftarrow \frac{\partial \sigma_{n+1}}{\partial \Delta\epsilon}(\bar{\sigma}_{n+1}, \Delta\epsilon^p, \hat{\alpha}_{dn+1}, \alpha_{dn+1}, \omega_{n+1}, \tilde{C}, \frac{\partial \bar{\sigma}_{n+1}}{\partial \Delta\epsilon})$ 
21:  for  $i = 1, \dots, M$  do
22:     $\gamma_{i,n+1} \leftarrow \gamma_i(\gamma_{in}, \Delta\bar{\sigma}, t_n, t_{n+1})$  ▷ Update viscoelastic state variables
23:  end for
24:   $\sigma_{n+1} \leftarrow (1 - \omega_{n+1}) \bar{\sigma}_{n+1}$  ▷ Update nominal stress
25:  return  $\sigma_{n+1}, \alpha_{pn+1}, \alpha_{dn+1}, \omega_{n+1}, \gamma_{1n+1}, \dots, \gamma_{Mn+1}, \frac{\partial \sigma_{n+1}}{\partial \Delta\epsilon}, \frac{\partial \sigma_{n+1}}{\partial \bar{\alpha}_{dn+1}}, \frac{\partial \alpha_{dn+1}}{\partial \Delta\epsilon}$ 
26: end function

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5 Experimental Program on Young Shotcrete and Calibration of the SCDP model

For determining the material properties of young shotcrete and for calibrating the SCDP model, an experimental program on a modern wet-mix shotcrete composition was conducted, as reported in (Neuner, Cordes, Drexel, and Hofstetter, 2017). The program focused on

- the evolution of the Young’s modulus,
- the evolution of the uniaxial compressive strength,
- the creep behavior under sealed conditions,
- and the shrinkage behavior under sealed conditions.

Additional tests for determining the specific mode I fracture energy by means of wedge splitting tests were conducted and reported in (Dummer, 2019).

Based on the obtained results, the SCDP model was calibrated. In the following, the experimental program and the respective predictions by the calibrated SCDP model are summarized. For a detailed explanation of the calibration procedure, the reader is referred to (Neuner, Cordes, Drexel, and Hofstetter, 2017; Neuner, Schreter, and Hofstetter, 2018).

The specimens for the investigated shotcrete composition with the manufacturer designation *Sp C25/30/(56)/ÜK3/J2/XC4/XF3/GK8* according to the Austrian guidelines for shotcrete (ÖVBB, 2009) were sampled directly at the construction site of the Brenner Base Tunnel. In contrast to normal concrete, shotcrete is compacted by means of the impact on the surface. Hence, the spraying process considerably influences the resulting material properties. Accordingly, shotcrete specimens must be sampled directly at the construction site using the robotic spraying equipment in order to obtain samples of the same quality as used during tunnel construction. For the present composition, *Sp C25/30/(56)* characterizes the shotcrete strength class, *ÜK3* is the monitoring class, *J2* is the early age strength class, *XC4* and *XF3* are the exposure classes, and *GK8* denotes the maximum aggregate size of 8 mm. Details of the composition are summarized in Table 1.

Property	Quantity	Unit
Water content	203	kg/m ³
Cement content <i>CEM I 52.5N(sb)</i>	380	kg/m ³
Limestone sand (0/4)	1031	kg/m ³
Crushed limestone aggregates (4/8)	694	kg/m ³
Additive <i>Fluasit</i>	40	kg/m ³
Accelerator <i>Mapequick 043 FFG</i>	7.5 – 8.5	%

Table 1: Wet-mix shotcrete composition investigated in the experimental program.

With exception of the wedge splitting tests, all tests were performed on cylindrical specimens with the diameter of 100 mm and height of 200 mm (Young’s modulus and uniaxial compressive strength) or 300 mm (creep and shrinkage tests). Wedge splitting tests for determining the specific mode I fracture energy were performed on cubic specimens with the side length of 150 mm.

The specimens were obtained at the construction site by two different techniques, i.e., (i) sampling from spray boxes with dimensions of 450 mm × 300 mm × 250 mm, which is the commonly employed

type of specimens for determining material properties of shotcrete, and (ii) directly from molds, denoted as *sprayed* specimens.

For the specimens sampled from spray boxes, drill cores (cylindrical specimens) or sawn cubes were taken from spray boxes at material ages of 24 hours or later.

Sprayed specimens were employed for tests on shotcrete younger than 24 hours, since specimens cannot be sampled from spray boxes at such a young age. For the sprayed specimens, slotted PVC tubes mounted on wooden rigs (cylindrical specimens) or wooden molds (cubic specimens) were used (Figure 3). The quality of the sprayed specimens was assessed by visual inspection of

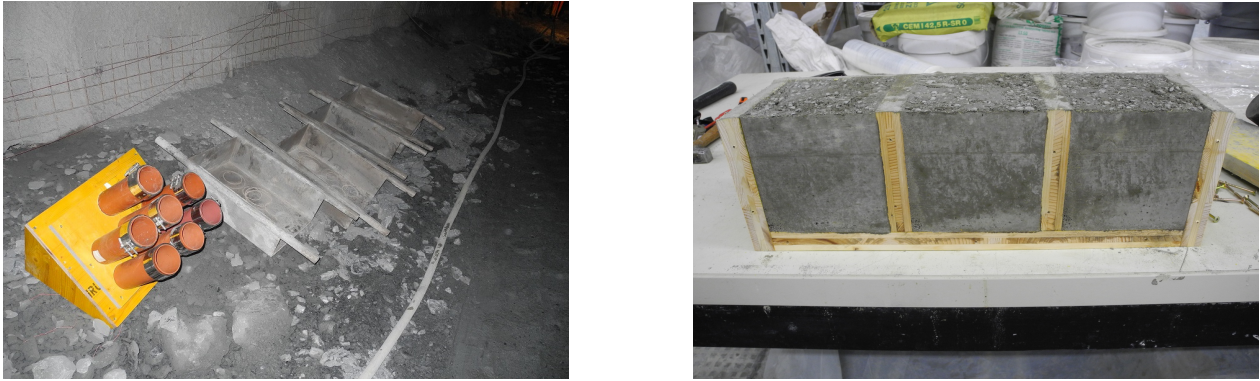


Figure 3: Specimen types: Tubular molds for cylindrical specimens and spray boxes placed at the construction site immediately before spraying (left). Cubic specimens for wedge splitting tests in a wooden mold with removed front panel during unmolding (right).

the surface, and faulty specimens due to shotcrete rebound and air pockets were sorted out. For the present shotcrete composition, sprayed specimens were unmolded safely already 5 hours after spraying (Figure 4).



Figure 4: Sprayed shotcrete specimen during an uniaxial compression test, 6 hours after spraying.

Most experiments require smooth top and bottom surfaces of the specimens, which are parallel to each other. Since cutting and grinding is impossible at young shotcrete ages, compensation layers consisting of a fast-curing epoxy resin were applied onto the top surfaces of the specimens. No treatment of the bottom surfaces was necessary, as those surfaces were already smooth due to the employed molds or spray boxes.

5.1 Evolution of the Young's Modulus and the Uniaxial Compressive Strength

The evolution of the Young's modulus was determined between the material age of 24 hours and 28 days, and the uniaxial compressive strength was tested between 6 hours to 28 days. Both tests were performed according to the guideline (ONR 23303, 2010). Figure 5 shows the experimentally determined evolution of the Young's modulus, and the predicted evolution by means of the calibrated SCDP model. Similarly, Figure 6 shows the experimental results on the evolution of uniaxial compressive strength, and the prediction by the SCDP model. It can be seen clearly that the calibrated SCDP model represents the experimentally observed evolutions very well.

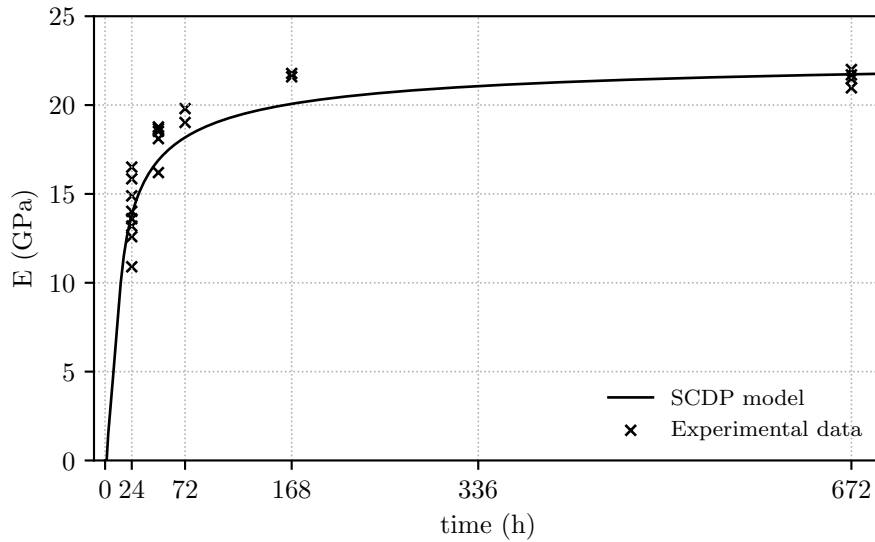


Figure 5: Time-dependent evolution of the Young's modulus between the material age of 24 h and 28 days, and prediction by means of the calibrated SCDP model.

5.2 Shrinkage and Creep Behavior of Young Shotcrete

Shrinkage and creep tests on sealed specimens were started at three different ages of the material, i.e., 8 hours, 24 hours and 27 hours after spraying, and they lasted for 56 days each. During the shrinkage and creep tests, the time-dependent strain at the center of the cylindrical specimens was determined by measuring the time-dependent length change over a distance of 200 mm, using three displacement transducers which were arranged at equal distances along the perimeter of each specimen. All tests were conducted in a climate chamber with an ambient temperature of 20 ± 1 °C and a relative humidity of 65 ± 5 %. Creep tests were conducted using a hydraulic creep test bench (Figure 8), and the magnitude of the applied compressive load was chosen to ensure linear creep behavior only.

Shrinkage tests were conducted simultaneously to the creep tests on sealed, load-free specimens. From the recorded time-dependent length changes measured on the three specimens, the respective

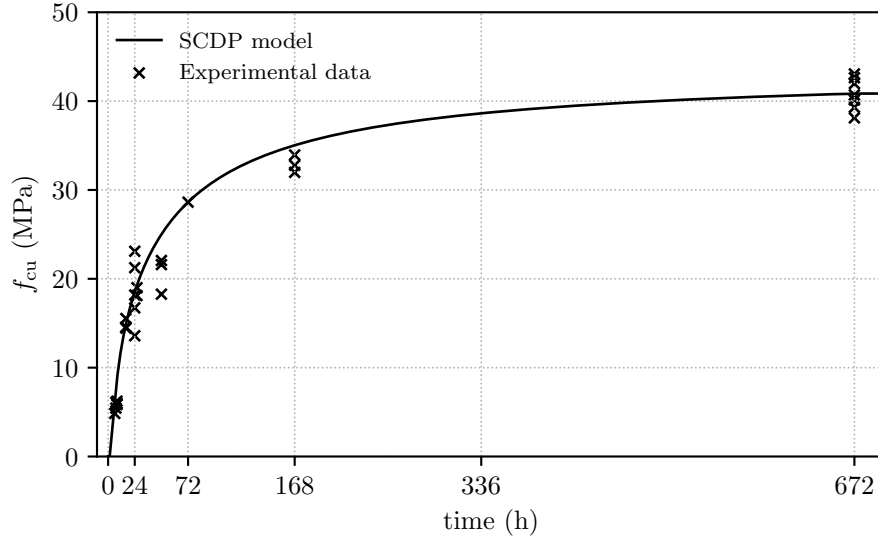


Figure 6: Time-dependent evolution of the uniaxial compressive strength between the material age of 6 h and 28 days, and prediction by means of the calibrated SCDP model.

time-dependent total strain is computed. It consists of the autogenous shrinkage strains and, at a very early age of the material, the thermally induced strain due to hydration heat. The evolution of the determined time-dependent total strain is shown for the three specimens together with the prediction based on the calibrated SCDP model in Figure 7. Note that the SCDP model does not represent the thermally induced strain. It can be seen that the calibrated SCDP model represents the experimentally observed material behavior very well.

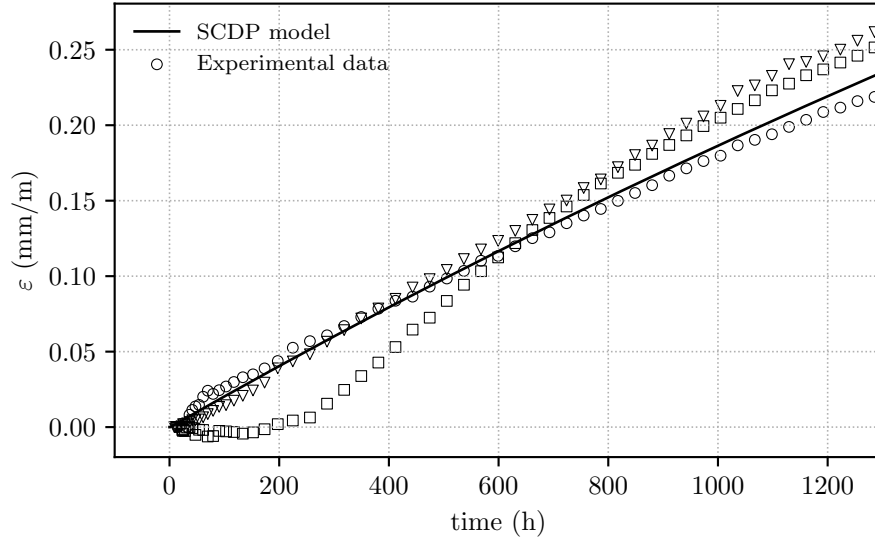


Figure 7: Time-dependent evolution of the total strain ε determined on sealed, load-free specimens simultaneously to the creep tests: Experimental results and predictions by means of the calibrated SCDP model.

The creep tests were started at shotcrete ages of 8 hours, 24 hours and 27 hours by loading the specimens with compressive stresses of 1.9 MPa, 3.9 MPa and 3.7 MPa, respectively. The creep tests started at 8 hours, 24 hours and 27 hours were loaded with compressive stresses of 1.9 MPa, 3.9 MPa and 3.7 MPa. Similar to the shrinkage tests, the time-dependent total strains are determined from the measured time-dependent length changes recorded during the creep tests. It consists of the basic

creep strain due to the applied load, the autogenous shrinkage strain, and the thermally induced strain due to hydration heat. The determined time-dependent total strain is shown for the three tests in Figure 9, together with the numerical predictions by means of the calibrated SCDP model. Similar to the shrinkage tests, the experimentally observed behavior is represented very well by the calibrated SCDP model.



Figure 8: Sealed specimen during a creep test in the hydraulic creep test bench with three displacement transducers arranged at equal distances along the perimeter of the specimen for determining the time-dependent evolution of the longitudinal strain.

5.3 Specific Mode I Fracture Energy of Shotcrete

For calibrating softening material behavior in constitutive models for quasi-brittle materials, the specific mode I fracture energy is a commonly employed, important material property. It is defined as the work per unit area required for separating a specimen into two parts in tension. For determining this material property, wedge splitting tests were conducted on cubic shotcrete specimens at a material age of 8 hours, 1 day, 7 days and 28 days. To this end, cubes with the side length of 150 mm were tested using a wedge splitting device and a hydraulic test bench. The test setup is shown in Figure 10.

For mounting the wedge splitting device, metal jaws were bonded on top of the cubes. In addition, small starter notches of 15 mm depth were sawn into the cubic specimens. During a test, the crack mouth opening displacement (CMOD) was measured directly at the shotcrete specimen using several displacement transducers. For determining the load-CMOD curves, the machine piston reaction force was recorded, and from the wedge geometry and the machine force, the crack mouth opening load was computed. For normalizing the curves, the fracture areas were measured precisely after

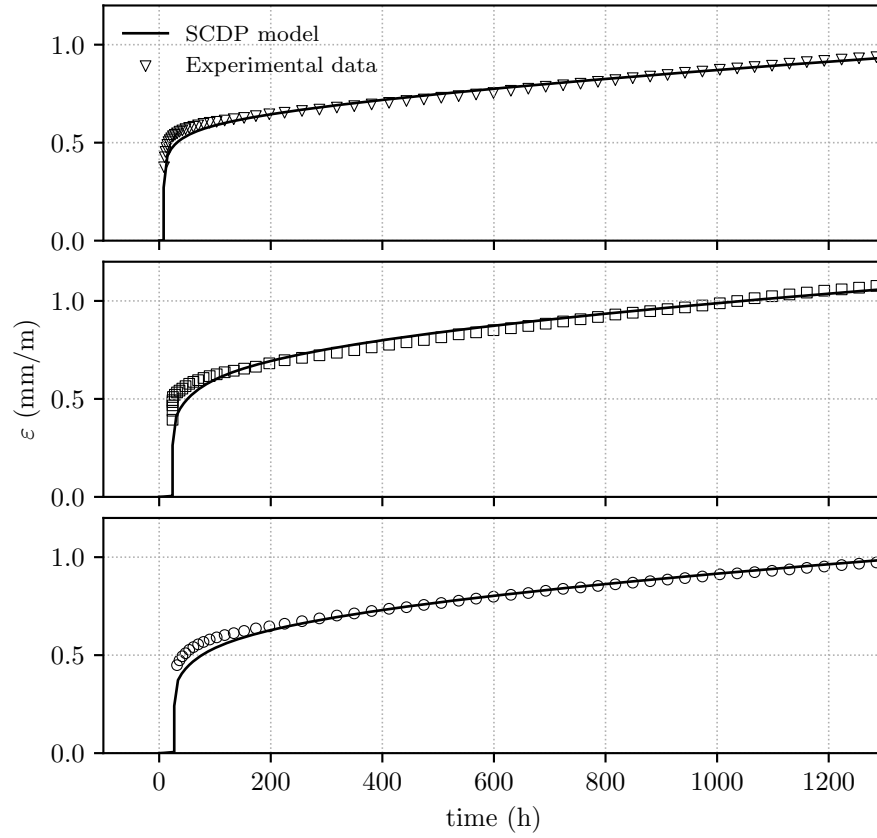


Figure 9: Evolution of the total strain ε during creep tests on sealed specimens loaded with sustained constant compressive stresses at material ages of 8 hours with 1.9 MPa (top row), 24 hours with 3.9 MPa (center row) and 27 hours with 3.7 MPa (bottom row): Experimental results and predictions by means of the calibrated SCDP model.

splitting of the specimens. From the obtained normalized load-CMOD curves, the specific mode I fracture energy is computed by means of numerical integration. The results are listed in Table 2.

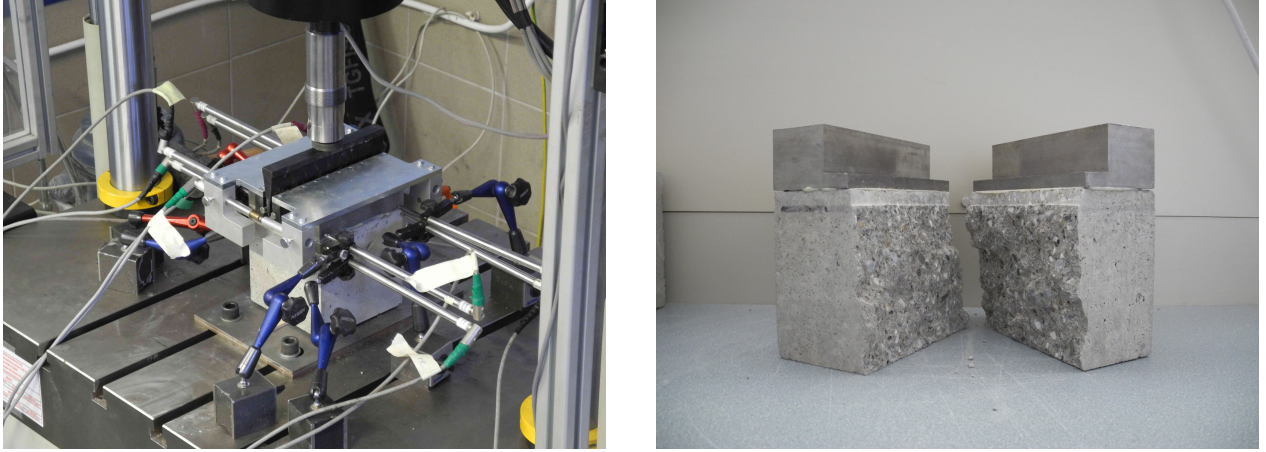


Figure 10: Wedge splitting tests: Cubic shotcrete specimen during a wedge splitting test together with the mounted wedge splitting device in a hydraulic test bench, recording the CMOD by means of displacement transducers (left). Cubic specimen after testing, displaying the fracture area, the starter notch and the bonded steel jaws for mounting the wedge splitting device (right).

Table 2: Experimental results on the specific mode I fracture energy

Material age (h)	G_{fl} (N/m)	$G_{fl,mean} \pm SD$ (N/m)
8 h	74.99	-
24 h	128.94	-
168 h	126.63	122.18 ± 4.50
	116.02	
	123.90	
672 h	130.34	114.30 ± 11.50
	103.94	
	108.63	

Modeling softening material behavior in the gradient-enhanced SCDP model requires calibration of the softening modulus ϵ_f in Eq. (40) and the nonlocal radius l in Eq. (41). Clearly, the question arises whether length parameter l is considered as a material parameter, related to characteristic internal micromechanical length scales of the material (Bažant, 1991; Bažant and Pijaudier-Cabot, 1989), or if it is considered as a model parameter in the spirit of the gradient-enhanced formulation acting as a purely regularizing framework.

In the present contribution, the latter approach is adopted for practical reasons: The size of l imposes a restriction on the minimal resolution of the finite element mesh, since the nonlocal interaction must be represented sufficiently accurately by the numerical discretization. However, for numerical simulations, the employed finite element size is usually related to the dimensions of the investigated structure in order to maintain feasible computation times. Accordingly, for the subsequently presented numerical study, the size of l is chosen a priori sufficiently large in accordance with employed finite element mesh, and ϵ_f remains as the only parameter to be adjusted based on

the experimental results. For calibrating ϵ_f for a chosen value of l , a simple direct uniaxial tension test is simulated, and from the numerically obtained load-displacement curve, the specific mode I fracture energy is computed by means of numerical integration. Parameter ϵ_f is adjusted by a back analysis to fit the computed specific mode I fracture energy.

Since the current version of the SCDP model does not consider the evolution of the softening modulus, it is fitted to the experimental value of the specific mode I fracture energy measured at the age of 1 day. Based on the respective value of $G_{\text{ff}}^{(1)} = 128.94 \text{ N/m}$ in Table 2, for the chosen value of $l = 75 \text{ mm}$ and a nonlocal weighting parameter $m = 1.01$, the softening modulus is determined as $\epsilon_f = 3.7 \times 10^{-4}$.

5.4 Summary of the Material Parameters of the SCDP model

The material parameters for the subsequently presented numerical study are summarized in Table 3. Parameters which have not been calibrated based on the presented experimental program have been adopted from (Neuner, Schreter, and Hofstetter, 2018) and (Schädlich, Schweiger, Marcher, and Saurer, 2014b), based on values from the literature. The latter are the Poisson ratio ν , the ratios $f_{\text{cy}}/f_{\text{cu}}$, $f_{\text{tu}}/f_{\text{cu}}$ and $f_{\text{cb}}/f_{\text{cu}}$, and the plastic strain at peak strength in uniaxial compression at material ages 1 hour, 8 hours and 24 hours, respectively.

Table 3: Material parameters for the SCDP model representing the investigated wet-mix shotcrete composition.

$E^{(1)}$ (MPa)	$E^{(28)}$ (MPa)	q_4 (MPa $^{-1}$)	ν (—)	e^{flow} (—)	$f_{\text{cu}}^{(1)}$ (MPa)	$f_{\text{cu}}^{(28)}$ (MPa)	$f_{\text{cy}}/f_{\text{cu}}$ (—)
13 943	21 537	34×10^{-6}	0.21	1.22	18.56	40.85	0.1
$f_{\text{cb}}/f_{\text{cu}}$ (—)	$f_{\text{tu}}/f_{\text{cu}}$ (—)	$\epsilon_{\infty}^{\text{shr}}$ (—)	t_{50}^{shr} (h)	$\epsilon_{\text{cpu}}^{\text{p}(1)}$ (—)	$\epsilon_{\text{cpu}}^{\text{p}(8)}$ (—)	$\epsilon_{\text{cpu}}^{\text{p}(24)}$ (—)	ϵ_f (—)
1.16	0.2	−0.0019	8645	−0.03	−0.002	−0.002	3.7×10^{-4}

6 Application of the SCDP Model in a Numerical Study

The influence of the shotcrete shell on the predicted mechanical behavior of a tunnel structure and the performance of the SCDP model are demonstrated by means of a 2D finite element study of deep tunnel advance. Although 2D finite element models represent a simplification of the actual three-dimensional mechanical problem, they allow for an estimate of the arising deformations and stresses in the shotcrete shell, as shown in (Neuner, Schreter, Gamnitzer, and Hofstetter, 2020) based on a comparative study of 2D and 3D numerical simulations of deep tunnel advance.

The tunnel stretch under consideration is derived from an emergency tunnel, which is part of the Brenner Base Tunnel project. It was discussed previously in (Neuner, Dummer, Cordes, Schreter, and Hofstetter, 2020), employing the SCDP model. Therein, the load bearing capacity of the constructed single permanent shotcrete shell was assessed by means of a back analysis, calibrated by geodetic measurement results.

The tunnel stretch was built to upgrade an existing bypass tunnel, constructed between 1989 and 1993, in order to meet modern safety standards. The excavation profile with a cross sectional area of 47 m^2 is shown together with a longitudinal view in Figure 11. The stretch of 9.1 km length is characterized by an overburden of approximately 350 m, It is located parallel to the bypass tunnel at a distance of 30 m, and connections to the latter are established by means of cross passages every 333 m.

The surrounding rock mass is classified as Innsbruck quartz phyllite with intercalations of limestone, dolomitic marble and green schists. In the predominant phyllite zones, which are passed through by the tunnel, the schists break up easily into thin layers which consist of layered silicates and quartz. Nevertheless, it is a dense rock due to its high proportion of mica, which results in a low water permeability and minor water inflows into the tunnel. For the design of the emergency tunnel, it was assumed that due to drainage of the existing bypass tunnel water ingress will be negligible. In case of dripping water ingress, the water was drawn off before shotcrete was applied. Chemical analyses of the water samples indicated very low sulphate ion concentrations, for instance, a sulphate content of 41.0 mg/l was measured. Due to a calcite saturation of the mountain water, deterioration of the shotcrete shell is assumed negligible.

Since the requirements for emergency tunnels are less demanding than those for railway tunnels, single permanent shotcrete shells provide an economic alternative to commonly employed inner shells made from cast-in-place concrete. The durability requirements for the emergency tunnel are met using an increased shotcrete cover of 55 mm, the monitoring class 3 ($\ddot{U}K3$) and by limiting the maximum crack opening to 0.3 mm in the geologically homogeneous sections. For meeting the safety requirements, in the design a single permanent shotcrete shell with a thickness of 250 mm and two layers of reinforcement was specified. In fault zones and sections exhibiting larger convergences, space for a potential secondary shell was reserved within the cross-section.

For determining the load-bearing capacity of the shotcrete shell by means of a back analysis, convergence measurements were conducted, using 5 monitoring points arranged along the cross-section. The initial measurements were carried out immediately after application of the first layer of shotcrete. In addition, 3D laser scanning measurements at two construction stages were performed for monitoring the excavation profile and the actual thickness of the shotcrete shell, measuring up to 450 mm at the crown and 500 mm at the side walls. However, the structural behavior of a shotcrete shell with an irregular distribution of the thickness is strongly influenced by the regions with a lower

thickness. Hence, in the subsequently presented numerical study a uniform thickness of 300 mm is assumed for the shotcrete shell.

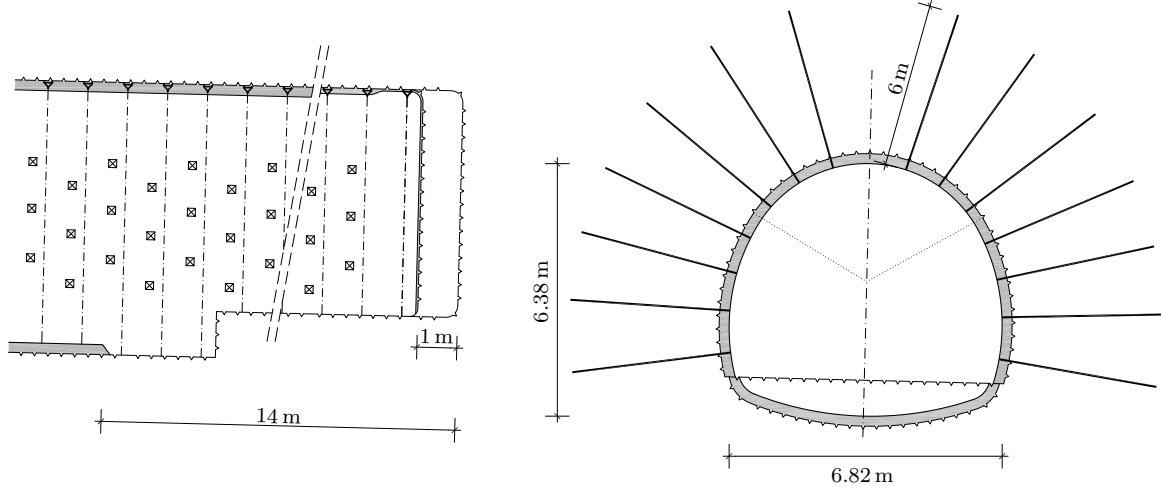


Figure 11: Longitudinal view and cross section of the investigated tunnel stretch, together with the dimensions and the employed tunnel support.

6.1 Finite Element Simulation of Tunnel Advance

The numerical study is based on the 2D finite element model illustrated in Figure 12. A domain of $200\text{ m} \times 200\text{ m}$ is considered, assuming plane strain conditions. At the construction site, the top heading and the bench were excavated simultaneously, whereas longer stretches of the invert were excavated a few days later. However, as shown in (Neuner, Dummer, Cordes, Schreter, and Hofstetter, 2020), the excavation of the invert has virtually no influence on the structural response. For this reason, and for the sake of simplicity, in the following study full face excavation is considered, i.e., the top heading, the bench and the invert are excavated simultaneously.

Regarding the boundary and initial conditions, displacements perpendicular to the model boundary are fixed, and an initial geostatic stress state, characterized by the coefficient of pressure at rest K_0 , is assumed. From the geometry, the initial conditions and the boundary conditions follows symmetry with respect to the vertical axis passing through the tunnel crown, which is exploited in the numerical model.

The material behavior of the shotcrete is considered by means of the SCDP model, employing the material parameters listed in Table 3. The shotcrete shell is discretized by fully integrated 4-node quadrilateral elements, using 16 elements across the thickness. In addition to the shotcrete shell, 14 rock bolts of 6 m length with a cross sectional area of 654.5 mm^2 each are modeled using linear truss elements. For those rock bolts, linear-elastic perfectly-plastic material behavior is assumed, with the Young's modulus of 210 000 MPa and the yield stress of 225 MPa. Rock mass is represented by means of the linear-elastic perfectly-plastic Hoek-Brown model, using fully integrated 4-node quadrilateral elements. The employed finite element mesh is shown in Figure 13.

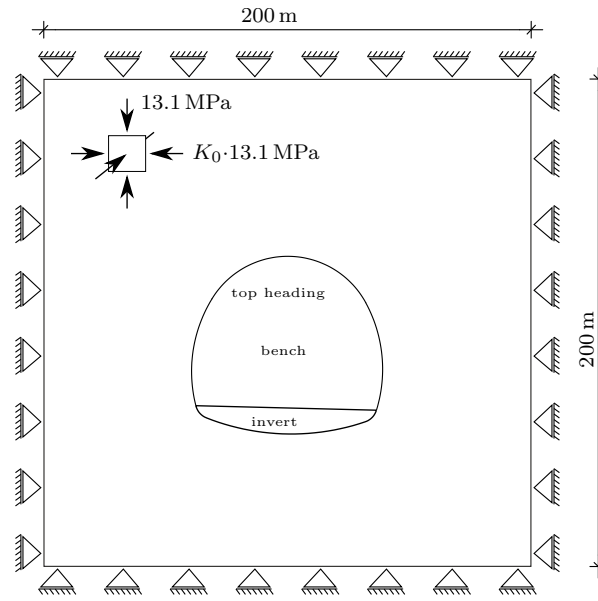


Figure 12: 2D plane strain model of the emergency tunnel with a single permanent shotcrete shell, showing boundary and initial conditions, and the excavation profile: top heading, bench and invert.

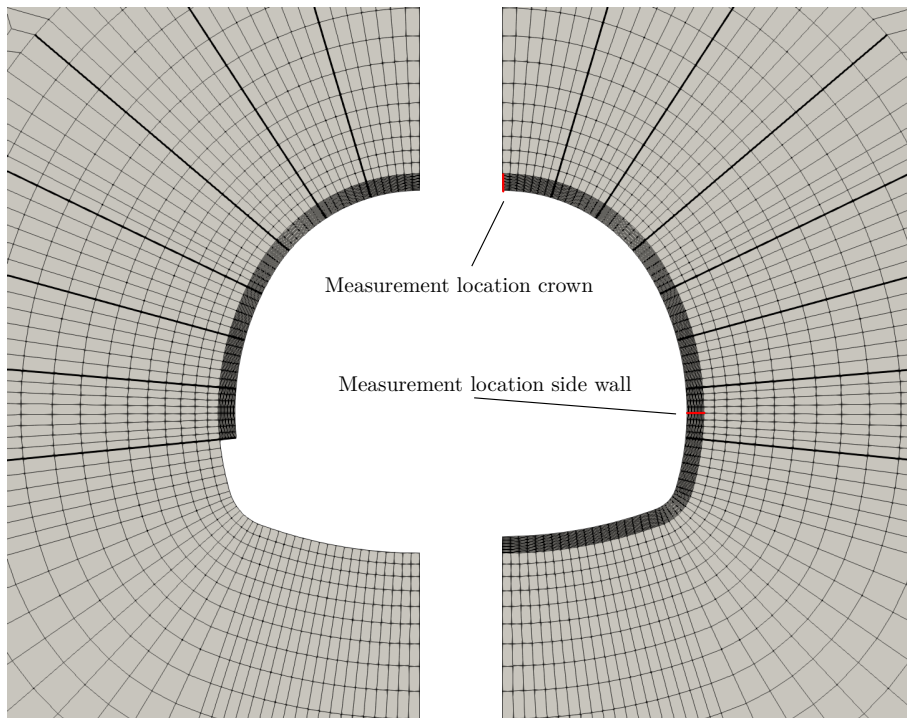


Figure 13: Finite element mesh after placement of the upper part of the shotcrete shell (left), and after placement of the lower part of the shell (right).

6.2 Time-dependent construction process

For assessing the structural response of the shotcrete shell, the time-dependent construction process is represented in the numerical model. As mentioned, full face excavation is considered, i.e., the top heading, the bench and the invert are excavated simultaneously. However, for the shotcrete shell, the actual two-stage construction process is taken into account: Immediately after excavation of the investigated cross section, the upper part of the shotcrete shell and the rock bolts are installed, whereas the lower part of the shotcrete shell is placed 3.5 d (84 h) later.

For simulating the time-dependent excavation process and its influence on the tunnel structure in a 2D finite element model, the stress release method is employed. It consists of a sequence of 5 steps:

- (i) Initial equilibrium is established by applying the geostatic stress state, and the equilibrium balancing internal forces are applied along the excavation boundary.
- (ii) The equilibrium balancing internal forces are released partially according to an initial stress release ratio λ_0 .
- (iii) The upper part of shotcrete shell and the rock bolts are installed in a stress-free state.
- (iv) The remaining equilibrium balancing internal forces are released to zero within a time period of 108 h according to a step-wise stress release scheme.
- (v) Simultaneously to step (iv), 84 h after placement of the upper part of the shotcrete shell, the lower part of the shotcrete shell is installed in a stress free state, completing the shotcrete shell.

The assumed step-wise time-dependent stress release function in step (iv) follows a combined linear-exponential function, which was introduced in (Neuner, Dummer, Cordes, Schreter, and Hofstetter, 2020), and it is illustrated in Figure 14. Alternative schemes were employed for instance in (Neuner, Schreter, Unteregger, and Hofstetter, 2017) using a parabolic function, or in (Neuner, Schreter, Gamnitzer, and Hofstetter, 2020) assuming a purely exponential decline. Taking the round length of 1 m for each excavation and securing cycle (lasting 6 h each), and assuming a negligible influence of the tunneling process on the considered cross section once the tunnel face has progressed by a distance 18 m, the equilibrium balancing internal pressure is fully released after 18 excavation cycles, i.e., 108 h after installation of the upper part of the shotcrete shell.

6.3 Linear-Elastic Perfectly-Plastic Hoek-Brown Model for the Rock Mass

The rock mass is considered as an isotropic continuum employing a linear-elastic perfectly-plastic material model. For delimiting the elastic domain, the Hoek-Brown yield function (Hoek and Brown, 1980) in the smoothened formulation by Menetrey and Willam (1995) given as

$$f(\sigma_m, \rho, \theta) = \left(\sqrt{\frac{3}{2}} \frac{\rho}{f_{cu}} \right)^2 + \frac{1}{f_{cu}} m_0 \frac{m_b}{m_i} \left(\sigma_m + \rho \frac{r(\theta, e)}{\sqrt{6}} \right) - s, \quad (138)$$

is used, formulated in the Haigh-Westergaard coordinates of the stress tensor, i.e., σ_m , ρ and θ . f_{cu} represents the uniaxial compressive strength, m_0 is the friction parameter, $r(\theta, e)$ is the Willam-Warnke function (Willam and Warnke, 1975) with the eccentricity parameter e chosen as 0.51. In

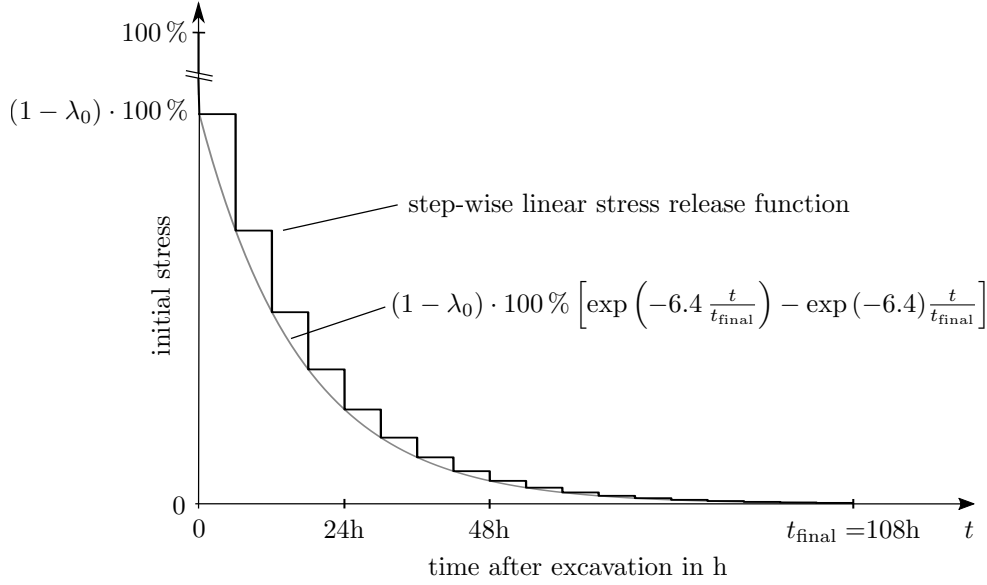


Figure 14: Assumed time-dependent stress release layout after installation of the upper part of the shotcrete shell in the 2D finite element model.

this way, the strength predicted by the original Hoek-Brown criterion is approximately matched on both the tensile and compressive meridian of the yield criterion. Parameters m_b/m_i and s are two reduction factors for representing the influence of discontinuities in rock mass, depending on the Geological Strength Index GSI and the disturbance factor D . The latter are explained in detail in (Hoek, Carranza-Torres, and Corkum, 2002). The Geological Strength Index GSI considers the prevailing geological conditions depending on the structure and surface conditions of the rock mass, as described in (Hoek and Brown, 1997). The disturbance factor D takes into account the disturbance of the in situ stress in the surrounding rock mass due to the excavation process. A tabular guideline is provided in (Hoek, Carranza-Torres, and Corkum, 2002) for estimating the latter. For excellent quality controlled blasting or mechanized tunneling a value of $D = 0$ is recommended. In addition to the parameters m_b/m_i and s , the influence of discontinuities on the elastic stiffness is considered by an empirical relation for reducing the Young's modulus of rock mass compared to intact rock, denoted as E_{rm}/E_i . To this end, a slightly modified form of the empirical relation by (Hoek and Diederichs, 2006) is used.

For describing the evolution of the plastic strain, a non-associated flow rule

$$\dot{\epsilon}^p = \frac{\partial g(\sigma_m, \rho, \theta)}{\partial \sigma} \quad (139)$$

based on a Mohr-Coulomb type plastic potential function (Menetrey and Willam, 1995)

$$g(\sigma_m, \rho, \theta) = \sqrt{(\epsilon \tan \psi)^2 + \left(r(\theta, e_g) \frac{3 - \sin \psi}{4 \cos \psi} \rho \right)^2} + \sigma_m \tan \psi \quad (140)$$

is employed, with ψ denoting the angle of dilation and e_g denoting the deviatoric eccentricity, defined as

$$e_g = (3 - \sin \psi) / (3 + \sin \psi) \quad (141)$$

for describing the shape of g in deviatoric planes. The meridional eccentricity ϵ (chosen as 0.1)

defines the fillet of the vertex of the plastic potential function for predominantly hydrostatic stress states. For improving the numerical stability in the plastic regime, a viscoplastic extension of the rock model is considered, using a Duvaut-Lions formulation (Simo, Kennedy, and Govindjee, 1988). The relaxation time is chosen as 1×10^{-3} s, acting as a numerical stabilization scheme with negligible influence on the structural response.

The parameters for the Hoek-Brown model for representing Innsbruck quartz phyllite are taken from (Schreter, Neuner, Unteregger, and Hofstetter, 2018). They were calibrated based on a series of triaxial compression tests on small scale rock specimens. The resulting material parameters are listed in Table 4. Figure 15 shows the failure envelope in the meridional plane, together with experimental results from triaxial compression tests.

Table 4: Material parameters for the Hoek-Brown model.

E (MPa)	ν (—)	f_{cu} (MPa)	m_0 (—)	ψ (°)	e (—)
56 670	0.21	42	12	11.6	0.51

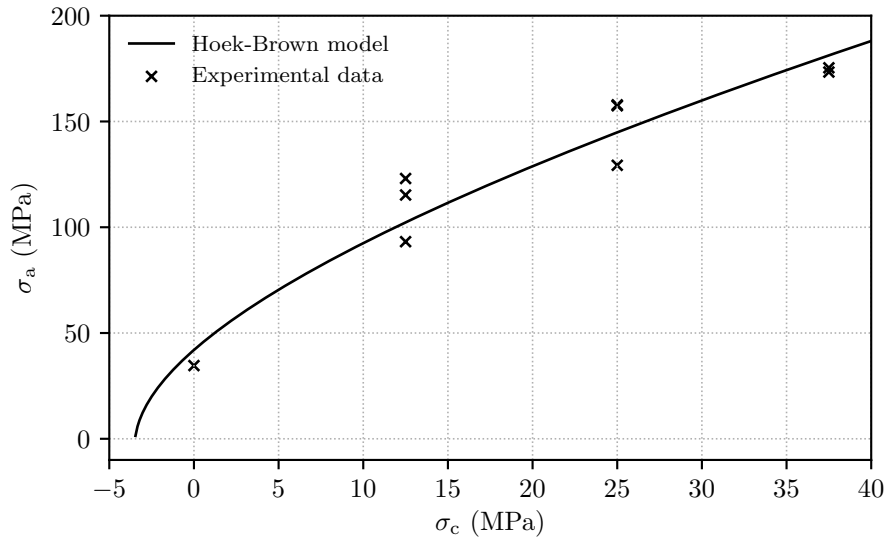


Figure 15: Predicted material strength by means of the calibrated Hoek-Brown model in the meridional plane and experimental results from triaxial compression tests, characterized by the axial stress σ_a and the confining pressure σ_c .

6.4 Calibration of the Finite Element Model

The finite element model is calibrated to fit the time-dependent displacements measured geodetically at the construction site. In detail, the model is adjusted to fit (i) the vertical crown displacement and (ii) the horizontal side wall displacement. Both locations are indicated in Figure 13.

To this end, the initial geostatic stress state, the initial stress release factor λ_0 (Figure 14), and the quality of the rock mass are calibrated, the latter by means of the Geological Strength Index (GSI) and the disturbance factor D of the Hoek-Brown model. Good agreement with the geostatic measurement records is obtained for the vertical component of the initial geostatic stress of 13.1 MPa with a lateral stress coefficient K_0 of 0.9, the initial stress release factor $\lambda_0 = 80\%$, and the GSI of

25. The disturbance factor $D = 0.5$ is assumed for the rock mass up to the depth of 6 m around the excavation boundary.

In addition, for improving the ratio between the predicted vertical crown displacement and the horizontal side wall displacement, deviating from the material parameters in Table 4, slightly reduced values of the Poisson ratio ($\nu = 0.14$) and the dilation angle ($\psi = 8.6^\circ$) are assumed for the rock mass.

In the following, 4 case studies are discussed, investigating various aspects of the predicted influence of the shotcrete shell on the mechanical response using the SCDP model:

- In **case study I**, the predicted mechanical response of the calibrated finite element model is discussed.
- In **case study II**, the influence of the construction procedure of the shotcrete shell on the mechanical response is demonstrated.
- In **case study III**, the influence of nonlinear creep behavior of shotcrete compared to linear creep behavior is discussed.
- In **case study IV**, it is shown that a potential collapse of the tunnel structure can be captured.

6.5 Case study I: Predicted Time-dependent Displacements and Utilization of the Shotcrete Shell

In case study I, the mechanical response of the calibrated finite element model is discussed. Figure 16 shows the evolution of the vertical crown displacement and the horizontal side wall displacement, obtained from the calibrated model, together with the mean value and the standard deviation of the respective geodetic measurement records. The range of the geodetically observed displacements is represented reasonably well by the finite element model: While the vertical crown displacement is slightly underestimated, a good agreement is obtained for the horizontal side wall displacement. It can be seen that after the installation of the lower part of the shotcrete shell in the finite element model 84 h after installation of the upper part, virtually no further displacements are predicted. This is slightly deviating from the geodetic observations, which indicate an increase of deformations for approximately 120 h after excavation of the cross section. The influence of time-dependent material behavior can be seen as the displacements slightly increase during the idle periods between the first excavation steps due to creep and shrinkage of shotcrete.

The principal stress components in the shotcrete shell approximately coincide with the tangential stress, the radial stress and the longitudinal stress. As shown in (Neuner, Schreter, Unteregger, and Hofstetter, 2017) and (Neuner, Schreter, Gamnitzer, and Hofstetter, 2020), the magnitude of the tangential stress is considerably higher compared to the one of the radial stress and the longitudinal stress, and hence, only the tangential stress is considered in the following.

The stress state in the shotcrete shell is assessed by the tangential stress predicted at the two measurement points located at the crown and the side wall. Figure 17 depicts the evolution of the tangential stress at the rock-shotcrete interface and the free surface at the respective points. Considerably higher compressive stresses are predicted at the crown compared to the side wall. Furthermore, it is apparent that the tangential stress is not distributed uniformly across the thickness of the shotcrete shell, i.e., distinct bending moments are prevailing.

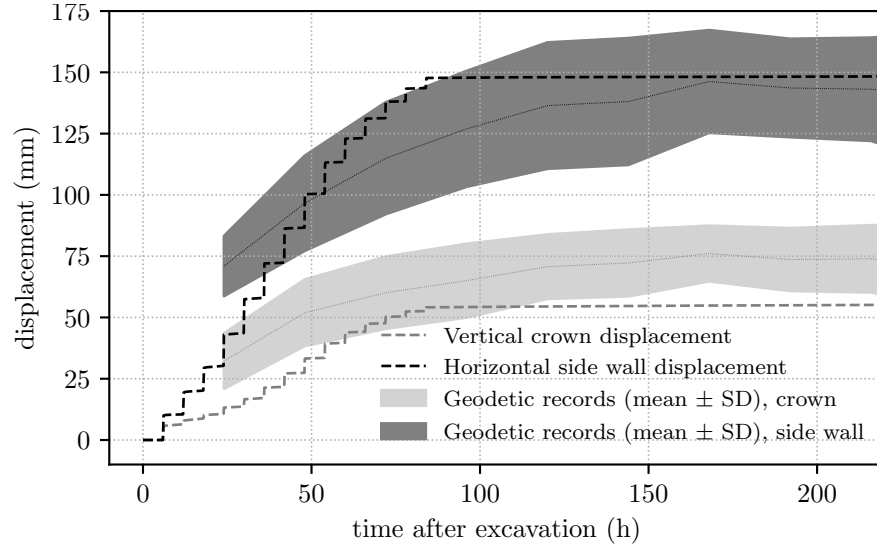


Figure 16: Predicted evolution of the vertical crown displacement and the horizontal side wall displacement by means of the calibrated finite element model, together with the range of the respective geodetically measured displacements.

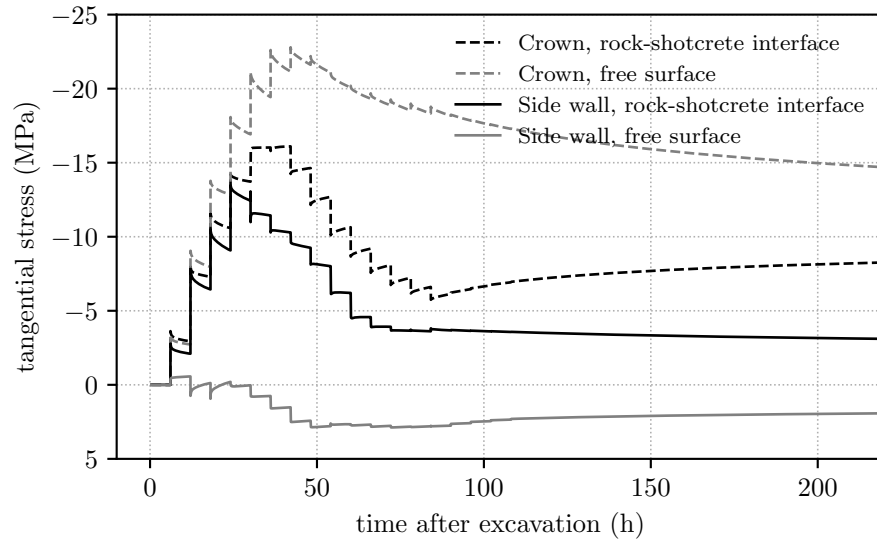


Figure 17: Evolution of the tangential stress, predicted at the rock-shotcrete interface and the free surface of the crown (dashed curves) and the side wall (solid curves), respectively.

Two interesting further aspects can be identified: First, for the crown a higher compressive stress is predicted at the free surface, whereas for the side wall the higher compressive stress is predicted at the rock-shotcrete interface, together with a tensile tangential stress at the free surface. Second, compressive stress components in Figure 17 are decreasing after the 5th excavation step. This is due to stress redistribution in the surrounding rock mass, resulting from inelastic material behavior.

For the shotcrete shell, the strain-like hardening variable α_p is used as an indicator for assessing the degree of utilization. It ranges from 0, i.e., the yield stress is not yet attained, to 1, i.e., the peak stress is attained, and softening material behavior is initiated. Figure 18 shows the contour plot of α_p in the shotcrete shell, together with the magnitude of the displacement of the surrounding rock mass. Higher degrees of utilization of the upper part from the side wall to the crown can be observed, indicating inelastic material behavior of the latter. In contrast, no inelastic behavior is predicted for the lower part of the shotcrete shell, which is a consequence of the strongly delayed closure of the shotcrete shell.

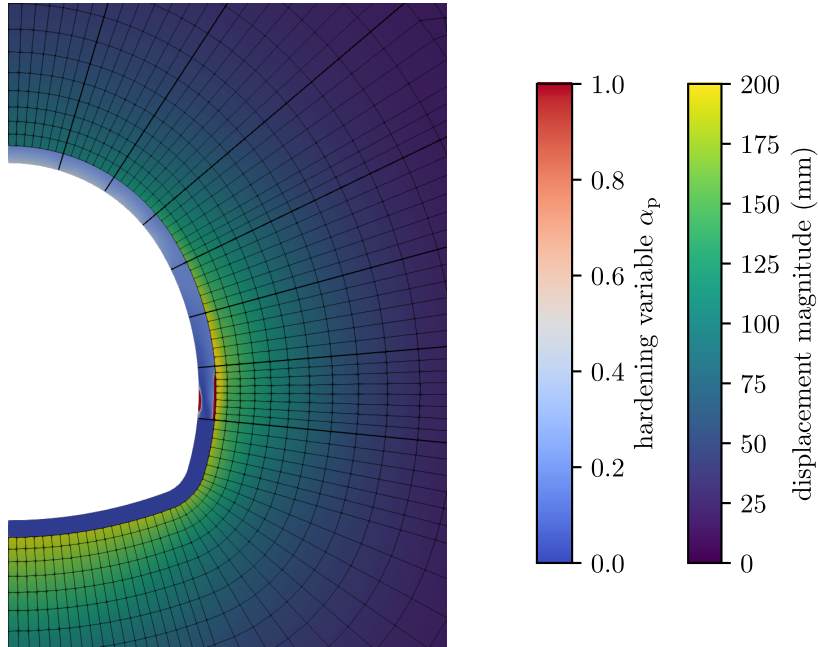


Figure 18: Distribution of the hardening variable α_p in the shotcrete shell, serving as an indicator for its utilization in the range of inelastic material behavior, and the magnitude of the displacement of the surrounding rock mass.

Figure 19 shows the predicted damage ω in the shotcrete shell, together with the accumulated equivalent inelastic strain $\tilde{\epsilon}^p$ in the surrounding rock mass. The latter is defined as

$$\tilde{\epsilon}^p = \sqrt{2/3 \epsilon_{ij}^p \epsilon_{ij}^p} \quad (142)$$

and serves as an indicator for inelastic behavior. Only minor damage occurs at the bottom of the upper part of the shell, mainly at the rock-shotcrete interface. This is due to large inelastic deformations predicted in the adjacent rock mass, resulting locally in large shear deformations imposed on the shotcrete shell. However, the predicted damage is only local, and it does not result in collapse of the tunnel structure. No damage is predicted in the lower part of the shotcrete shell. It can be concluded due to the strongly delayed construction of the lower part of the shotcrete shell an ineffective utilization of the shotcrete shell is obtained.

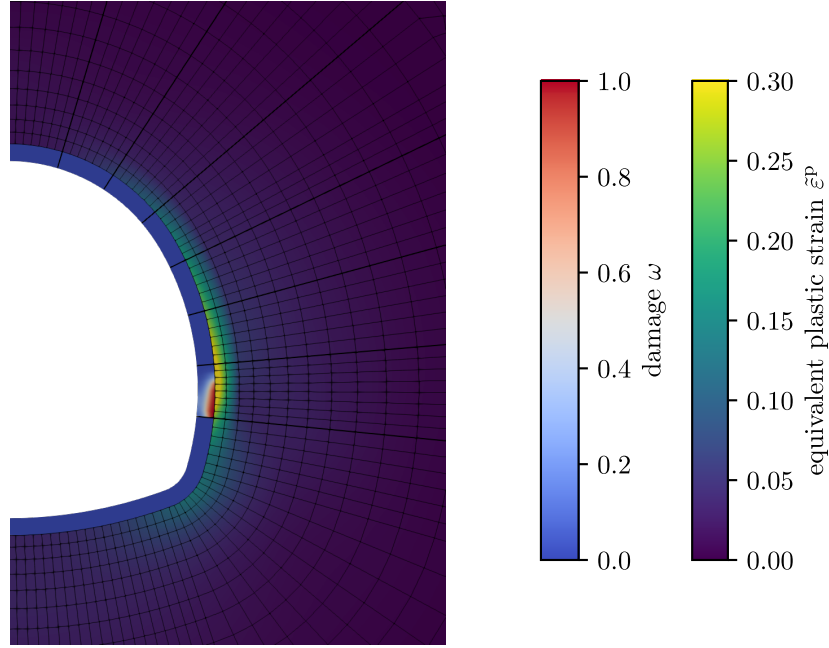


Figure 19: Predicted damage ω in the shotcrete shell and equivalent inelastic strain $\tilde{\epsilon}^p$ in the surrounding rock mass.

6.6 Case study II: Influence of the Construction Procedure of the Shotcrete Shell on the Predicted Time-dependent Structural Response

For demonstrating the influence of the shotcrete shell on the predicted structural response, in case study II the installation of the complete shotcrete shell is assumed in one step, i.e., the lower part of the shotcrete shell is installed simultaneously with the upper part in simulation step (ii).

Figure 20 shows the predicted vertical crown and horizontal side wall displacements, compared to those from case study I. Considerably smaller deformations are predicted in case study II, i.e., a stiffer response of the shotcrete shell is obtained. The stiffer response is a consequence of the fast closure of the invert, resulting in a closed load bearing ring, which allows carrying the imposed loads by the surrounding rock mass in an efficient manner.

Figure 21 shows the predicted tangential stress in the shotcrete shell at the rock-shotcrete interface and the free surface of the crown and the side wall, respectively. Different from case study I, the stress state in the shotcrete shell is characterized by a nearly uniform distribution of tangential stresses across the thickness. Furthermore, the tangential stress at the crown is similar to the one at the side wall.

For comparison of the degree of utilization with case study I, Figure 22 depicts the contour plot of α_p in the shotcrete shell. A nearly uniform degree of utilization is predicted for case study II without attaining the material strength of the shotcrete. Accordingly, compared to case study I a lower degree of utilization is predicted, and no damage occurs. Hence, the performance of the shotcrete shell, constructed with a fast closure of the invert, is significantly superior to the one with a delayed two-stage construction sequence, as employed in case study I.

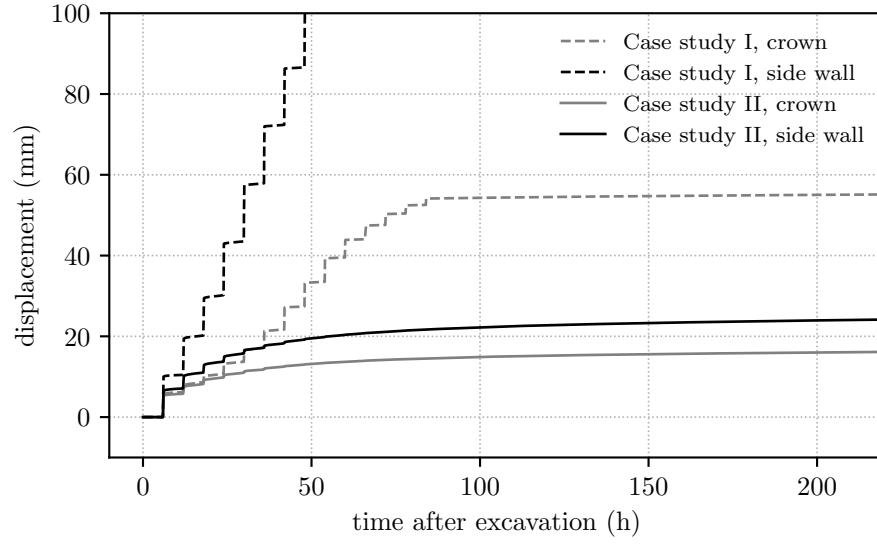


Figure 20: Evolution of the vertical crown displacement and the horizontal side wall displacement: Case study I, considering the sequential construction scheme of the shotcrete lining with the lower part installed 84 h after the upper part (dashed curves), and case study II, considering the complete construction of the shotcrete shell in one step (solid curves).

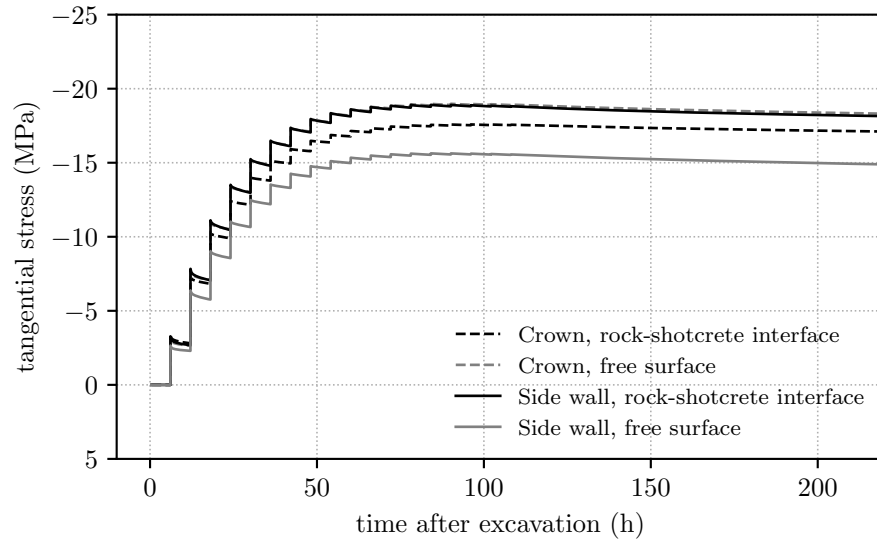


Figure 21: Evolution of the tangential stress, predicted at the rock-shotcrete interface and the free surface of the crown (dashed curves) and the side wall (solid curves), respectively.

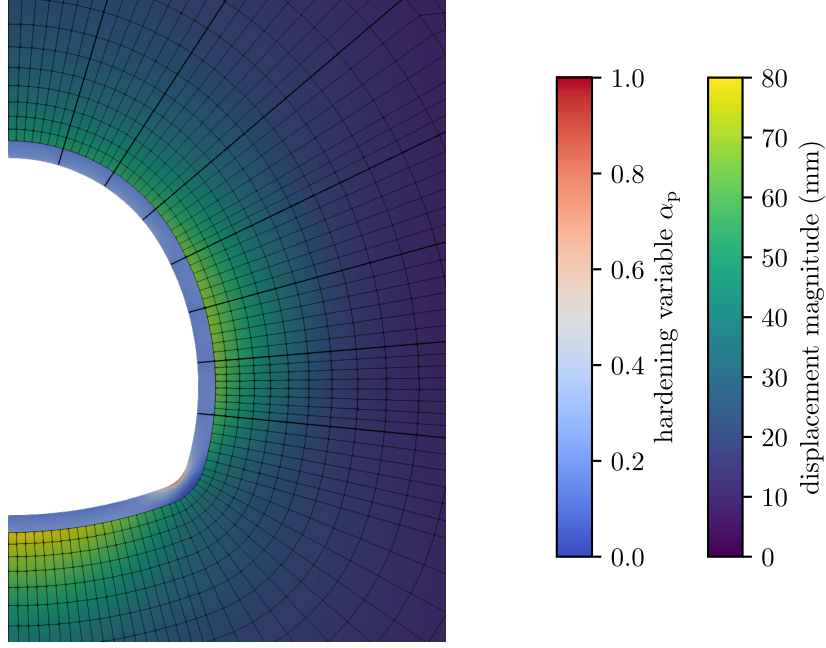


Figure 22: Distribution of the strain-like hardening variable α_p in the shotcrete shell and the magnitude of the displacement of the surrounding rock mass for case study II.

6.7 Case study III: Influence of Nonlinear Creep Behavior of Shotcrete

In case studies I and II, nonlinear creep behavior of shotcrete was considered by means of the amplification function $F(\bar{\sigma}(t), t)$ in the SCDP model, defined in Eq. (16), which enters Eqs. (3) and (4). In case study III, the influence of nonlinear creep behavior of shotcrete on the predicted structural response is demonstrated. To this end, the model from case study II is considered, together with restricting creep of shotcrete to linear creep behavior only by assuming $F(\bar{\sigma}(t), t) = 1$.

Figure 23 depicts the predicted time-dependent vertical crown and horizontal side wall displacements, together with respective results considering nonlinear creep of shotcrete from case study II. It can be seen that considerably smaller displacements are predicted if nonlinear creep of shotcrete is neglected.

Figure 24 shows the time-dependent evolution of the tangential stress at the crown and the side wall. In contrast to the substantial difference in the predicted displacements between case study III and case study II, only slightly higher compressive stresses are predicted compared to case study II (Figure 21).

As expected, neglecting nonlinear creep results in a higher degree of utilization, and in the present case the material strength is attained, as shown in Figure 25. Therein, it can be observed that the material strength is attained in the shotcrete shell at the location of maximum curvature, i.e., $\alpha_p \geq 1$. However, only minor damage occurs ($\omega < 1 \times 10^{-2}$). This case study demonstrates the importance of considering nonlinear creep behavior of shotcrete in numerical simulations, as nonlinear creep results in substantially larger deformations.

It is emphasized that nonlinear creep results in the growth of micro cracks in the material, and accordingly, the increase of material damage, which is not represented by the current version of the SCDP model. An extension of the SCDP model for representing coupling effects between nonlinear creep and material damage is still pending.

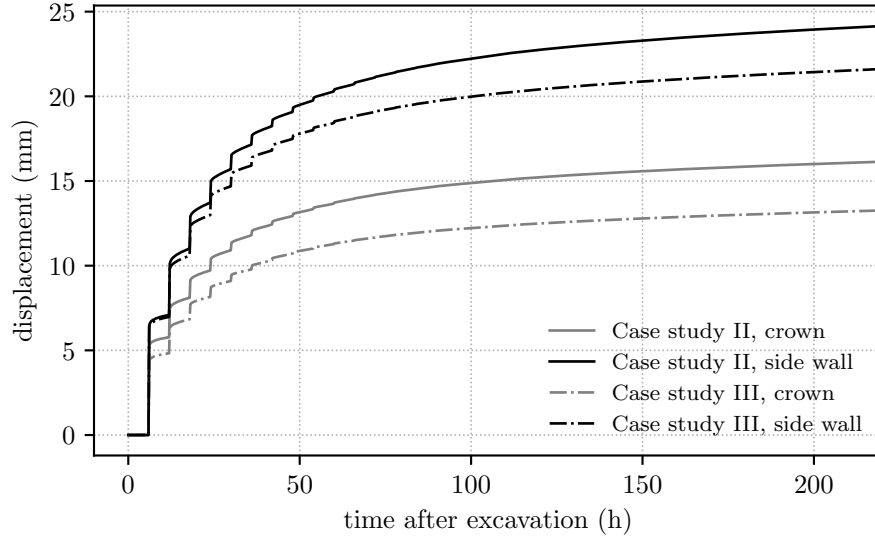


Figure 23: Time-dependent evolution of the vertical crown displacement and the horizontal side wall displacement, considering nonlinear creep of shotcrete (solid curves) and linear creep only (dash-dotted curves).

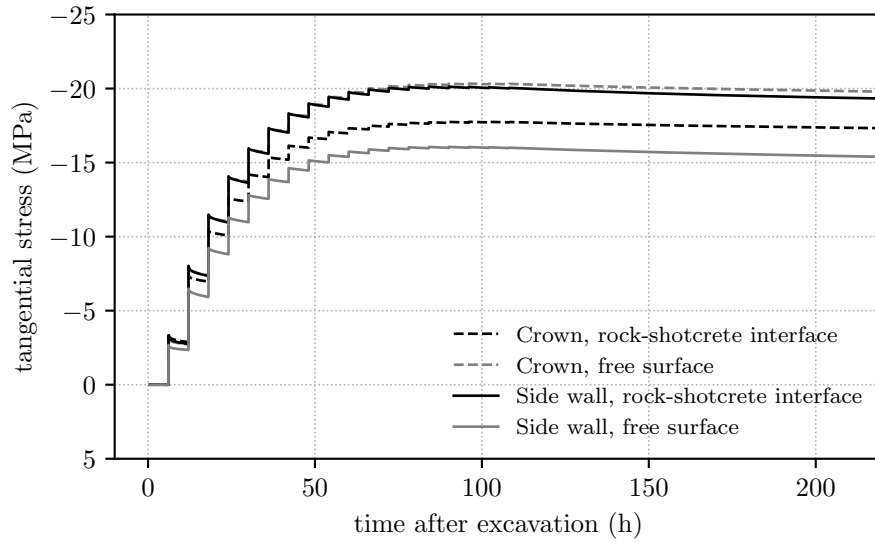


Figure 24: Evolution of the tangential stress considering linear creep only in case study III, predicted at the rock-shotcrete interface and the free surface of the crown (dashed curves) and the side wall (solid curves), respectively.

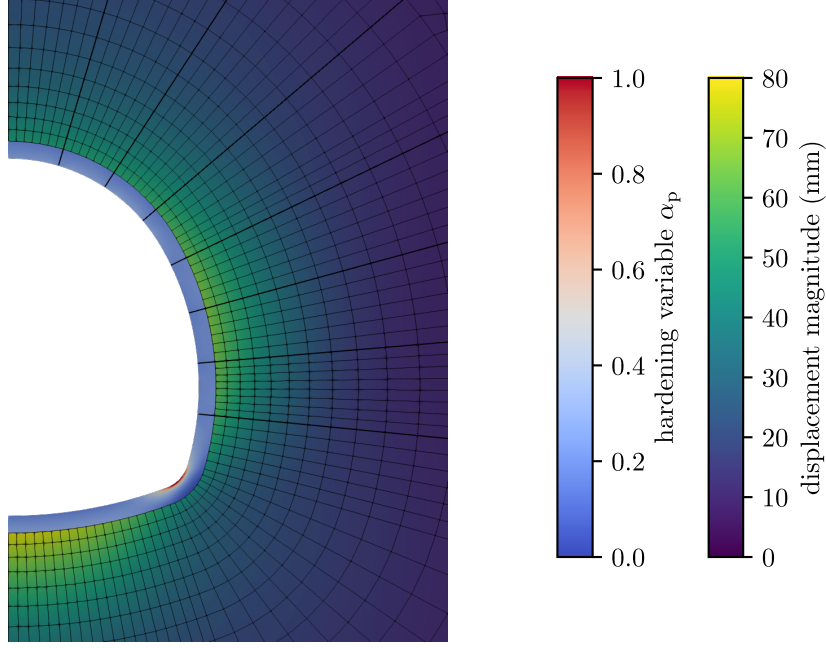


Figure 25: Distribution of the strain-like hardening variable α_p in the shotcrete shell and the magnitude of the displacement of the surrounding rock mass for case study III.

6.8 Case study IV: Prediction of Potential Structural Collapse

By means of the SCDP model, potential collapse of the tunnel structure can be captured. This is demonstrated by considering again case study II, characterized by placing the shotcrete lining in one step, together with the geostatic stress increased by 70 %. Hence, considerably larger deformations are imposed on the shotcrete shell. Consequently, failure of the tunnel structure is predicted in the 6th excavation step, i.e., at the shotcrete age of 36 h.

Figure 26 depicts the evolution of the respective vertical crown and horizontal side wall displacements for case study IV. A sharp increase of the displacements is predicted for the excavation steps before collapse, accompanied by pronounced creep behavior during the idle steps.

Figure 27 shows the distribution of the hardening variable α_p in the shotcrete shell immediately before the predicted collapse. It can be seen that the material is utilized to a considerably higher extent in large areas of the shotcrete shell. In particular, at the crown and the invert the material strength is attained.

Figure 28 shows the predicted damage ω in the shotcrete shell immediately before the predicted collapse of the tunnel structure. The initiation of failure of the shotcrete shell is predicted at the invert, manifested by the formation of a diagonal shear band across the shotcrete shell. This case study confirms the importance of advanced constitutive models, representing softening material behavior, which is indispensable for predicting potential failure of the shotcrete lining.

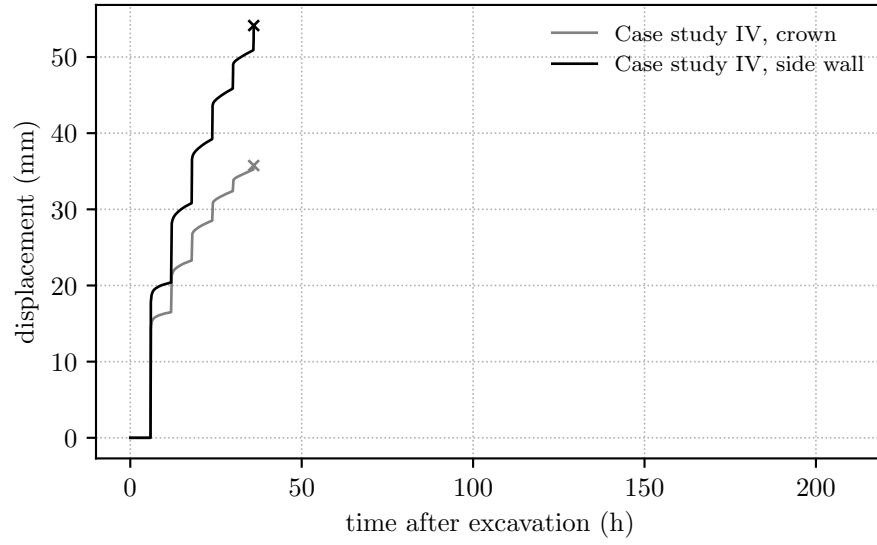


Figure 26: Evolution of the vertical crown displacement and the horizontal side wall displacement considering an increased geostatic stress in case study IV, resulting in failure of the tunnel structure in the 6th excavation step.

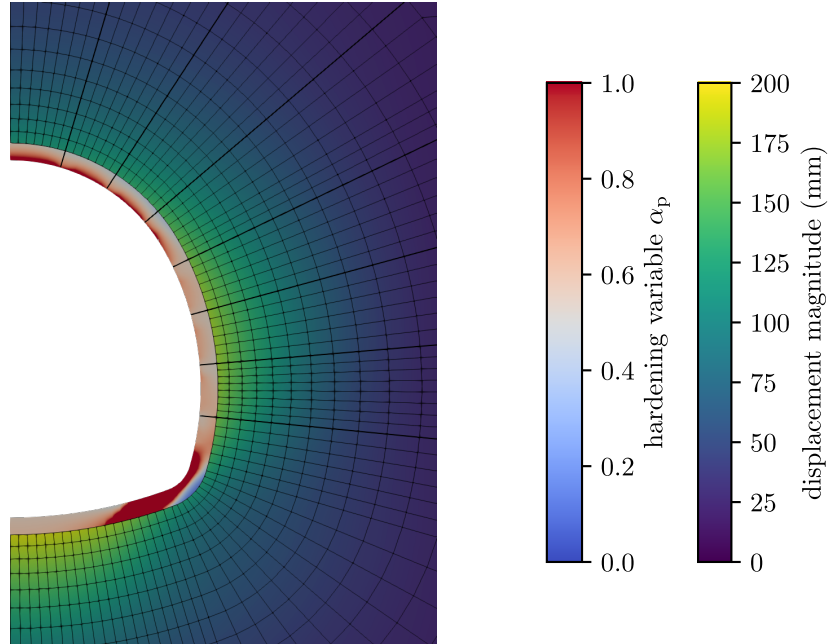


Figure 27: Distribution of the strain-like hardening variable α_p in the shotcrete shell and the magnitude of the displacement of the surrounding rock mass for case study IV.

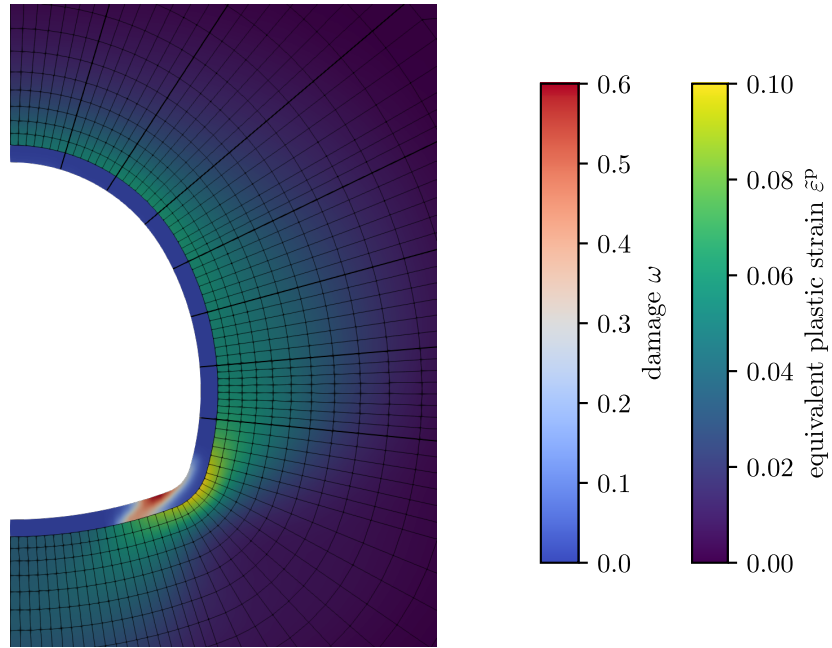


Figure 28: Predicted damage ω in the shotcrete shell and equivalent inelastic strain $\tilde{\epsilon}^p$ in the surrounding rock mass immediately before the predicted collapse of the tunnel structure in case study IV.

7 Summary, Conclusions and Outlook

The present review paper addressed the complex, time-dependent material behavior of shotcrete by the synthesis of experimental investigations and numerical modeling. Central focus was on the SCDP model, a damage plasticity model representing time-dependent material properties, hardening and softening behavior, inelastic behavior, stiffness degradation, and shrinkage and creep of shotcrete. It is based on three constituents, comprising the damage plasticity model by Grassl and Jirásek (2006a), the solidification theory for concrete creep by Bažant and Prasannan (1989a), and semi-empirical shrinkage models according to (ACI Committee 209, 1992) or (Bažant and Panula, 1978). For regularizing softening material behavior in finite element analyses, an advanced regularization scheme is employed. To this end, the over-nonlocal implicit gradient-enhanced formulation proposed by Poh and Swaddiwudhipong (2009b) for the concrete model by Grassl and Jirásek (2006a) was adopted for the SCDP model.

The SCDP model was calibrated based on the results of a comprehensive experimental program on a modern wet-mix shotcrete composition employed in tunnel construction. In particular, the evolution of the Young's modulus and the uniaxial compressive strength, and the shrinkage and creep behavior were determined on shotcrete specimens. Additionally, wedge splitting tests were conducted for investigating the specific mode I fracture energy. The specimens for the lab tests were sampled directly at the construction site of the Brenner Base Tunnel. Based on a comparison with the test results, it was shown that the calibrated SCDP model represents the experimentally observed material behavior very well.

Next, a finite element study of deep tunnel advance was presented. It focused on an application in engineering practice, i.e., a stretch of the Brenner Base Tunnel with a single permanent shotcrete lining. A 2D finite element model was developed, accounting for the individual steps of the construction process and the nonlinear, time-dependent material behavior of shotcrete using the SCDP model. The finite element model was calibrated by geodetic measurement data from the construction site, such that the computed deformations largely coincide with the measured ones.

The presented numerical study highlighted various aspects of the predicted influence of the shotcrete shell on the mechanical response of the tunnel structure. As shown by a comparative study, fast closure of the shotcrete shell allows carrying the imposed load by the surrounding rock mass in an efficient manner, resulting in considerably smaller deformations compared to a sequential construction of the shotcrete shell with a late closure of the invert. Furthermore, it was demonstrated that nonlinear creep has a considerable influence on the predicted deformations, i.e., substantially smaller deformations are predicted if nonlinear creep of shotcrete is neglected. Finally, it was shown that potential collapse can be predicted by means of the SCDP model. For capturing such potential collapse of the tunnel structure, advanced constitutive models representing softening material behavior are indispensable.

The presented experimental and numerical results may serve for stimulating future research activities. Concerning future experimental programs, the following proposals are made:

- The discussed technique of using sprayed specimens for material tests on shotcrete has proven as a simple and effective alternative to drill cores. Since no preparation of the specimens by cutting or grinding is necessary, experiments can be started at very early shotcrete ages.
- Nonlinear creep of shotcrete shells is considered as an important issue in tunneling: Due to

the imposed deformations, high degrees of loading can arise, resulting in turn in nonlinear viscoelastic material behavior of shotcrete. As an experimental study on nonlinear creep of shotcrete is not yet available in the literature, such an investigation by performing creep tests with different levels of the applied sustained stress is strongly proposed.

- For validating the deformations of shotcrete shells predicted in numerical simulations, detailed in-situ measurements of both displacements and strains are required. Routine geodetic measurements performed during tunnel advance allow only for a very limited comparison and validation of the those deformations. More sophisticated techniques, e.g., based on optical fibers installed in the shotcrete shell, may allow a deeper insight into the actual mechanical response of the shotcrete lining during tunnel advance.

In this regard, a follow-up experimental program on shotcrete is currently conducted at the University of Innsbruck. It comprises laboratory tests for determining (i) the nonlinear creep behavior, (ii) the temperature evolution due to hydration, (iii) the desorption isotherm, (iv) the evolution of porosity, as well as (v) a series of wedge splitting tests focusing on the evolution of the specific mode I fracture energy.

Regarding constitutive modeling of shotcrete, in particular regarding the SCDP model, and the application of advanced shotcrete models in numerical simulations of tunneling, the following prospects are given:

- The SCDP model represents nonlinear creep behavior by means of the amplification function of the solidification theory by Bažant and Prasannan. Because of the lack of experimental results on nonlinear creep of shotcrete, the validation of this formulation is still pending, and no coupling effects between nonlinear creep and material damage are currently considered. Novel experimental results may facilitate the development of an improved formulation, considering such coupling effects between creep and damage.
- Due to the lack of experimental data, a time-independent softening modulus is currently assumed in the SCDP model. Hence, a more sophisticated model, considering time-dependent softening material behavior of shotcrete will be developed, accompanied by a new experimental program on the evolution of the specific mode I fracture energy during hydration of shotcrete. In this context, Lackner and Mang (2004) proposed a thermodynamic framework for considering the evolution of the specific mode I fracture energy dependent on the degree of hydration.
- Due to potential localized material failure in the shotcrete shell, large stress and damage gradients may occur in numerical simulations. For representing such large gradients with classical finite elements, sufficiently fine meshes are required, which might be computationally very demanding especially in 3D. In this regard, isogeometric methods (T. J. Hughes, Cottrell, and Bazilevs, 2005; C. V. Verhoosel, Scott, T. J. Hughes, and De Borst, 2011; Beer, S. Bordas, et al., 2015; Nguyen, Anitescu, S. P. Bordas, and Rabczuk, 2015) may be employed for representing such large gradients more accurately while maintaining feasible computation times.
- Tunnel Information Modeling (TIM) (Ninić and Meschke, 2015; Koch, Vonthron, and König, 2017; Stascheit, Ninić, Meschke, Hegemann, and U. Maidl, 2018; Ninić, Bui, Koch, and

Meschke, 2019) facilitates the development of digital twins of tunnel constructions over the complete life span, i.e., the design and planning phase, the construction phase and/or the service phase. Advanced constitutive models like the SCDP model may enhance such digital twins by supporting the design and construction of tunnels as well as also for long term safety assessments during the service time of deep tunnels.

In the presented numerical study, isotropic material behavior was assumed for the sake of simplicity. However, stratified rock mass, such as Innsbruck quartz phyllite, exhibits distinct orthotropic or transversely isotropic material behavior resulting in, e.g., asymmetric deformation patterns with respect to the vertical symmetry plane of the tunnel, which cannot be represented by isotropic rock models. Thus, the development of a novel continuum-based model for describing orthotropic material behavior of rock mass is currently in progress, enabling together with the SCDP model improved predictions of the deformation and the stress state in the tunnel structure.

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